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Large-scale clustered attractor network model evokes persistent activity in vitro through novel network clamp method

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A thesis submitted in partial fulfillment of the requirements for the Doctor of Philosophy degree
in Neuroscience

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Abstract

Ring attractor models have been instrumental in theorizing the neural basis of working memory, with the seminal work of Compte et al. (2000) demonstrating the role of recurrent excitation and slow NMDA dynamics in sustaining persistent activity. However, these models possess significant limitations: they implement a connectivity profile based on short-range excitation that does not reflect the known topology of cortical microcircuits in the prefrontal cortex. Additionally, the ring model framework cannot fully capture the irregular firing patterns observed *in vivo*. We hypothesize that because of the scale of this canonical ring model, the synaptic signals operate at a larger amplitude than is biologically plausible and would not evoke a physiological response in a real neuron.

To test this hypothesis, we employed our novel "network clamp" technique, injecting simulated synaptic conductances from the model into biological neurons *in vitro*. This revealed that the excitation-dominated input from the classical model drives the neuron into a non-physiological, overstimulated state characterized by sustained depolarization block and failure to generate action potentials, thus failing to replicate the model's predicted output.

To address these limitations, we developed a novel spiking neural network that includes a more biologically realistic, clustered excitatory connectivity at much larger scale. In this clustered model, persistent activity emerges from coordinated increase in AMPA, NMDA, and GABA conductances. Those three conductance signals interact to generate a fluctuation-driven regime, which, when injected into real neurons using network clamp, evoked a physiological spiking activity that aligned closely with the model's prediction.

The difference between the two models is that while the persistent activity in the ring model is dominated by slow excitation that maintains suprathreshold input, the persistent activity in our model is maintained by input fluctuations that drive spiking while voltage remains below threshold on average, a dynamic that avoids depolarization block in biological neurons.

More broadly, this work introduces our network clamp method to test the biological validity of computational models in living neurons. This approach provides a powerful tool for studying the complex interplay between single neuron properties and emergent network dynamics, establishing a crucial bridge between theoretical and experimental neuroscience.

Keywords

spiking neural networks, whole-cell patch-clamp, network clamp, prefrontal cortex,
working memory

Summary for Lay Audience

The brain is made of cells called neurons that communicate by sending signals to each other. Large groups of these interconnected neurons influence each other in complex ways, and their combined activity is what produces thought and memory. Researchers build computer simulations of these networks of neurons to understand how their interactions give rise to cognitive abilities. An important example of such cognitive ability is working memory, the ability to hold information, like a phone number, in mind for a short period of seconds. The precise way information is encoded in working memory is still debated.

A leading theory suggests that working memory is maintained by circuits of neurons which self-sustained the activity, carrying the information even after the stimulus disappears from view. Computer simulations can be extremely valuable to model network of neurons, by comparing these simulations to real brain recordings, researchers can see which kinds of network designs are most likely to produce the activity patterns observed during a working memory task. However, existing models have limitations. Indeed, they do not perfectly mimic certain aspects of the brain's natural activity.

To address this, we have developed a new network model with an architecture that more closely mirrors the cortex at the surface of the brain. Our model operates in a way that more closely matches the neural activity pattern thought to underlie working memory. To show the plausibility, we took the signals received by a neuron within our model and delivered them to a real, living neuron in a lab dish. The neuron responded with a pattern of electrical activity that closely captured what our model predicted.

This approach provides a valuable new method to assess theories about the circuits of neurons in the brain. Such an experiment can help us understand how the properties of single neurons and the signals generated by the wiring of networks interact to create brain function, which is incredibly difficult to study directly in an intact brain.

Epigraph

“Ever present, never twice the same. Always changing, never less than whole”

Robert Irwin

Co-Authorship Statement

This thesis is presented in monograph format. It is centered on a primary research manuscript, currently in preparation, which is extended and adapted into Chapters 3 and 4.

Chapter 1 is an introductory chapter that provides background, defines key terms, states the research problem and aims. It is my own work, and it outlines the methodology and structure of the thesis.

Chapter 2 is a review of the cortical microcircuits underlying persistent activity and the neural network models developed to study the mechanisms of persistent activity and is my own work.

In chapter 3 and 4, I share co-first authorship with Sam Mestern and Lawrence Oprea. Sam developed network clamp and performed preliminary network clamp and participated in the development of our large-scale clustered model. Lawrence participated in running scans of simulations for our model and was involved in the writing of the methods section. Both Sam and Lawrence provided feedback on the figures. I led the patch-clamp experiments, wrote the chapters as presented here, and created the figures.

Lyle Muller supervised the research and contributed to conceptualization, investigation, and writing of the manuscript draft and provided feedback to the analysis and figures. Wataru Inoue supervised the research and provided feedback for the experiments.

Acknowledgments

I would like to thank my supervisors, Dr. Lyle Muller and Dr. Wataru Inoue, for giving me the opportunity to work with them as well as for their guidance and patience over the last 5 years. I would like to express my sincere gratitude to fellow students in both Inoue lab and Muller lab that have taken the time to help me with this project, Sam, Gabriel, Lawrence and Chris. I also thank all the other members of both labs that have supported me in ways both big and small.

I also wish to thank all my colleagues and friends who made my time in London memorable, and whose encouragements helped me immensely in bringing this thesis to completion. Aish, Gabriel, Roberto, Sam, Xin, Mehrzad, Pria, Moe, Alex, Julia, Aoi, and all my friends in the Departments of Philosophy and the Department of Physics and Astronomy.

I would like to thank my family and friends in France for being so supportive all the time from across the ocean.

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Chapter 1

1 Introduction

This introductory chapter provides background, defines key terms, states the research problem and aims, we also outline the methodology and structure of the thesis. First, we present the concept of working memory (WM) and the role of dorsolateral prefrontal cortex (dlPFC) in spatial WM as well as the seminal theory of persistent activity mechanism. This chapter also explains the basis of whole-cell patch-clamp, the electrophysiological method used throughout this thesis. We describe its use for measuring the electrical intrinsic properties of single neurons, used to inform researchers of their integrative properties, and how this technique can be extended via dynamic clamp to inject biological neurons with more complex and realistic stimulations. We then introduce the concept of spiking neural network, and the theory of attractor network for modeling persistent activity. Finally, we describe the goals and organization of this thesis.

1.1 Working memory and persistent activity

Working memory (WM), the ability to maintain and manipulate information over short periods of time, forms the core basis of higher cognition (Baddeley, 1992). During visual WM tasks, positional information about a visual cue must be remembered during a delay period, of a few seconds in the absence of sensory input. During this delay period, neurons in the PFC of primates have been shown to exhibit sustained firing that outlasts the initial visual stimulus (Bauer & Fuster, 1976; Funahashi et al., 1989; Fuster & Alexander, 1971; Kubota & Niki, 1971). It is generally accepted that this persistent activity represents the sensory information encoded in the neural circuits, as it correlates with WM task success. The PFC area serves as the central executive for WM, integrating multi-modal information to support higher cognitive functions such as attention, decision-making, and planning (Friedman & Robbins, 2022; Funahashi, 2017; Miller & Cohen, 2001). Its critical role in WM has been demonstrated through different methods (Bauer & Fuster, 1976; Courtney et al., 1998; Niki & Watanabe, 1976).

Yet many aspects of the underlying mechanism by which persistent activity is generated, maintained and modulated are still debated (Fiebig & Lansner, 2017; Lundqvist et al., 2016; Mongillo et al., 2008).

1.2 Dorsolateral prefrontal cortex role in spatial working memory

The dlPFC plays a central role in spatial WM and attention. There has been a large body of research recording the neural activity in the dlPFC of non-human primate performing spatial WM tasks (Fig. 1.1 B; for review see Constantinidis et al., 2018; Funahashi, 2017; Riley & Constantinidis, 2016). To test spatial WM in animal models, researchers often use the task called oculomotor delay response task (ODR).

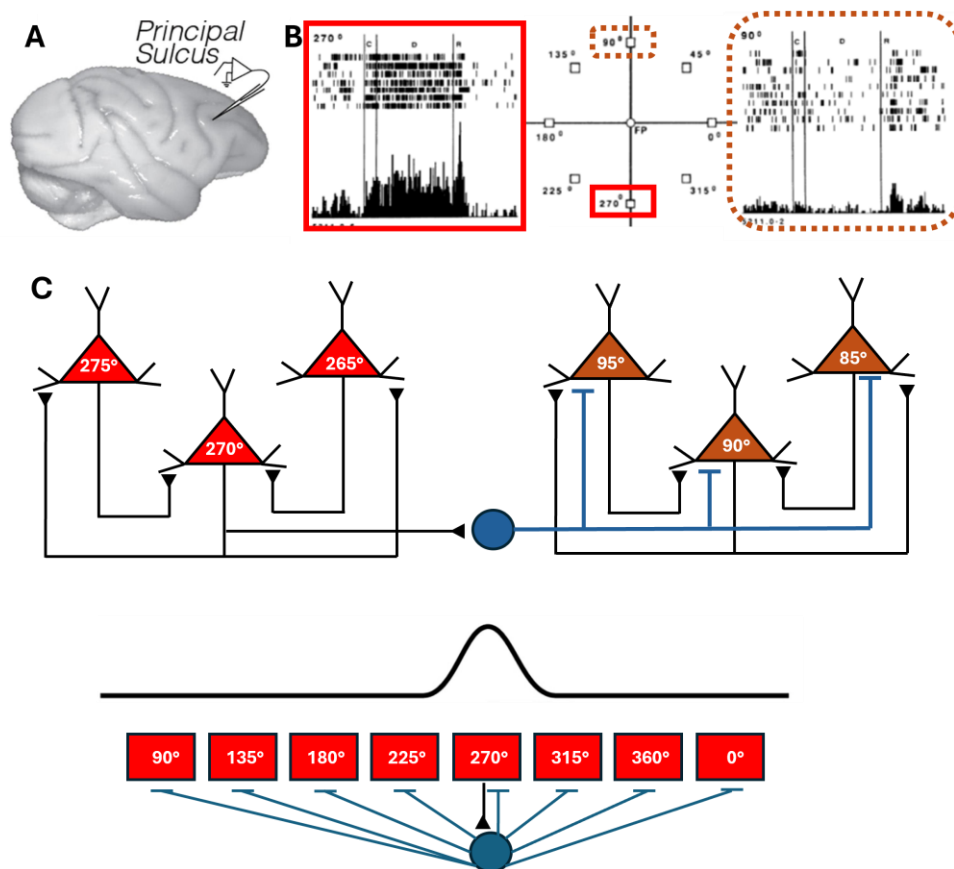


Figure 1-1. Canonical theory of persistent activity.

(A) Seminal extracellular recordings of single units in the dlPFC of non-human primates performing an oculomotor delayed response task (ODR) report sustained activity during the delay period, adapted from (Curtis & Sprague, 2021). B. Neurons are tuned to respond to a specific position of the cue in the visual field, adapted from (Funahashi et al., 1989). C. It is proposed that activity is maintained by recurrent connections between similarly tuned excitatory neurons while lateral inhibition suppress activity in neurons tuned to far away locations to maintain a stable activity pattern (Goldman-Rakic, 1995).

In the ODR task, the subjects must memorize the position of a transient visual stimulus in one of eight locations, maintain this spatial information in memory during a short delay period without the visual stimulus, and then accurately recall the position by making a saccadic eye movement to the target location after the delay. During the delay period of ODR tasks, dlPFC neurons exhibit an increase in spiking activity that is thought to encode information about the presented stimulus in WM (Fig. 1-1; Funahashi et al., 1989, 1993; Kubota & Niki, 1971; Niki & Watanabe, 1976; Wilson et al., 1993). This delay activity is present in neurons specifically tuned to respond to one position of the visual cue. The spatial modality of WM has been a central focus of research because it presents robust stimulus-specific activity that is highly correlated with task performance and easily measurable in animal models (Chafee & Goldman-Rakic, 1998; Goldman-Rakic, 1995; Malloy, 2011; Romo et al., 1999). Therefore, the dlPFC provides a key experimental paradigm for dissecting the neural basis of WM.

Since the seminal work recording the persistent activity *in vivo* in non-human primates, researchers have worked to elucidate what physiological properties allow the dlPFC microcircuit to exhibit persistent activity (Compte, 2006; Riley & Constantinidis, 2016; X.-J. Wang, 2021; Zylberberg & Strowbridge, 2017). To achieve this, numerous characteristics must be investigated at different scales, using various experimental methods (Chafee & Goldman-Rakic, 1998; Finn et al., 2019; Henze et al., 2000; Melchitzky et al., 2001; Watakabe et al., 2023).

1.3 Neuronal integration of synaptic inputs

Cognition emerges from the activity of interconnected neurons in the brain (Park & Friston, 2013). The dynamics of this activity depend on the structure of this interconnectivity, the synaptic properties and the intrinsic properties of neurons (Timofeev et al., 2000). Indeed, neural populations in the brain present a wide diversity of properties that determine how they each integrate inputs differently. Effort has been carried out for decades to classify neurons into classes, based on morphological, electrophysiological and molecular properties, with the objective to link their integrative properties to their function (Gouwens et al., 2019).

Neurons integrate synaptic inputs to determine whether to generate an action potential (AP), also referred to as spike. This process begins when pre-synaptic neurons transmit signals to the dendrites of a postsynaptic neuron via a synapse (Kandel et al., 1991). Each input triggers a change in the postsynaptic neuron's membrane potential, or voltage, known as a post-synaptic potential (PSP). The PSPs which depolarize the membrane potential, are termed excitatory PSPs (EPSPs), while the PSPs that hyperpolarize the membrane potential are termed inhibitory PSPs (IPSPs). For an AP to be initiated, the cumulative effect of EPSPs and IPSPs must bring the membrane potential at the axon hillock to a critical threshold value. As PSPs propagate through the dendritic tree, they interact through temporal and spatial summation (Spruston et al., 2016). The morphology of the dendritic tree is crucial, as it determines how PSPs summate and propagate toward the soma. A larger dendritic tree not only supports a greater number of synapses but also influences the spatial and temporal integration of inputs (Kandel et al., 1991). Beyond simple addition or subtraction, synapses can also modulate the amplitude of EPSPs generated nearby through dendritic integrative mechanisms, such as shunting inhibition and active dendritic properties. These nonlinear interactions within the dendritic arbor represent a crucial computational step, shaping whether the neuron reaches the threshold for AP initiation (Spruston et al., 2016). While the number of synapses can be estimated by the number of dendritic spines, it is difficult to fully grasp their precise interaction experimentally.

The next section describes how electrophysiological properties of a neuron can be measured at the soma using the method called whole-cell patch-clamp electrophysiology. However, it is important to keep in mind that while these techniques capture the integrative properties of a neuron as a whole, they do not fully reveal the complex spatial dynamics of integration occurring within the dendritic arbor.

1.4 Characterization of intrinsic properties with whole-cell patch-clamp electrophysiology

The intrinsic electrophysiological phenotype of a neuron is determined by the specific type of ion channels expressed on its membrane, and their density. Those channels dictate the flow of current across the neuron's membrane, governing how the neuron integrates synaptic inputs (PSPs), generate APs, and produce characteristic firing patterns.

Whole-cell patch-clamp electrophysiology is the primary tool for quantifying these properties, as it provides direct electrical access to the neuronal membrane. In this framework, the neuron's passive voltage properties is modeled as an RC circuit (Molleman, 2002). The capacitance (C_m) reflects the lipid bilayer's charge-separating insulation. The membrane resistance (R_m) determines magnitude of voltage change in response to current stimulation. Their product defines the membrane time constant ($\tau = R_m * C_m$), which describes how long a neuron's voltage will take to get to steady state at stimulation current onset and how long the neuron will decrease before reaching the neuron resting potential after the stimulation is over.

Another key membrane property is the resting membrane potential (V_{rest}), set by passive leak currents. But neurons are also defined by more dynamic properties of their membrane. These active properties, such as its AP threshold, shape and rheobase—the minimum current to elicit an AP—emerge from the specific voltage-gated channels expressed on the neuron membrane.

Whole-cell patch-clamp recordings are conducted in two primary modes. The voltage-clamp configuration holds the membrane at a commanded potential to isolate and measure specific ionic currents, providing data on channel density and kinetics. The current-clamp configuration is the mode used throughout this thesis. In current-clamp configuration, a defined current is injected into the patched neuron to observe the native voltage response.

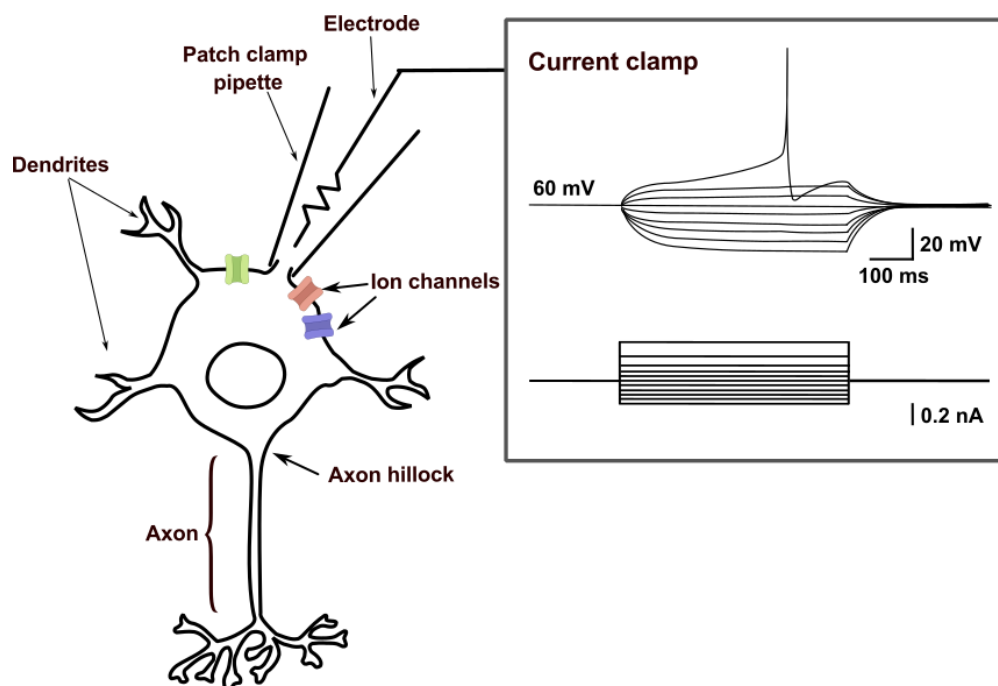


Figure 1-2. The whole-cell patch-clamp configuration.

In the whole-cell patch-clamp configuration, a section of the neuron soma in contact with the hollow patch clamp pipette gets opened by applied pressure. This creates an interface between the intracellular medium in the soma of the neuron and the artificial intracellular solution in the patch clamp pipette. The electrode in the filled pipette records and inject current stimulation. The recorded current is amplified to adapted scale and get digitized to be interpretable by computer software.

In this thesis, we use a standard square-pulse stimulation protocol in current-clamp (Fig. 1-4). This involves injecting transient current pulses of increasing amplitude. This protocol reliably characterizes intrinsic property, measuring how voltage responses scale from subthreshold integration to repetitive firing, and allows for neuronal classification. However, these artificial, step-like inputs differ fundamentally from the complex, dynamic synaptic bombardment neurons experience *in vivo*.

To understand realistic synaptic input, we first examine spontaneous activity recorded in the awake brain.

1.5 Spontaneous activity *in vivo*

For decades, it has been possible to record the spiking activity in the brain of animal models under anesthesia or in the awake state (Evarts, 1964; Hubel & Wiesel, 1968; Wurtz, 1968). Such *in vivo* recordings have revealed that neural activity is characteristically noisy and variable across trials of behavioral tasks (Faisal et al., 2008). How can variable activity generate reliable behavioral response? Additionally, this stochastic activity, often resembling a Poisson process (Softky & Koch, 1993), is also present spontaneously without being evoked by any sensory stimulus (Timofeev, 2000). This raises another important question; why does the brain allocate energy in maintenance of noisy neural activity in the brain that could hinder reliable signal detection and information transmission (Stein et al., 2005). Rather than being detrimental, this random and irregular activity is now understood to play a critical role in supporting cognitive processes (McDonnell & Ward, 2011).

It is thought that irregular interspike intervals (ISIs) allow neurons to achieve a dynamic range of responses, enhancing the ability of neural circuits to encode, integrate, and compute information (Ringach, 2009; Shadlen & Newsome, 1998). The role that spontaneous activity plays for computation of neural circuits has long been debated due to its stochastic nature, but key findings have been reported. First, it has been shown that when a neuron *in vitro* is repeatedly injected with the same fluctuating stimulus, the neuron produces a sequence of spikes with highly reliable and precise timing (Mainen & Sejnowski, 1995). This contrasts with the irregularity recorded in the intact brain when presented repeatedly with identical sensory stimulus (Arieli et al., 1996; Faisal et al., 2008). Subsequent studies have shown that noisy input plays a role in detection of weaker signals (McDonnell & Ward, 2011).

This phenomenon, known as stochastic resonance, suggests that a certain level of background noise can actually help in detecting signals by improving the signal-to-noise ratio (McDonnell & Abbott, 2009). But it has also been recognized that spontaneous activity is can be a representation of the brain's state (Liu et al., 2022; Sadaghiani, 2010). In parallel, the use of multi-electrode arrays recordings and two photon microscopy *in vivo* at large scale reported patterns across time and correlated neural activity across millimeters of cortex, revealing structure of the activity that was not visible at smaller scale (Barson et al., 2020; Busch et al., 2024; Davis et al., 2020; Ringach, 2009).

Spontaneous activity is increasingly viewed as a core substrate for cognition (Vidaurre et al., 2018). While the investigation of its role in cognition is a central and complex question of neuroscience, another major topic of neuroscience is the question of its emergence and maintenance in the cortex, as well as its impact at the single neuron level (Blankenship & Feller, 2010; Destexhe et al., 2003; Uddin, 2020). This spontaneous background activity is an emergent property of recurrently connected networks where the activity is maintained stable by a balance between excitation and inhibition property that emerges from the immense number of recurrently interconnected neurons (Shadlen & Newsome, 1998; Stein et al., 2005; Van Vreeswijk & Sompolinsky, 1996). Intracellular recordings *in vivo* have supported this idea (Haider et al., 2006).

Sanchez-Vives and colleagues have shown that, by using specific extracellular medium in the recording chamber *in vitro*, it is possible to replicate the voltage dynamic of up and down states seen *in vivo* under anesthesia (Sanchez-Vives & McCormick, 2000). However, while pharmacological manipulations in such preparation confirm that the activity pattern emerges from network activity involving both excitation and inhibition, they do not allow for the quantitative characterization of the specific input that generates this voltage response.

In a seminal study on how noisy input affects neuronal firing, Mainen & Sejnowski (1995) used whole-cell patch-clamp to inject neurons with a random current waveform and found that this evoked reproducible firing patterns across trials, in contrast to the trial-to-trial variability observed with constant current stimulation.

This study has been influential for showing the reproducibility of spike patterns when noise signal is used as stimulus. But similarly to classical characterization protocols this study used predefined current stimulation which did not account for the evolution of the voltage of the neuron membrane. *In vivo*, a synaptic input does not cause predetermined amount of current to flow into the neuron. Instead, it triggers opening of ion channels on the membrane of the post-synaptic neuron, and current flow through the membrane according to the instantaneous voltage of the membrane and the amount of ion channel opening at the synapse. This number of ion channels opening dictates the membrane's permeability to current, a property known as conductance.

The constant synaptic bombardment received by neurons *in vivo* not only generates unpredictable spike timing output, but this spontaneous background also has an influence on the intrinsic properties of neurons, changing their integrative properties compared to the context of quiescence *in vitro* (Chance et al., 2002; Fellous et al., 2003). By putting neurons in what has been named a high conductance state, background activity changes how neurons respond to subsequent inputs (Destexhe et al., 2003; Destexhe & Paré, 1999). To demonstrate this effect, researchers used the technique of dynamic clamp to present neurons with controllable input injected *in vitro* in a manner that captures the effect of synaptic stimulation (Prinz et al., 2004).

1.6 Dynamic clamp

It has long been possible to record spiking activity in the brain. More recent *in vivo* recordings methods such as high-density electrode arrays allow to record the neural activity from hundreds of points in the brains of awake behaving animals (Franke et al., 2012), but it is difficult to make a direct link between the activity recorded extracellularly with the input received by the neuron or the class of the neuron (Griffen, 2014).

While it is possible to perform whole-cell patch-clamp electrophysiology *in vivo* in awake or anesthetized animal brain, it remains extremely difficult, and the low yield makes systematic characterization difficult (Haider et al., 2006; Steriade et al., 2001; Timofeev et al., 2000).

These two *in vivo* approaches present the difficulty of interpreting results due to neural variability and the limited perspective caused by experimental constraints.

On the other hand, *in vitro* patch-clamp electrophysiology can be used to repeatedly and systematically collect essential information about the integrative properties of neurons. It has allowed systematic classification of the variability between cell types in a controlled environment (Gouwens et al., 2019). Moreover, standard stimulation protocols in voltage or current-clamp modes do not replicate the voltage dependent input neuron receive through their synapses from their network in the intact brain (Chance & Abbott, 2009).

In dynamic clamp, neither the voltage nor the injected current is maintained at a predetermined value. Instead, the injected current is calculated at each time step of the experiment to reproduce the effect of a specific membrane conductance value (Robinson & Kawai, 1993; Sharp et al., 1993). The intensity of the current stimulation will depend on the desired conductance value as well as the driving force, the difference between the present voltage of the neuron and the reversal potential of the channels being simulated. This simulates the conductance dynamics happening in the membrane *in vivo* (Prinz et al., 2004).

A key application of dynamic clamp has been to show how signals modeling synaptic background activity maintains the input conductance of neurons at a high value, which in turn influences their integrative properties (Chance et al., 2002; Destexhe et al., 2003; Fellous et al., 2003). These studies have used synthetic stochastic signals, approximating the characteristics of spontaneous activity, to study the effect of more realistic input on single neurons *in vitro* using dynamic clamp (Destexhe et al., 2001; Fellous et al., 2003). This type of experiment shows the possibility to study effect of complex conductance input signals on single neurons. Although dynamic clamp has seen increasing adoption in recent years, there have been no reports to date of this use to inject signals derived from network models. With the exception of project carried out in the lab (Ichiyama et al., 2022). In this thesis we injected biological neuron *in vitro* with conductance signals generated in two models of WM.

The next two sections present the key concepts and models used in the literature to model WM and provide essential knowledge to contextualize this result.

1.7 Spiking neural network models

Spiking neural networks (SNNs) are models that use continuous differential equations to describe the internal state of neurons, representing their voltage. The information is transferred between those interconnected neurons through discrete spikes. Because such model explicitly includes a temporal dimension, they offer a more biologically realistic framework for studying neural computations, synaptic mechanism and dynamics of microcircuits compared to traditional artificial neural networks (ANNs) (Yamazaki et al., 2022).

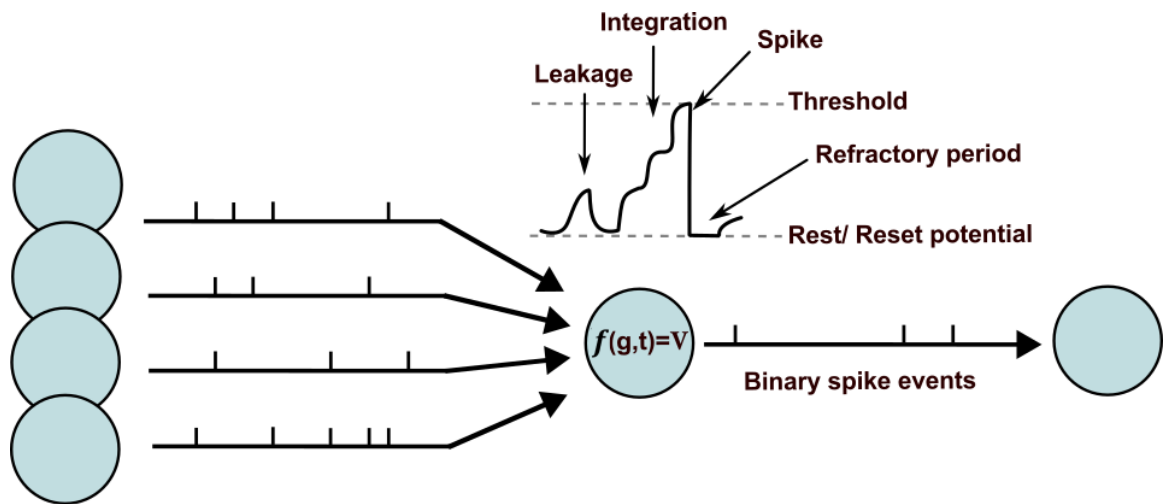


Figure 1-3. Leaky integrate and fire neuron (LIF) in a spiking neural network (SNNs).

SNNs take into consideration the time variable. Action potentials are discrete events that cause the increase of the post-synaptic neuron's voltage. Evolution of LIF neuron voltage is governed by a differential equation that increases at the time of an incoming spike or decreases passively to represent the dynamics of leakage current. Once the voltage reaches threshold, an action potential is generated, and the voltage is maintained to reset potential value for the duration of the refractory period.

This makes neural network models a more adapted tool for investigating how architecture and other network parameters participate in the emergence of different neural activity patterns (Brunel, 2000). Thus, SNNs are crucial tool for understanding the processes happening in cortical circuits (Gerstner et al., 2014; Maass, 1997).

Different types of SNN exist on a wide spectrum of biophysical detail. Hodgkin-Huxley (HH) models describe in fine detail the dynamics of different types of voltage-gated ion channels. Such implementation is the more detailed representation of single neuron dynamics, notably for replicating the full shape of APs, but it is more computationally expensive.

In the present thesis, we use the leaky-integrate and fire (LIF) model (Fig. 1-3) (Gerstner et al., 2014). This implementation captures synaptic integration and passive membrane properties while simplifying the intrinsic conductance dynamics that govern spike generation mechanism in more biophysically detailed Hodgkin-Huxley models, making it more efficient for simulation of large-scale networks.

SNN can also be implemented with different synaptic transmission mechanisms. In network with current-based synapses, synaptic inputs are independent of the post-synaptic neuron's voltage, leading to post-synaptic currents (PSCs) that are invariant to the neuron's membrane potential and previous inputs. Meanwhile, in networks with conductance-based synapses, the post-synaptic currents depend on the difference between the instantaneous voltage and the value of reversal potential. This difference, the driving force of the current, can greatly alter the effective current stimulation. Importantly, despite both types of models often exhibiting similar first-order statistics like average firing rates, models with conductance-based synapses are often better at capturing higher order statistics, particularly cross correlation (Cavallari et al., 2014).

The present thesis focuses on network model of LIF neurons with conductance-based synapses, not only to improve the replication of biophysical regime of *in vivo* activity, but also with the objective to generate conductance synaptic signals that can be used in our dynamic clamp based method.

1.8 Attractor network theory and simulations

A network model is said to reach an attractor state when its dynamics consistently converge to a stable pattern of activity. In other words, the network is drawn to an activity pattern and remains in it overtime without strong enough perturbation.

Attractor network theory provides a conceptual explanation for how neural circuits can maintain patterns of activity through their topology. Analytical, or mean-field, models describe this behavior by averaging the firing of many neurons into a few global variables. These models looked at topology and synaptic mechanisms (Amit & Brunel, 1997a; X.-J. Wang, 1999). Other focused more the excitation–inhibition balance and the irregular aspect activity (Moody et al., 1998; Tsodyks & Sejnowski, 1995).

While analytical approaches such as mean field and neural field equations provide valuable insights into population-level dynamics, they rely on averaging over the interactions between individual neurons (La Camera, 2022). Because SNNs explicitly simulate individual neuron dynamics, they are more adapted to study the effect of noise on temporal structure of spiking activity (Brette et al., 2007).

SNNs can be used to model attractor dynamics in a more biophysically grounded context (Compte, 2000). But because it requires to solve differential equations for individual neurons at each time step, the cost in computer power limits the study of these models to relatively small scale (below 5000 neurons).

Indeed, it requires running simulation many times to search for a set of network parameters that results in the dynamic of interest. The ring attractor model, introduced by Compte et al. (2000) is the best-known example, and will serve as the canonical reference in this thesis. While there is a distinction between the metastable attractor state presented by most SNNs models and the mathematical definition of a stable attractor, as explained in section 2.3.3, we adopt the conventional terminology used throughout the literature and refer to both states as ‘attractors states’.

1.9 Rationale

Canonical attractor ring models generate bistable activity patterns at a firing rate reflecting persistent activity *in vivo*. Because those models rely on a small number of neurons, persistently firing neurons receive excitatory inputs that seem unrealistically high, from the dense local excitatory connectivity.

Additionally, it has been hypothesized that such operational regime pushes neural activity toward regular firing patterns in those models, a condition inconsistent with *in vivo* observations.

1.10 Hypotheses

In this thesis, we hypothesize that a clustered attractor model with sufficiently large scale (50,000 neurons) would overcome the limitations presented in the previous section. A key challenge in computational neuroscience is testing model predictions against biological reality. To validate our model and study the interaction of network activity with single neuron intrinsic properties, we use dynamic clamp and inject the synaptic signals generated within our model, to evoke similar voltage response in biological neurons.

We have the following predictions in mind for this thesis:

- 1- Synaptic signals from canonical ring models will fail to evoke the predicted voltage responses in biological neurons due to their unphysiologically strong excitation.
- 2- A large-scale clustered model will maintain persistent, at physiological firing rate, that better replicates *in vivo* irregular firing. This activity will emerge through weaker excitatory synaptic inputs and more cohesive AMPA and GABA dynamics and will be more biologically plausible compared to the NMDA dominated persistent activity in the ring model. Strong fluctuating GABA inputs will be able to evoke the predicted response in biological neurons, as fluctuation driven regime allows reactivation of ionic channels.
- 3- The fluctuation-driven synaptic signals generated by our large-scale model will successfully evoke relevant voltage responses in biological neurons

1.11 Objectives

The first objective of this thesis is to develop a large-scale clustered SNN model that would better capture the irregularity of *in vivo* activity. To address this goal, we developed a network model with clustered connectivity at larger scale than existing models.

In Chapter 3, we first describe our approach for exploring our model's parameter space. We then compare the firing statistics of our model to *in vivo* activity. Finally, we characterize the synaptic regime emerging in our model in relation to the ring attractor model.

The second objective of this thesis is to test the ability of the ring model and our large-scale clustered model to evoke their respective predictions in the voltage response of biological neurons. To do so, we used our novel network clamp method to inject synaptic conductance signals into patch-clamp neurons *in vitro* and explored the potential of our model and method to study unanswered questions the role of dopamine in the emergence and maintenance of persistent bump of activity (Chapter 4).

Chapter 2

2 Literature Review

2.1 Introduction

Working memory (WM) is a core cognitive function sustained by persistent neural activity in the prefrontal cortex (PFC). Decades of experimental work have identified a range of anatomical and physiological specializations, such as synaptic kinetics, cell type diversity and complexity of microcircuit architecture that are specific to this brain region, and that are thought to underlie this ability. The prevailing hypothesis is that persistent activity in the PFC is maintained by network level mechanisms (Curtis & Sprague, 2021). Computational models offer a unique perspective on how such activity might emerge from network structure and synaptic dynamics (X.-J. Wang, 2021).

In this chapter, we review evidence from both experimental and theoretical research to provide context for the design choices of our model. We argue that while existing models capture essential aspects of persistent activity, they often lack crucial biological constraints and operate in synaptic regimes that fall outside biologically plausible scales.

As we show in the next section of this chapter, many neurophysiological characteristics reviewed here have been shown to correlate with WM performance. Whether these features are causally necessary for generating persistent activity in a plausible operational regime remains an open question. Addressing this fully would require implementing each feature in a model, which is beyond the scope of the present work, but represents an important future direction. We argue instead that the primary substrate for generating persistent activity and metastable bump attractor dynamics in a fluctuation-driven regime is large-scale, clustered recurrent connectivity—building upon the crucial role of NMDA receptors already established by previous models (X.-J. Wang, 1999). Our approach therefore prioritizes the features we hypothesize to be core requirements for the phenomenon, while acknowledging that additional layers of biological realism (interneuron diversity, morphology, neuromodulation) should be added in future work to investigate more complex computations.

2.2 Neurophysiological evidence of dlPFC characteristics underlying persistent activity

A large body of research has been conducted to understand how the PFC, and the dorsolateral PFC (dlPFC) in particular, can sustain persistent activity and support WM (Constantinidis & Procyk, 2004; Friedman & Robbins, 2022; Goldman-Rakic, 1987, 1995). The dlPFC exhibits anatomical and physiological adaptations that distinguish it from sensory and other association cortices, including its intrinsic microcircuit organization (Kritzer & Goldman-Rakic, 1995; Levitt et al., 1993; Pucak et al., 1996), synaptic composition (Van Vugt et al., 2020; M. Wang et al., 2013), and long-range connectivity profile (Watakabe et al., 2023). These features are built on the canonical layered and columnar architecture shared across the neocortex, while showing important specificity.

In this section, we examine these characteristics in detail, alongside the role of neuromodulation, and compare persistent activity in the dlPFC with that observed in other cortical regions. Our goal is to delineate the biological constraints essential for a network model to generate a persistent bump of activity at realistic firing rates and with synaptic signals that fall within plausible physiological ranges.

2.2.1 Laminar and columnar organization - Shared features of all cortices

To understand the specialized mechanisms that enable the dlPFC to sustain persistent activity, one must first appreciate the fundamental architectural plan it shares with the other cortical regions. The PFC is considered to be at the center of executive function (Friedman & Robbins, 2022), but shares many organizing principles that govern cortical structure. It is built upon a canonical six-layered laminar structure and a columnar functional unit, this canonical functional unit is thought to allow the flow of information and integration processes (Douglas & Martin, 2004; Sholl, 1967). Establishing this common framework is crucial, as the unique computational capabilities of the dlPFC arise not from a fundamentally different design, but from systematic variations and specializations imposed upon this shared substrate (Murray & Constantinidis, 2024).

The laminar organization defined by up to six layers, distinguishable by their neural density and cellular composition, constitutes a recurring circuit motif where a neuron's depth determines its input sources and output targets resulting in its functional role. Layer 1 (L1), the outermost layer, consists primarily of the apical dendritic tufts of deep-layer pyramidal neurons and serves as a primary target for long-range, top-down inputs from other cortical and subcortical areas, carrying contextual and modulatory signals (Huang et al., 2024; Shen et al., 2025).

Layers 2 and layer 3 (L2-3) are the principal intracortical processing and association layers. In this layer, neurons form dense local horizontal connections and provide the major source of feedforward projections to higher-order cortical areas. They integrate "bottom-up" sensory information—relayed via Layer 4—with "top-down" contextual information arriving at their apical dendrites in L1 (Shipp, 2007). Layer 4 (L4), the granular layer, is the primary recipient of thalamocortical input, acting as the main gateway for sensory information into the cortex (Harris & Shepherd, 2015; Herkenham, 1980). Its prominence varies across regions; it is prominent in granular cortices, including dIPFC, and is reduced or absent in agranular areas like the motor cortex, reflecting a fundamental difference in how information is received (Shipp, 2005).

Layer 5 (L5) is the major output layer in the neocortex, sending projections to subcortical structures, such as striatum, brainstem, and other cortical areas (Moberg & Takahashi, 2022). The target of such projection is often used to classify the pyramidal neurons in two classes. Intratelencephalic (IT) neurons project to other cortical areas and the striatum, facilitating cortical communication and integration with the basal ganglia for complex functions like planning and learning. On the other hand, extratelencephalic (ET) neurons project to subcortical motor centers, including the thalamus, brainstem, and spinal cord, translating cortical computations into direct motor commands and premotor signals (Moberg & Takahashi, 2022).

Layer 6 (L6) provides top-down control through feedback regulation mechanisms (Thomson, 2010). Part of L6 pyramidal neurons project back to the thalamus to modulate gain of sensory input. Another part of L6 neurons project to local L4 to gate activity of neurons receiving thalamic inputs (Lam & Sherman, 2010). Through those two pathways, L6 fine-tunes the sensory and contextual information flow based on behavioral goals, expectations, and attention (Thomson, 2007).

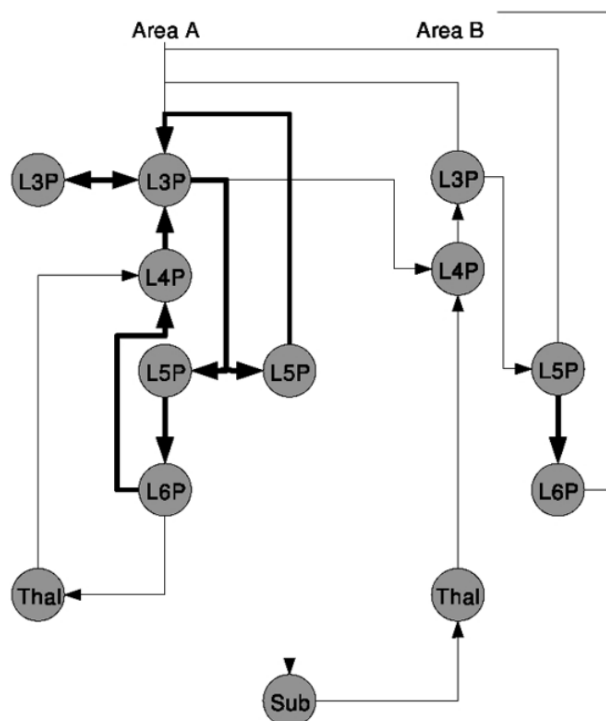


Figure 2-1. Canonical cortical circuit based on visual cortex organization.

The graphic describes the interaction between groups of cortical excitatory (P: pyramidal neuron) neurons and their subcortical relations. The nodes of the graph are organized vertically based on their cortical layer (L) and horizontally distributed between two cortical area.

Superimposed on this horizontal layering is a vertical, columnar organization. In cortical columns, groups of neurons, spanning all layers, seem to present functional similarity and are hypothesized to be fundamental repeating units of processing (Mountcastle, 1997). This columnar motif, along with the laminar connectivity rules, forms the basis of the "canonical microcircuit" based on study of the visual cortex (Fig. 2-1; Douglas & Martin, 2004).

This canonical structure proposes a template of a functional unit that can be generalized to cortical integration process and help understand how cortical regions operate. The following sections will explore the systematic deviations from this template between regions as their structure, composition and connectivity is thought to underlie their specific function. Notably, associative areas such as PFC present specific characteristics in synaptic physiology, cell types, connectivity patterns, and neuromodulation which enable their unique integration process. Specifically, there seems to be variations along the cortical hierarchy that reflect each area's function in processing the flow of information as it goes from sensory cortices to associative ones.

2.2.2 The graded features variations between sensory and associative cortices

The neocortex is functionally organized along a hierarchy, with sensory and associative areas distinguished by their function and structure. Sensory cortices, such as the visual, auditory, and somatosensory areas, receive and relay unimodal input but also carry out the initial stages of processing. In contrast, associative cortices, including the PFC and posterior parietal (PPC) cortices, integrate information from multiple modalities and exert top-down, feedback control over sensory processing. This bidirectional flow of information defines a symmetry of functional influence: feedforward projections carry information up the cortical hierarchy for successive processing steps, while feedback projections return signals that modulate perception and attention (Felleman & Van Essen, 1991; Mesulam, 1998). Across this hierarchy, several anatomical and physiological features vary gradually, forming functional gradient.

Notably, the neural density is higher in sensory cortices and get lower toward the associative cortices (Atapour et al., 2019; Barrett & Simmons, 2015; Goulas et al., 2018). Despite being sparser, the neurons in associative cortices have more complex dendritic arborescence and more dendritic spines which allow them to form more connections (Elston, 2003; Elston & Fujita, 2014).

Along the cortical hierarchy, there is evidence for systematic differences in temporal integration, with longer intrinsic timescales and slower NMDAR-mediated synaptic currents in associative cortices such as PFC compared to early sensory areas (Murray et al., 2014; H. Wang et al., 2008). These properties are consistent with distinct functional roles: association cortices integrate converging inputs over extended temporal windows, whereas neurons in primary sensory cortices operate on narrower temporal windows, enabling them to act as coincidence detectors that require near-simultaneous inputs to fire (Murray et al., 2014; L. Pinto et al., 2022). This slower synaptic decay in associative regions is partly attributable to a higher contribution of GluN2B-containing NMDA receptors at prefrontal synapses, and to a broader gradient of GRIN2B expression that is lowest in primary sensory cortex and higher in association and limbic areas (Burt et al., 2018; Scherzer et al., 1998; H. Wang et al., 2008).

While prolonged NMDA currents provide the essential temporal substrate for integrating converging inputs and enabling sustained reverberation, such recurrent loops of excitation necessitate a robust and dynamically tuned inhibitory system to balance the activity. Inhibition has also been shown to sculpt neuronal selectivity and stabilize the activity patterns for WM maintenance (Constantinidis & Goldman-Rakic, 2002; Rao et al., 1999).

Anatomical studies have shown a systematic shift in interneuron composition along the cortical hierarchy. Sensory cortices, such as primary visual cortex (V1), prioritize fast, reliable signal transmission and sharp tuning (Hu et al., 2014; S.-T. Yang et al., 2018). This is reflected in a microcircuit dominated by parvalbumin-expressing (PV+) interneurons, which provide powerful, fast-spiking perisomatic inhibition ideal for gain control and precise temporal coding. In contrast, associative cortices like the PFC require dynamic, context-dependent integration and sustained activity (González-Burgos, 2005). This is supported by a more balanced interneuron expression with a relatively greater prominence of somatostatin-positive (SST+) and vasoactive intestinal polypeptide-positive (VIP+) interneurons (DeFelipe et al., 2002). SST+ cells provide dendritic-targeting feedback inhibition crucial for stabilizing network attractors, while VIP+ cells mediate disinhibition, allowing top-down signals to selectively modulate network states.

This shift is evidenced by studies in marmoset and other primate species showing PV+ neurons constitute ~60% of interneurons in V1 but a significantly lower proportion in the PFC, where the relative contribution of SST+ and VIP+ populations is enhanced (DeFelipe et al., 2002; Gonchar, 2008; Zaitsev et al., 2009).

From the evidence presented in this section we can conclude that the systematic shifts in intrinsic microcircuit organization along the sensory-associative gradient reflect the range of cortical integration processes and computational specialization. Thus, functional role of those brain areas is not only defined by their connectome, but also through their specific intrinsic characteristic (Park & Friston, 2013; Wig et al., 2011).

2.2.3 Parcellation of the PFC

The PFC is a major part of the frontal lobe, distinct from the primary motor cortex, and the premotor cortex, also called supplemental motor cortex (Jacobsen, 1935).

Historically, the different subregions within the PFC were delineated based on cellular composition and the presence of a granular L4 (Brodmann, 1909). This work was later expanded by providing more detailed subdivisions and refinements using modern techniques (Paxinos, 2012). Paxinos integrated additional data from cytoarchitecture, receptor binding techniques, and functional studies. Definition of PFC subregions are based on cytoarchitectural characteristics of the primate (Walker, 1940). But studies of non-human primates often attempt to establish a correspondence with human PFC subregions based on shared structure or function (Cruz-Rizzolo et al., 2011; Petrides & Pandya, 1999).

Initially, functions of the different parts of the PFC were inferred by conducting lesions studies and assessing which behavior showed sign of cognitive impairment as a result (Bauer & Fuster, 1976; Friedman & Robbins, 2022; Milner et al., 1985; Szczepanski & Knight, 2014). From this work, the primate PFC is segmented into 4 subregions: the medial PFC (mPFC), the rostral PFC, the orbital Prefrontal Cortex (OFC) and the dlPFC.

The mPFC is composed of portions of Brodmann areas 8, 9 and all of 24 in its dorsal part. While the ventral section is composed portions of Brodmann areas 10, 12, 14 and all of 25, with no clear boundary between ventral and dorsal. Lesion studies have shown that this area is associated with social perception (Mah et al., 2004), and lesions also cause emotional disturbance (Barrash et al., 2022). The mPFC is also involved in introspective and self-referential thought processes (D'Argembeau, 2013). The ventral part of the mPFC is particularly important for emotional regulation and guidance of social interaction (Rolls, 2004).

Rostral PFC, often considered part of the mPFC, corresponds to Brodmann area 10. It supports strategic planning and prospective memory and multitasking (Burgess et al., 2000), and is thought to be crucial in cognitive flexibility, that is, the ability to switch between cognitive contexts (Badre & D'Esposito, 2009).

The OFC, situated just above the eyes contains Brodmann area 11,12,13 and 47. It is important for decision-making and evaluation of rewards and punishments. Damages to the OFC causes both reinforcement-learning deficits and behavioral disinhibition (Ghashghaei et al., 2007).

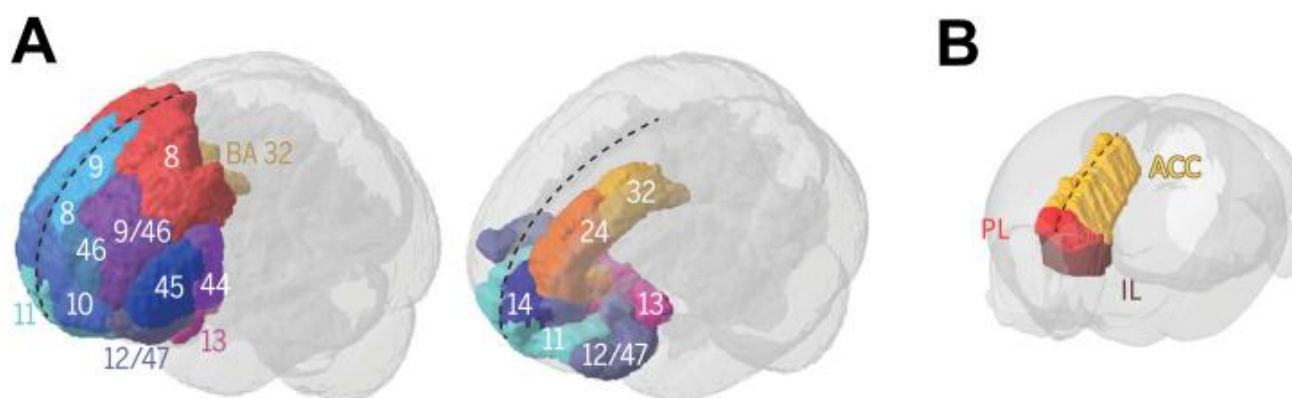


Figure 2-2. Comparative anatomy of the prefrontal cortex (PFC) in human and rodent.

(A) Schematics of illustration of Brodmann areas considered to be part of the PFC in the human brain (tilted lateral view) including the anterior cingulate cortex (ACC). The second illustration (right) presents medial and ventral areas less visible in left illustration. (B) Schematic of medial PFC (mPFC) subregions in the rodent brain (tilted lateral view), including the ACC. The rodent mPFC, which lacks a granular layer IV, is often considered a functional analogue to primate PFC networks. Its major subdivisions include the ACC, prelimbic area (PL), and infralimbic area (IL). Adapted from Carlén (2017).

Finally, the dlPFC is composed of the Brodmann areas 9 and 46. It has attracted a lot of interest for its critical role in most executive functions including abstract reasoning, cognitive flexibility and especially WM. Lesions studies to the dlPFC have shown the resulting impairment of memory guided behaviors involving WM (Bauer & Fuster, 1976; Funahashi et al., 1989, 1993; Fuster & Alexander, 1971; Wilson et al., 1993).

The initial definition of PFC by Jacobsen (1935) emphasized the importance of granular layer as a key characteristic of the areas belonging to the brain region. This definition has since been broadened based on connectivity and function, with a widely used criterion being the projection zone of the mediodorsal thalamus (Rose & Woolsey, 1948). This framework supports the view that the rodent mPFC serves as a functional analogue to aspects of the primate PFC, despite the rodent frontal lobe being entirely agranular (Fig. 2-2; Carlén, 2017; Le Merre et al., 2021). A large body of important work on WM mechanisms has been conducted in rodents, often focusing on the mPFC (Anastasiades & Carter, 2021; Dudchenko, 2004; Kim et al., 2021; Kupferschmidt et al., 2022; Zahrt et al., 1997). However, the extent to which findings from rodents can be applied to primates remains an area of active debate (Laubach et al., 2018).

2.2.4 The intrinsic characteristics of PFC

The expression of NMDA receptors is particularly high in the PFC (Burt et al., 2018; Scherzer et al., 1998; H. Wang et al., 2008). This results in a slower decay time constants that are crucial for persistent activity and WM (M. Wang et al., 2013). In a purely AMPA receptor-mediated network, achieving bistability through self-sustained activity requires firing rates exceeding 100-200 Hz to avoid network collapse as membrane potential decays to zero with rapid (2-5ms) decay time constant (X.-J. Wang, 1999). However, persistent activity recorded in the primate PFC *in vivo* typically exhibits firing rates of only 10-40 Hz (Funahashi et al., 1989; Fuster et al., 1982; Kubota & Niki, 1971). The prolonged depolarization provided by NMDA receptors enables temporal summation that maintains self-sustained activity at these lower rates, unlike network relying solely on AMPA transmission (X.-J. Wang, 2001). Experimental results support this idea as the blockage of NDMA receptors greatly degrade persistent activity (Driesen et al., 2013; Verma & Moghaddam, 1996; M. Wang et al., 2013).

This supports the idea that the relatively slow decay time of NMDA receptor mediated EPSPs allows for the maintenance of self-sustained activity through a reverberation mechanism (X.-J. Wang, 1999).

In the primate dlPFC, the primary source of sustained excitation is thought to be reverberation of the activity through L2-3 recurrent excitatory connections (X.-J. Wang, 1999). Anatomically, such intra-layer recurrent excitatory connections is considered to be present in the primate dlPFC as pyramidal neurons form local projections in the shape of stripe patterns (Goldman-Rakic, 1984; Levitt et al., 1993; Melchitzky et al., 1998; Pucak et al., 1996). These stripes, measuring 275-300 μm in width and 0.2-1mm in length, are proposed to provide the substrate for reverberation of sustained activity (Melchitzky et al., 1998).

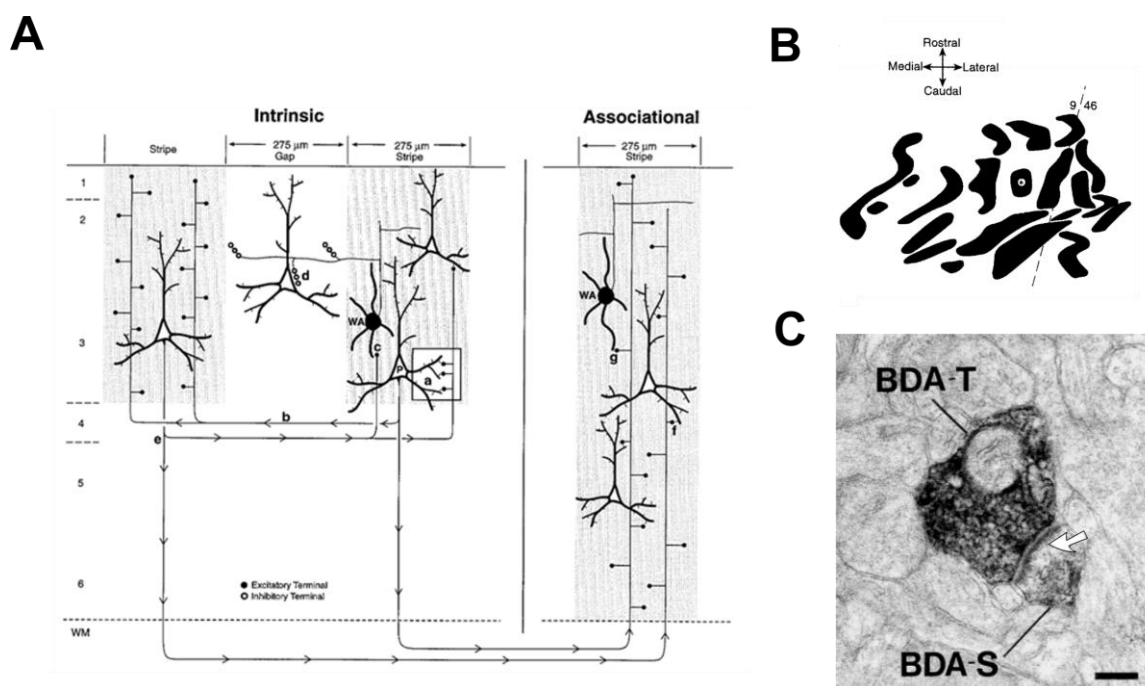


Figure 2-3. Electron microscopy reveals single neuron synaptic targets form local connectivity stripe patterns.

(A) Pyramidal neuron inter cluster target are 90% other excitatory neuron, while intra-cluster targets are other pyramidal neurons. adapted from Melchitzky et al. (2001). (B) Pyramidal neurons intrinsic projections form stripes patterns. Adapted from Melchitzky et al. (1998). (C) Electron micrograph of biotinylated dextran amine (BDA) labeled axon terminals (BDA-T) and retrogradely labeled dendritic spines (BDA-S) forming synapse (curved arrow). Scale bar 0.40 μm . Adapted from Melchitzky et al. (1998).

Melchitzky et al. (1998) has shown that within its own stripe a pyramidal neuron will project in equal proportion to pyramidal neurons and interneurons, while the same pyramidal neuron's projections to a distinct stripe will target pyramidal neurons 95% of the time.

The stripe like pattern contrasts with patch like patterns in other cortical regions (Watakabe et al., 2023). Importantly, this intrinsic local connectivity differs from sensory cortices such as primary visual cortex (Kritzer & Goldman-Rakic, 1995).

The intrinsic stripe pattern of the dlPFC is similar to the interdigitated cortico-cortical afferent terminals to this region, reported by (Goldman-Rakic, 1984) to be 300-750 μm wide and spaced 1.0-1.5 mm center to center.

2.2.5 The debated nature of persistent activity in the PFC

Despite the long period over which WM has been studied, many debates remain about the neural basis of this cognitive ability. In the dominant theory, the PFC maintains sensory information through persistent activity relying on intrinsic characteristics presented in the previous section (Riley & Constantinidis, 2016). Despite the relative consensus on the presence of persistent activity, two major points of contentions remain. First, while persistent activity is a hallmark of the dlPFC, it is often less prominent or exhibits less stable patterns in other subregions of PFC (Curtis & Sprague, 2021).

For example, the OFC, which is densely connected to the striatum for reward processing typically exhibit transient responses to rapidly updating values rather than sustained tonic firing (Rolls, 2004). Meanwhile, the ACC integrates signals related to conflict, effort, and action monitoring (Jin et al., 2018; Yee et al., 2021). In this area, information seems more likely to be encoded in phasic bursts or oscillatory synchronization rather than continuous activity (Womelsdorf et al., 2010). Finally, the vmPFC is a hub for memory-emotion integration via its connections with the amygdala and hippocampus (Benoit et al., 2014). This area operates on a longer, more variable timescales suited to affective evaluation, and not the second-by-seconds maintenance of discrete sensory cue.

This discrepancy has led to significant theoretical divergence. Some researchers conclude that the PFC does not store content but instead represents abstract rules or exerts top-down control, with the actual memory trace distributed in sensory cortices or posterior parietal cortices (D'Esposito & Postle, 2015).

The leading theory remains that in contrast to dlPFC, other prefrontal areas conduct more dynamic computation rely less on rigidly maintained persistent activity and receive instead more dynamic, time-varying inputs from diverse brain regions (Curtis & Sprague, 2021; Mante et al., 2013; Riley & Constantinidis, 2016).

Indeed, PFC subregions exhibit distinct brain-wide connectivity profiles that underpin their specialized functional roles in cognition. But this raises a second point of debate about the emergence and maintenance of persistent activity. Specifically, based on the finding that the dlPFC forms recurrent connections with regions like PPC, and lateral intra-parietal area (LIP; Goldman-Rakic et al., 1984; Katsuki et al., 2014), it has been proposed that persistent activity in the dlPFC is supported by reverberation through those long-range recurrent connections (Guo et al., 2017; Hazy et al., 2007; Riley & Constantinidis, 2016). But modeling work suggests that distributed activity patterns still strongly rely on intrinsic structure, specifically from areas at the top of cortical hierarchy (Mejías & Wang, 2022).

2.2.6 Comparison with other persistently firing cortical areas

Persistent activity has been observed in other cortical areas, but its underlying mechanism and functional roles diverge. In the PPC, neurons display delay-related activity during WM tasks, as observed by (Chafee & Goldman-Rakic, 1998; Constantinidis & Steinmetz, 1996). This delay-period activity is functionally significant, as evidenced by studies showing that PPC inactivation affects WM task performance (Chafee & Goldman-Rakic, 1998; Quintana et al., 1989).

Constantinidis & Steinmetz (1996) found that PPC neurons are more susceptible to interference from distracting stimuli during the delay period of WM tasks.

This points to the idea that PPC is involved in the encoding of every incoming input, while dlPFC is more robust to distractors and maintains information that is task-relevant and goal oriented (Suzuki & Gottlieb, 2013).

Additionally, PPC encodes different aspects of sensory information. It tends to be less generalized (Sarma et al., 2016) and more specific to certain features of the stimuli (Qi, 2010; Qi et al., 2015). Both PFC and PCC are implicated in inhibition of irrelevant movements, but the responses to stimuli appear earlier in the PPC. The PPC activity also seems to decay with a shorter time constant than the dlPFC (Zhou et al., 2016).

The dlPFC and PPC share several structural and functional similarities. In terms of cytoarchitecture, both regions exhibit comparable cellular organization (Beul et al., 2017; Goulas et al., 2018), but more intricate differences have been reported (Katsuki et al., 2014). Notably, both areas exhibit different proportions of bursting neurons (González-Burgos et al., 2019), suggesting it could be central for robustness of persistent bump maintenance in the PFC.

Pyramidal neurons in the limbic system, notably in the hippocampus and the entorhinal cortex, can also present persistent activity (Egorov et al., 2002; Taube, 1998). Head-direction cells, place cells, and grid cells, which support spatial orientation, exhibit sustained firing that outlast a transient sensory or self-motion input (Taube, 2003). The maintenance of this activity has been shown to depend less on recurrent excitation with a local circuit and more on an intrinsic mechanism. The calcium-activated non-specific cation current (I_{CAN}) enables neurons to enter a depolarizing plateau potential, allowing them to sustain spiking even in the absence of sustained synaptic drive. This intrinsic mechanism is activated by cholinergic neuromodulation (Egorov et al., 2002; Fransén et al., 2006), raising the question of the contribution of neuromodulators to persistent activity in the PFC. Although mechanisms modulating intrinsic excitability of single neurons can shape persistent activity in the PFC, they are generally not considered to be the primary driver, in contrast to the I_{CAN} described in the limbic system (Zylberberg & Strowbridge, 2017).

2.2.7 Neuromodulation

The PFC dynamically represents information in memory during short period of time in the absence of the triggering sensory stimulus. Such functions of PFC rely heavily on neuromodulatory systems (Arnsten, 2012). This neuromodulation not only regulates synaptic transmission but also intrinsic excitability of individual neurons. The dlPFC is sensitive to its neurochemical environment, receiving dense innervation from arousal systems including dopamine (DA), norepinephrine (NE), and acetylcholine (ACh) (Arnsten, 2012; Cool & Arnsten, 2022).

Seminal work by Sawaguchi & Goldman-Rakic (1991) established that D1 receptor antagonists impair memory-guided saccades. Additionally, *in vivo* studies have shown that DA enhances the firing rates of PFC neurons during the delay periods of WM tasks (Williams & Goldman-Rakic, 1995). While DA regulation is essential for normal WM performance, there exists an optimal level of DA necessary for efficient WM performance over which excessive DA levels impair WM, suggesting an inverted-U relationship between DA levels and cognitive function (Vijayraghavan et al., 2007).

In vitro studies have provided detailed insights into the cellular mechanisms by which DA modulates neuronal excitability (Henze et al., 2000), through enhancements of persistent sodium (Na⁺) and NMDA receptor-mediated conductances (Seamans et al., 2001; Seamans & Yang, 2004). DA plays a central role in regulating and shaping network activity but also modulating intrinsic excitability. But other neuromodulators have been shown to be important for WM (Arnsten et al., 2012; Cools & Arnsten, 2022). This has led to the suggestion that there could be an interplay between ACh, serotonin, NE and DA that could drive a mechanism of intrinsic bistability (Camperi & Wang, 1998). These neuromodulators could, along with sustained synaptic inputs originating from the network, help the maintenance of persistence bump of activity robust to noise and distractors. This would be similar to the mechanism of plateau potential in head direction system driven by the activation of muscarinic receptors by ACh in the limbic system (Taube, 2003). But in the case of PFC network, neuromodulators are considered to only provide flexibility in the generation of persistent activity and better robustness against irrelevant stimulus (Arnsten et al., 2012; Cools & Arnsten, 2022).

2.3 Modeling the structure and dynamics of persistent activity microcircuit

During WM task, cortical neurons present elevated firing rates, according to their preferred position of the visual cue. This activity persists throughout the delay period (Funahashi et al., 1989). There is accumulating evidence that this persistent activity is sustained through a network mechanism (X.-J. Wang, 2021). This section reviews the theoretical models that have been developed to formalize this mechanism and explain the emergence and maintenance of this persistent activity. Different computational approaches have been used to formalize empirical insights into models of WM, navigating the trade-offs between biological fidelity and theoretical simplicity (Compte, 2006). We examine key milestones in this effort, from mean-field continuum models to time-continuous numerical biophysical simulations, emphasizing each approaches' advantages and drawbacks and how they each gave a different perspective to our understanding of dlPFC circuit dynamics (X.-J. Wang, 2021).

Finally, we position our large-scale clustered model as advancing this trajectory by reconciling biological detail with the richer dynamics of large-scale.

2.3.1 Irregular firing is a key characteristic of *in vivo* activity made possible by E/I balance

Persistent activity during WM task is thought to encode sensory information (Goldman-Rakic, 1995; X.-J. Wang, 2021). One might expect that such encoding would require reliable activity in a precisely timed pattern. However, the persistent activity recorded *in vivo* during WM task is instead highly irregular and asynchronous (Compte, et al., 2003). This asynchronous irregular regime is characterized by a coefficient of variation (CV) of the interspike intervals (ISIs) that approach or exceeded value of 1 and an absence of correlation between neurons spike timing. This statistic indicates that the firing pattern approximates a Poisson process (Shadlen & Newsome, 1998). This stochastic behavior is a hallmark of cortical activity and its functional implications have been studied (Softky & Koch, 1993), notably in the context of persistent activity (Barbieri, 2007; Hansel & Mato, 2013).

Some propose that the irregular aspect of this activity allows networks to have the maximum range of information encoding capability (Ma et al., 2023; W. Zhang et al., 2021; Zhao et al., 2022). Theoretical work suggests that in cortical circuits irregular activity emerges from the excitation-inhibition balance (Doiron et al., 2016). To explain such state, many studies suggest that excitatory and inhibitory synaptic currents almost completely cancel each other (Renart et al., 2010; Shadlen & Newsome, 1994). Early work by Van Vreeswijk & Sompolinsky (1996) established that balanced asynchronous states can emerge in large recurrent networks with sparse, random connectivity. Such balanced randomly connected network has been shown to present asynchronous irregular regime (Brunel, 2000). In contrast, attractor network models of WM implement structured connectivity which is crucial for the emergence of stimulus-specific activity pattern through recurrent connections.

Reconciling the asynchronous and highly irregular spiking dynamics characteristic of cortical circuits with the multistability required for WM represents a crucial challenge for developing biologically plausible models (Barbieri, 2008).

2.3.2 Analytical implementations

In parallel to the experimental recordings of persistent activity *in vivo*, computational neuroscience has developed models that provide a network-level perspective on the mechanisms underlying WM. However, the immense complexity of the PFC microcircuit and its intricate interconnectivity with other brain regions poses significant challenges for modeling, as limited computational power restricts the biological detail that can feasibly be implemented. A critical question arises: What core biological constraints must a model incorporate to accurately replicate the firing statistics of persistent activity reported in the dlPFC?

Early work by Amari (1977) proposed the attractor dynamic as a general theory of cortical dynamics. This foundational study introduced a theoretical framework for continuous attractor networks using neural field equations.

The key feature of the model was a spatial structured connectivity kernel with a Mexican-hat profile: where neighboring neuron formed excitatory connections while distant neurons formed inhibitory connections. This framework enabled the analysis of spatial dynamics and stability, which allows the emergence of patterns such as bumps and waves within population-level scale.

However, as a neural field model, this framework describes population activity as a continuous variable, abstracting away the discrete nature of individual spikes and their precise timing (Amari, 1977). Moreover, the model's connectivity assumes local excitation and distal inhibition—the inverse of actual cortical circuitry, where long-range excitatory projections relay activity across regions and local inhibition regulate activity at short range (Kandel et al., 1991).

In order to understand the stochastic aspect of spontaneous activity, later implementation of analytical attractor models extracted statistical properties of spiking activity. A seminal study measured the average irregularity of firing patterns as well as its correlation (Amit & Brunel, 1997b). They reported a link between the emergence of asynchronous irregular regime to single neuron properties such as synaptic weight and time constant.

For this framework, they implemented the mean-field theory to statistically describe the average firing rates dynamics of neural population derived from self-consistent equations. Importantly, Amit & Brunel (1997a) describe the method in more detail; it consists in implicitly including noise via spiking variability in the mean-field approximation. This model relied on coupled ordinary differential equations (ODEs) to describe neural populations' dynamics. In contrast, full numerical implementation of SNNs preserves the discrete nature of individual spikes, leading to more biophysically grounded network models.

2.3.3 Biophysical model

By explicitly modeling individual neuron dynamics, finite network size, and synaptic stochasticity, numerical simulations of SNNs fundamentally alter the nature of attractor states compared to analytical mean-field approaches.

Most analytical models assume system with infinite scale and absence of noise, which produce mathematically stable attractors capable of maintaining persistent activity indefinitely (Camperi & Wang, 1998; Moody et al., 1998; Tsodyks & Sejnowski, 1995).

By contrast, numerical implementations capture the complex dynamics that emerge from noisy interactions in finite networks. These simulations reveal metastable attractor states, where persistent activity is maintained only over biologically relevant timescales. Such metastable attractor states provide mechanistic explanations and more plausible evidence for the emergence of transient persistent activity observed in physiological data.

Compte et al, (2000) was the first and most influential biophysically grounded attractor model with explicit excitatory and inhibitory population. This SNN composed of LIF neurons with ring topology included NMDA synaptic conductance which allows it to present bistability at a low, realistic rate. This ring attractor implementation became the canonical framework for modelling persistent activity.

The numerical simulation of a spiking neural network is more computationally expensive, but the model presents more detailed dynamics of the non-linear interaction between neurons. This is especially important to test the effect of noise on the robustness of the stable bump pattern of activity.

The initial objective of these models was to have a stimulus specific pattern of activity that recapitulates the rate of activity recorded *in vivo*. The model would also need to be a bistable network that could switch between baseline spontaneous activity rate to a bump pattern of activity with persistent activity rate (~ 20 Hz) and background activity rate (< 1 Hz to 5Hz), thus replicating the activity recorded *in vivo* (Funahashi et al., 1989).

Compte et al, (2000) was the first to present this activity pattern at those rates. But this model present limitation, notably for reproducing the high order statistics or *in vivo* activity (Barbieri, 2008).

2.3.4 Ring derived models

The ring model from (Compte, 2000) showed improvement in terms of the biophysical plausibility as it integrates biological constraints. This model used integrate-and-fire neurons with realistic synaptic conductance dynamics (AMPA, NMDA, and GABA) and structured connectivity such that neurons excited nearby cells with similar spatial preferences while inhibiting those with dissimilar preferences. With this topology organized around a 1-dimensional ring, the network produced localized bumps of persistent activity in response to transient localized stimulation. Global inhibition prevents the runaway excitation throughout excitatory population. This article was pivotal because it validated theoretical prediction from analytical attractor models in a biologically plausible framework.

Subsequent iteration of the ring attractor model studied the inclusion of additional biologically plausible characteristics. X.-J. Wang et al., (2004) proposed an implementation where different types of inhibitory neurons presenting specific tuning curves (Fig. 2-4).

Indeed, many ring attractor models often simplify inhibitory interneurons as a homogeneous population with uniform global connectivity. As mentioned previously, biological circuits exhibit diverse interneuron subtypes with distinct morphological targets and properties underlying different functional roles in computation.

It is difficult to associate tuning of interneurons with the different types of interneurons *in vivo* during persistent activity (Kilpatrick et al., 2013; Renart et al., 2003). But evidence from a neuromodulatory study suggest that dopamine depress inhibitory transmission between fast spiker (FS) interneurons and pyramidal neurons, while it enhance the inhibition between non-FS interneurons and pyramidal neurons (Gao et al., 2003). This suggests that dopamine sharpens the bump activity pattern by differentially modulating two inhibitory pathways.

It dampens inhibition from interneurons co-tuned with the persistently firing neurons, while concurrently enhancing inhibition targeting pyramidal neurons with different tuning. This actively suppress background activity to avoid run-away excitation (Fig. 2-4; Li et al., 2020). Accordingly, X.-J. Wang et al. (2004) proposed that one class of inhibitory neurons representing the calbindin-positive (CB) neurons exhibited inverted tuning curve relative to other neurons.

Dopamine has been shown to be crucial for the nominal WM function and for the underlying persistent activity. It has also been shown that dopamine affects the dopamine receptor expressing neuron both at the synapses (Arnsten et al., 2012) and their intrinsic excitability (Henze et al., 2000; C. Yang & Seamans, 1996) but it is difficult to isolate the two mechanisms. Studies have proposed implementation of the dopamine mechanism on ring networks (Durstewitz et al., 1999, 2000).

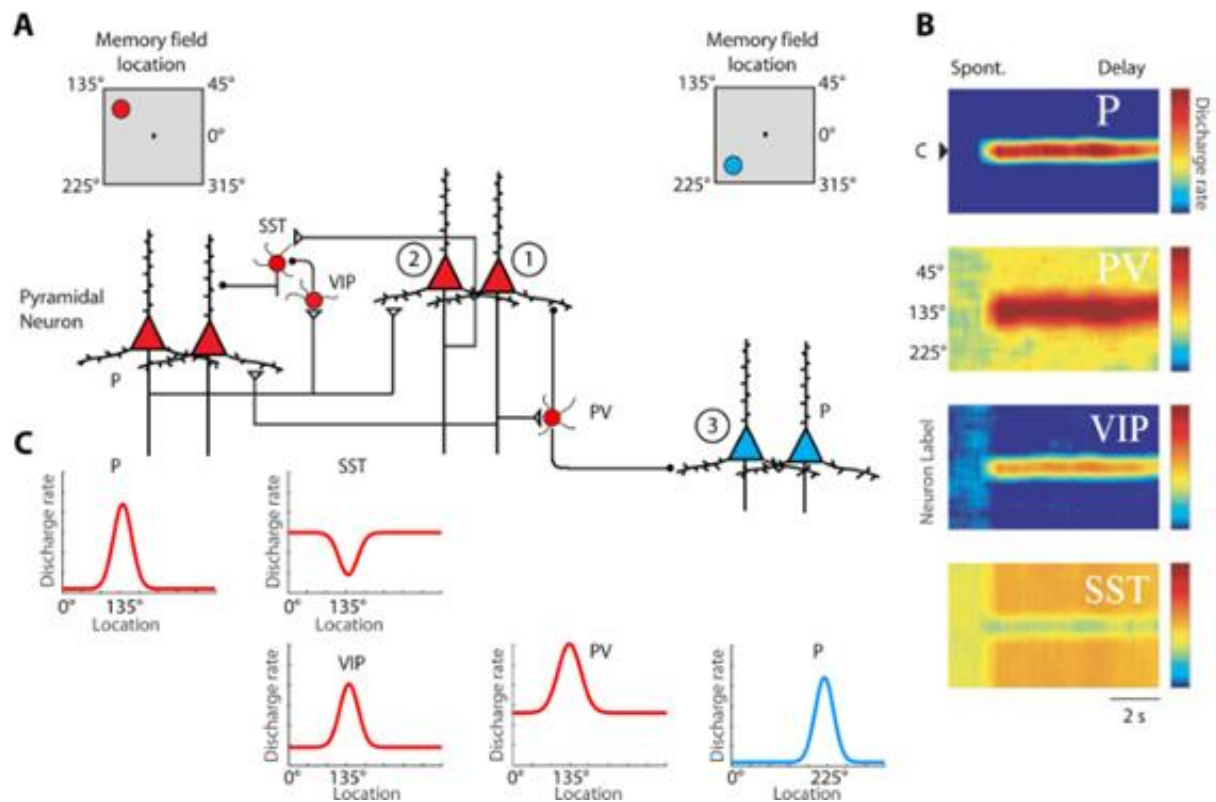


Figure 2-4. Ring model implementing different classes of interneurons proposed by Wang et al. 2004.

(A) In this proposed circuit, recurrent connections are formed between pyramidal neurons tuned to respond to the same position of a visual cue, but they also target interneurons which target both pyramidal neurons with same tuning (SST) and with different tuning (PV). (B) Activity of different cell types pyramidal (P), PV, VIP and SST. The activity pattern of each population represents their different function in maintenance of network activity. (C) Tuning curve of the neuron models from panel A depending of their cell type and preferred stimulus position. (Figure from Li et al., 2020)

The ring attractor framework has also been used to explain other instances of persistent activity beyond WM. Notably, in the limbic system, head direction cells maintain firing for their preferred head orientation, a mechanism crucial for spatial navigation (Taube, 2003). Indeed, the sense of direction is maintained even when an individual has their eyes closed (Goodridge et al., 1998; Taube et al., 1990), the attractor dynamics was proposed to support the maintenance of this activity while angular velocity from the vestibular system input updates internal representation head direction changes through drift of the bump of activity across the ring (Clark & Taube, 2012).

Analytical implementation of the ring attractor model were proposed specifically for the head direction system (Skaggs et al., 1995; K. Zhang, 1996). Numerical implementations have been proposed and also relies on recurrent excitation (Degrís et al., 2004). But many experimental studies of head direction system tend to support a mechanism of I_{CAN} which is lead to the emergence of plateau potential (Boucheny et al., 2005).

Similarly, 2D planar attractor networks have been employed to model spatial coding in hippocampal place cells and entorhinal grid cells (Couey et al., 2013; McNaughton et al., 2006). These models extend the conceptual framework of attractor dynamics to encode multidimensional variables, enabling explanations of grid cell periodicity and place field stability.

In a context more closely related to spatial WM, ring attractor has also been proposed as a model of decision-making through evidence accumulation (Kutschireiter et al., 2023; X.-J. Wang, 2008; Wong & Wang, 2006).

Despite the different iterations of the numerical implementation of the ring model, the core limitation of the model remains in the fact that it does not fully capture the irregularity of the cortical activity reported *in vivo*, where the CV of ISIs is centered around 1 (Softky & Koch, 1993), while ring model typically exhibit a CV around 0.5 or lower (Barbieri, 2008). In ring models, strong recurrence is needed to sustain a memory bump, and Barbieri (2008) suggested that the resulting supra-threshold synaptic inputs drives excessive regularity of firing.

Renart et al. (2007) theorized that in a model where both excitatory and inhibitory increase their rates in the persistent states, the input variance to both populations would increase significantly compared to the background state. In their mean-field analysis of recurrent networks, they found that achieving a fluctuation-driven persistent activity state requires significant parameter fine-tuning. Specifically, the degree of required fine-tuning increases with network size. This suggests that large-scale numerical network simulation finely tuned balance of excitation and inhibition that would be difficult to achieve.

2.3.5 Clustered network models

The classical ring model is defined by a continuous attractor architecture, where neurons are arranged along a smooth manifold based on a preferred feature, such as direction (Compte, 2000). In this framework, the connectivity profile of excitatory neurons is topographically smooth and continuous; each neuron connects to its neighbors with a strength that decays monotonically with distance. Thus, neurons inside the activity bump remain transsynaptically coupled to neurons located outside the bump via chains of intermediate neurons.

In contrast, clustered network models are defined by discrete, modular excitatory architecture (Litwin-Kumar & Doiron, 2012). Here, neurons are grouped into cluster with the probability of a connection within a cluster is high, creating a densely recurrent module. Conversely, the probability of a connection between neurons belonging to different clusters is significantly lower, but such connection can form between any cluster, not just neighboring ones. This structure leads to sparse long-range cross-cluster connectivity. Consequently, persistent activity in this framework manifests as sustained firing within one or more specific clusters, forming a discrete metastable attractor state. The work of Litwin-Kumar and Doiron (2012) has shown that clustered excitatory connectivity leads to emergence of attractor state and stochastic switching between activity patterns.

This study and other implementations report trial-to-trial activity rate variability (Gavenas et al., 2024; Klinshov et al., 2014; Schaub et al., 2015). While those studies do not report irregular firing patterns, we theorize that this connectivity framework can result in firing patterns replicating *in vivo* irregularity when simulated at larger scale.

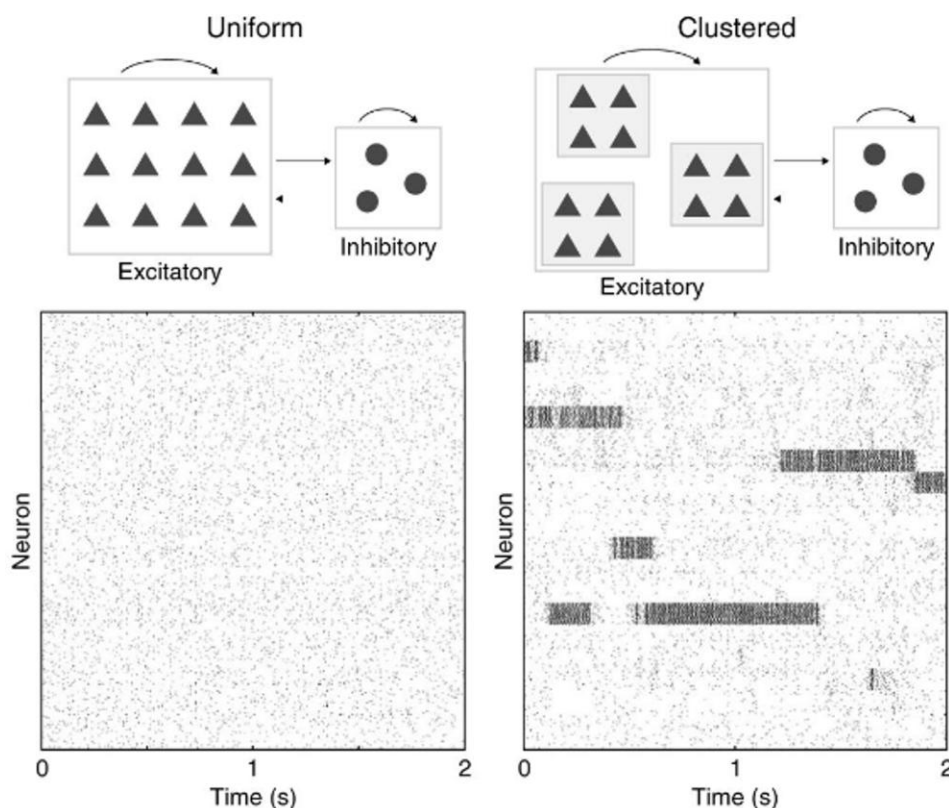


Figure 2-5. Clustered connectivity.

This figure presents the schematics and spike raster showing the spike times of excitatory neurons in networks where excitatory population presents uniform random connectivity (left) and clustered connectivity (right). As seen in the raster plot (bottom right), the activity of network with clustered connectivity present bump patterns emerging and switching between pattern spontaneously. Adapted from Litwin-Kumar & Doiron (2012).

This modular connectivity framework can more readily be scaled up to larger network size compared to the ring architecture, and we predict that larger scale network presenting a clustered connectivity will be able to present richer dynamics with smaller underlying synaptic signals.

Indeed, in small network models (e.g., a few thousand neurons), maintaining sufficient input to drive spiking requires artificially large synaptic signals. We hypothesize that as the size of our network gets larger, the number of connections each neuron receives (K) will also increase and activity rates can be maintained with sparser connectivity profile and synaptic signals at more reasonable plausible biological scale. Van Vreeswijk & Sompolinsky (1996) showed, in a randomly connected network, that when the network size increases, the strength of individual synapse required to maintain firing rate decreases by a factor $1/\sqrt{K}$.

The literature reviewed in this chapter establishes a detailed picture of the neural basis of persistent activity, and the important perspective that computational models provide on the underlying network-level dynamics. While the models of persistent activity have been crucial for investigating the plausibility of the attractor hypothesis, how a network can achieve a bump state while simultaneously maintaining the highly irregular firing patterns observed *in vivo* remains an open question.

The canonical ring model achieves persistence through a mean-driven regime driven by strong excitatory input that maintains voltage above threshold resulting in persistent activity. In contrast, *in vivo* recordings suggest that persistent activity relies instead on fluctuation-driven regime where mean voltage is below threshold on average and spikes are generated by voltage fluctuations.

We hypothesized that by scaling up a clustered network to a biologically relevant size, the network could achieve an attractor state that is both stimulus specific, highly irregular, and reliant on synaptic signals of plausible magnitude. This hypothesis forms the foundation of the project presented in this thesis. In Chapter 3, we introduce the modeling work of this thesis. In Chapter 4, we test the biological plausibility of this synaptic regime directly. Using a network clamp, we inject the synaptic signals from both the canonical ring model and our large-scale clustered model into biological neurons *in vitro*.

Chapter 3

3 Large scale clustered network generates plausible activity and synaptic input signals

3.1 Introduction

Working memory (WM) is the ability to hold and manipulate information in mind to guide cognitive tasks and goal directed behavior (Baddeley, 1986). Based on numerous experimental findings, persistent activity, the ability of neurons to maintain elevated activity without continuous sensory input, is the predominant theory explaining the neural basis of WM (Goldman-Rakic, 1995; X.-J. Wang, 2021).

How the brain maintains persistent activity is still not fully understood and likely involves interplay between specific properties of individual neurons and their connectivity with the circuit (Egorov et al., 2002; Elston, 2003; Rotaru et al., 2007; X.-J. Wang, 2001). It has been proposed that, at the circuit level, neural networks can settle into stable patterns of activity called attractor states (Amit, 1989; Hopfield, 1982; Rolls, 2010). In analytical models of persistent activity, this emerges from a structured network topology (Amit & Brunel, 1997a; Tsodyks & Sejnowski, 1995). Specifically, the combination of strong local recurrent excitatory connections and global feedback inhibition allows a network to stabilize into a bump attractor state, following a transient external input mimicking sensory stimulus (Amit & Brunel, 1997b; Camperi & Wang, 1998; Moody et al., 1998; Zhang, 1996).

Simulations of spiking attractor network models have been crucial for testing this theory under realistic biological constraints. A seminal advance was the implementation of the ring attractor model in a spiking network (Compte, 2000).

In this model, numerical simulation allowed the authors to demonstrate how a slow, 100 ms synaptic timescale replicating the dynamics of N-methyl-D-aspartate (NMDA) receptor was crucial in the maintenance of low-rate persistent activity through recurrent connections (Compte, 2000; X.-J. Wang, 1999).

Despite the advances, the ring model presents limitations. Primarily, the canonical ring models do not fully replicate the irregular aspect of neural activity in the brain (Barbieri, 2008). The cortical activity recorded *in vivo* has been shown to operate in an asynchronous irregular (AI) pattern (Compte, et al., 2003; Shadlen & Newsome, 1998; Softky & Koch, 1993). In this state, the activity is characterized by individual neurons firing in an irregular pattern, with a coefficient of variation (CV) of the interspike interval (ISIs) around 1, indicating a stochastic variability (Compte, et al., 2003). Additionally, the pairwise correlation between neurons firing patterns has been shown to be close to zero (Darshan et al., 2018; Schneidman et al., 2006).

Theoretical studies suggested that the ring attractor models are not able to replicate the irregular aspect of cortical activity due to their reliance on mean-driven regime and that a more coordinated increase of both inhibition and excitation would result in the more realistic fluctuation-driven regime (Ahmadian & Miller, 2021; Mongillo et al., 2012; Renart et al., 2007). We hypothesize that because of its scale, the ring model relies on strong excitation while its strictly local excitation results in similarly tuned neurons exhibiting significant input overlap.

To address these limitations, we developed a large-scale clustered model that would be able to present a persistent bump pattern of activity that better represents the irregularity of *in vivo* activity. To test this, we developed a network model composed of 50,000 excitatory and inhibitory neurons where the excitatory population features a clustered connectivity pattern. We show that instead of the overreliance on slow conductance of the canonical ring model, in our model, the transition to persistent activity state is driven by dynamic cohesive increase in input conductances. As a result, our network model operates in a fluctuation-driven regime and presents a more irregular persistent activity.

3.2 Methods

3.2.1 Compte et al. Ring model

A replication of established ring model in (Compte, 2000) was written using python using the Brian2 library (https://github.com/Erwan-Martin/Ring_model_persistent_activity.git).

Every neurons include three input conductance variables g_i , $g_{e_{slow}}$, $g_{e_{fast}}$ which present time courses of GABA, NMDA and AMPA conductances respectively

$$C_m \frac{dv}{dt} = GL[EL + V_m] + g_{e_{fast}}(t)[E_e + V_m] + g_i(t)[E_i + V_m] + g_{e_{slow}}(t)[E_s + V_m] * \theta + I_e$$

with the saturation of NMDA-like conductance modeled as: $\theta = \frac{1}{1 + \exp(-0.062 * 1000 * V_m) / 3.57}$

This equation was solved numerically using the second-order Runge-Kutta (RK2) midpoint method with time step 0.05 ms. We define a constant threshold value ($V_{th} = -50mV$). When $V_m(t) \geq V_{th}$: $V_m(t + 1) = V_{reset}$. With $V_{reset} = -60mV$ and $\tau_{ref} = 2ms$

3.2.2 NETSIM clustered Network Architecture

Spiking neural network simulations were implemented using the NETSIM framework, which is designed for large-scale, biophysically realistic spiking models. The network consisted of a total of $N = 50,000$ neurons, comprised of $N_e = 40,000$ excitatory neurons and $N_i = 10,000$ inhibitory neurons. Excitatory neurons were organized into clusters, allowing us to investigate how clustered connectivity affects network dynamics.

For each neuron, a fixed number of K synaptic connections were established. Within the excitatory population, these K connections were subdivided into three groups:

- (K_{in}): the number of intra-cluster (within-cluster) connections,
- (K_{out}): the number of inter-cluster (between-cluster) connections,

- (K_{E-I}): the number of connections to inhibitory neurons.

Each excitatory cluster contained (N_{epop}) neurons. Inhibitory neurons projected randomly to both excitatory and inhibitory populations without cluster-specific constraints.

Neurons were modeled using a conductance-based approach. The membrane potential $Vm_j(t)$ of neuron (j) was updated using Euler integration with a fixed time step ($\Delta t = 0.1, \text{ms}$) as follows:

$$Vm_j(t + \Delta t) = Vm_j(t) + \Delta t * \frac{ge_{fast,j}(t)[Ee + Vj(t)] + gi,j(t)[Ei - Vj(t)] + ge_{slow,j}(t)[Es - Vj(t)] + gL[EL - Vj(t)] + Ie}{Cm}$$

where $ge_{fast,j}(t)$, $gi,j(t)$ and $ge_{slow,j}(t)$ denotes the excitatory, inhibitory, and NMDA-like synaptic conductances, respectively. These conductances decay exponentially with time constants τ_e , τ_i , and τ_n and are augmented by synaptic events buffered with delays computed from the distances between neurons and a conduction speed parameter. When $Vm_j(t)$ reaches $V_{th} = -50\text{mV}$, $Vm_j(t + \Delta t) = V_{reset}$. With $V_{reset} = -60\text{mV}$ and $\tau_{ref} = 2 \text{ ms}$.

3.2.3 Network Activity and Statistical Measures

To verify that the network operated in an asynchronous irregular regime, we analyzed several statistical measures extracted from the spiking activity, including the firing rate, pairwise spike correlation, coefficient of variation (CV) of the interspike intervals (ISIs), and a measure termed the ratio.

Firing rate. The firing rate r_j of neuron (j) was computed as the total number of spikes produced by neuron (j) during the 1 second post-simulation time.

Pairwise Spike Correlation. The degree of synchrony between neurons was assessed using the Pearson correlation coefficient, defined as

$$C_{jk} = \frac{\langle (s_j(t) - \mu_j) (s_k(t) - \mu_k) \rangle}{\sigma_j, \sigma_k}$$

Coefficient of Variation.

The coefficient of variation (CV) of the inter-spike intervals was computed to characterize the irregularity of spiking:

$$CV = \frac{\sigma_{ISI}}{\mu_{ISI}}$$

Ratio. In addition, a measure termed ratio was defined to quantify the relative difference between firing rates of neurons outside the stimulated cluster and those within the stimulated (post-stimulation) period. Specifically, if (r_{in}) denotes the mean firing rate of neurons in the stimulated cluster after stimulation, and (r_{out}) represents the mean firing rate of all other excitatory neurons recorded during the same period, the ratio is given by

$$ratio = 1 - \frac{r_{out}}{r_{in}}$$

A higher value of ratio indicates a more pronounced difference between the stimulated cluster and the surrounding network activity

Parameter search methodology. We conducted extensive computational parameter searches, performing several tens of thousands of simulations using custom Python scripts executed on Compute Canada high-performance computing resources. Each simulation systematically explored parameters critical to network dynamics, particularly focusing on conditions that supported realistic persistent firing states.

The varied parameters included the total number of synaptic connections per neuron (K), the number of neurons per excitatory population (N_{epop}), excitatory synaptic conductance (g_e), inhibitory synaptic conductance (g_i), NMDA-like slow excitatory conductance (g_n), and the level clustering (lvl_{clust}) describing the connectivity compared to a randomly connected network defined as follows:

$$level_{\text{clustering}} = \frac{\frac{K_{in}}{K_e}}{\text{Number of clusters}}$$

Voltage steady state

The voltage steady-state (V_{ss}) represents the theoretical equilibrium membrane potential determined by a weighted average of synaptic conductances and their reversal potential.

$$V_{ss} = \frac{g_L E_L + g_e E_e + g_i E_i}{g_L + g_e + g_i}$$

where g_L is the leak conductance, E_L is the leak reversal potential, g_e is the excitatory conductance, E_e is the excitatory reversal potential, g_i is the inhibitory conductance, and E_i is the inhibitory reversal potential.

Voltage variance decomposition

The variance decomposition method separates the total membrane potential variance ($Var(V)$) into components arising from excitatory conductance fluctuations ($Var(V_{hyp}E)$), inhibitory conductance fluctuations ($Var(V_{hyp}I)$), their covariance, and a residual nonlinear contribution term ($\mathbf{R}$):

$$Var(V) = Var(V_{hyp}E(t)) + Var(V_{hyp}I(t)) + 2Cov(V_{hyp}E, V_{hyp}I) + \mathbf{R}$$

$$V_{hyp}E(t) = \frac{g_L E_L + g_e(t) * E_e + g_{i_{mean}} * E_i}{g_L + g_e(t) + g_{i_{mean}}}$$

$$V_{hyp}I(t) = \frac{g_L E_L + g_e(t) * E_e + g_{i_{mean}} * E_i}{g_L + g_e(t) + g_{i_{mean}}}$$

$$\mathbf{R} = Var(V) - (Var(V_{hyp}E) + Var(V_{hyp}I) + 2Cov(V_{hyp}E, V_{hyp}I))$$

The hypothetical voltages ($V_{hyp}E$ and $V_{hyp}I$) represent hypothetical voltage fluctuations obtained by taking one conductance type instantaneous value at each time point while fixing the other conductance type at its mean value. The term $\mathbf{R}$ represents the residual nonlinear effects that are not captured by the other terms.

3.3 Results

3.3.1 NMDA-dominated input of persistently firing neurons in ring model

We replicated the classic ring model from Compte et al. (2000) using the Brian simulator (Stimberg et al., 2019) and injected a one-second current stimulation into a localized patch of excitatory neurons. This evoked a localized bump of activity that persisted after stimulation offset, as expected (Fig. 3-1B). At the single-neuron level, this bump is reflected in persistent elevated spiking with statistics similar to the original model: a mean rate of 10 sp/s, moderately irregular firing patterns ($CV = 0.67$), and low pairwise spike correlations of 0.007 (Pearson correlation). The traces of synaptic conductances reveal that persistent activity in this model is primarily maintained by slow excitatory conductance (Fig. 3-1C). After the initial baseline period, the slow NMDA conductance, $g_{e_{slow}}$ increases to very high values (approximately 25 nS), while the other conductance variables remain relatively unchanged.

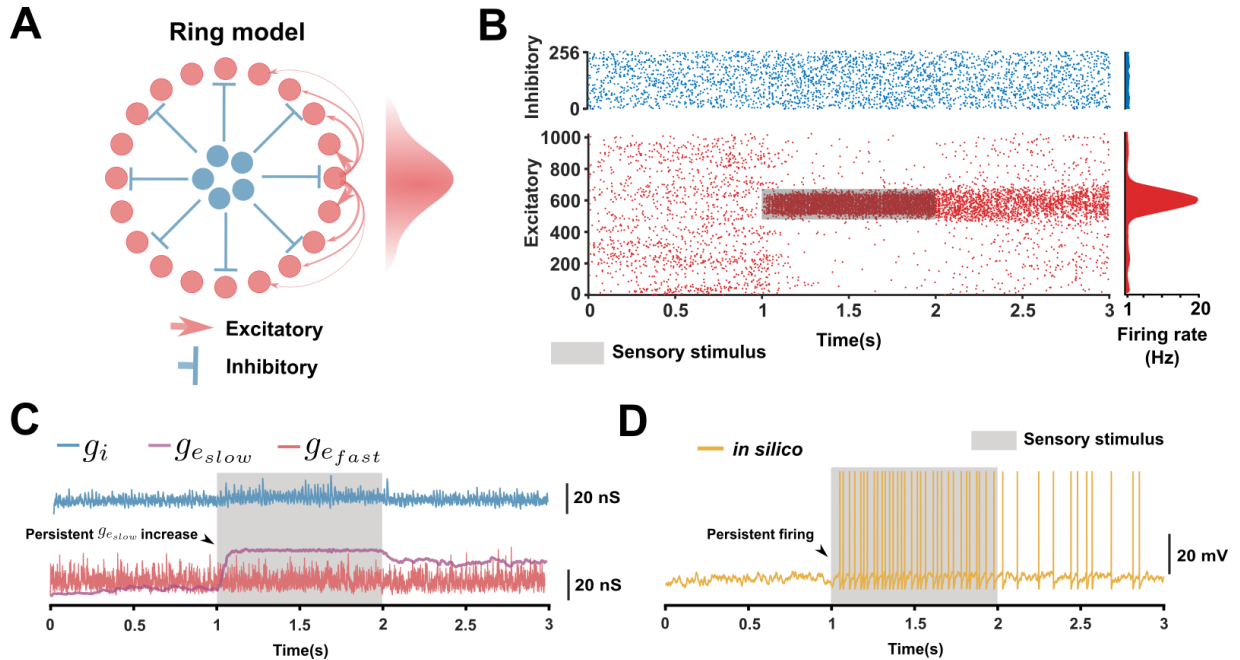


Figure 3-1. Ring model persistent activity relies on an unrealistically high NMDA dominated excitatory increase.

(A) The ring network model presents local E-E connections. The connection weights decrease with distance following a Gaussian distribution. (B) Raster plot representing inhibitory and excitatory spiking activity. Activity profile averaged over the self-sustained period (Right). (C) Conductance input signals received by a persistently firing proxy neuron model. g_i , $g_{e_{slow}}$, $g_{e_{fast}}$ representing time courses of GABA, NMDA and AMPA conductances respectively. (D) Voltage trace of neuron model receiving conductance inputs presented in panel C.

3.3.2 Parameter space characterization of clustered Netsim network

Taking these observations into account, we developed a large-scale clustered network at the scale of a single PFC stripe ($N = 50,000$ neurons, Fig. 3-3). This network model was implemented using *NETSIM*, a set of custom C code, developed in the Muller lab for simulating spiking neural networks at large scale (available at github.com/mullerlab/NETSIM; Davis et al., 2021).

While certain parameters are fixed based on experimental data, others are unknown and require running numerous simulations with different set of parameter values to find a set of values that generate the desired activity pattern and characterize the parameter space. Characterizing the parameter space of our model is crucial for understanding how the implementation of clustered connectivity affects the operation regime of the network.

As explained previously, numerical implementation requires solving numerous differential equations at each time step of the simulation; the computational cost scales with simulation time and the size of the network being simulated.

For this reason, we started simulating a network of size $N=10,000$. We performed 15,000 simulations across possible sets of parameters values being examined. Figure 3-2 present the result of this scanning process. Each heatmap shows the firing rate of the stimulated neurons for different values of synaptic weights. On the x-axis are presented the different values of excitatory weight, while on the y-axis are the values of inhibitory weight. Each heatmap represents simulations with different level of clustering and different densities of excitatory to inhibitory connectivity ($E \rightarrow I$ in-degree; Fig. 3-2). The level of clustering compares the number of in-cluster connections to the number of connections there would be in random, uniform connectivity framework. The $E \rightarrow I$ in-degree represent the average number of excitatory inputs received by each inhibitory neuron. An $E \rightarrow I$ in-degree of 230 connections reflects the expected value under unbiased interconnectivity, where excitatory neurons connect to the two population in proportion to their sizes. When this variable is increased additional $E \rightarrow I$ connections are added but other in-degree remain fixed. Thus, the top-left panel, represents a network with sparse $E \rightarrow I$ connections and random connectivity, as it has no clustering of its excitatory population.

This is a critical point of reference, as this configuration recreates the extensively studied random network model and its well-established dynamical regimes (Kumar et al., 2008). In this condition, strong inhibition combined with weak excitation leads to quiescence (blue; firing rate = 0sp/s), whereas strong excitation with weak inhibition results in runaway activity (white).

As demonstrated by Kumar et al. (2008), in randomly connected networks, a part of this stable state region is characterized by exhibits asynchronous irregular (AI) activity—marked by low, variable firing rates and low synchrony—which is considered a hallmark of *in vivo* cortical activity. In the top row, where $E \rightarrow I$ connectivity remains sparse, an increase in the clustering coefficient induces a pronounced leftward shift in the stable activity region.

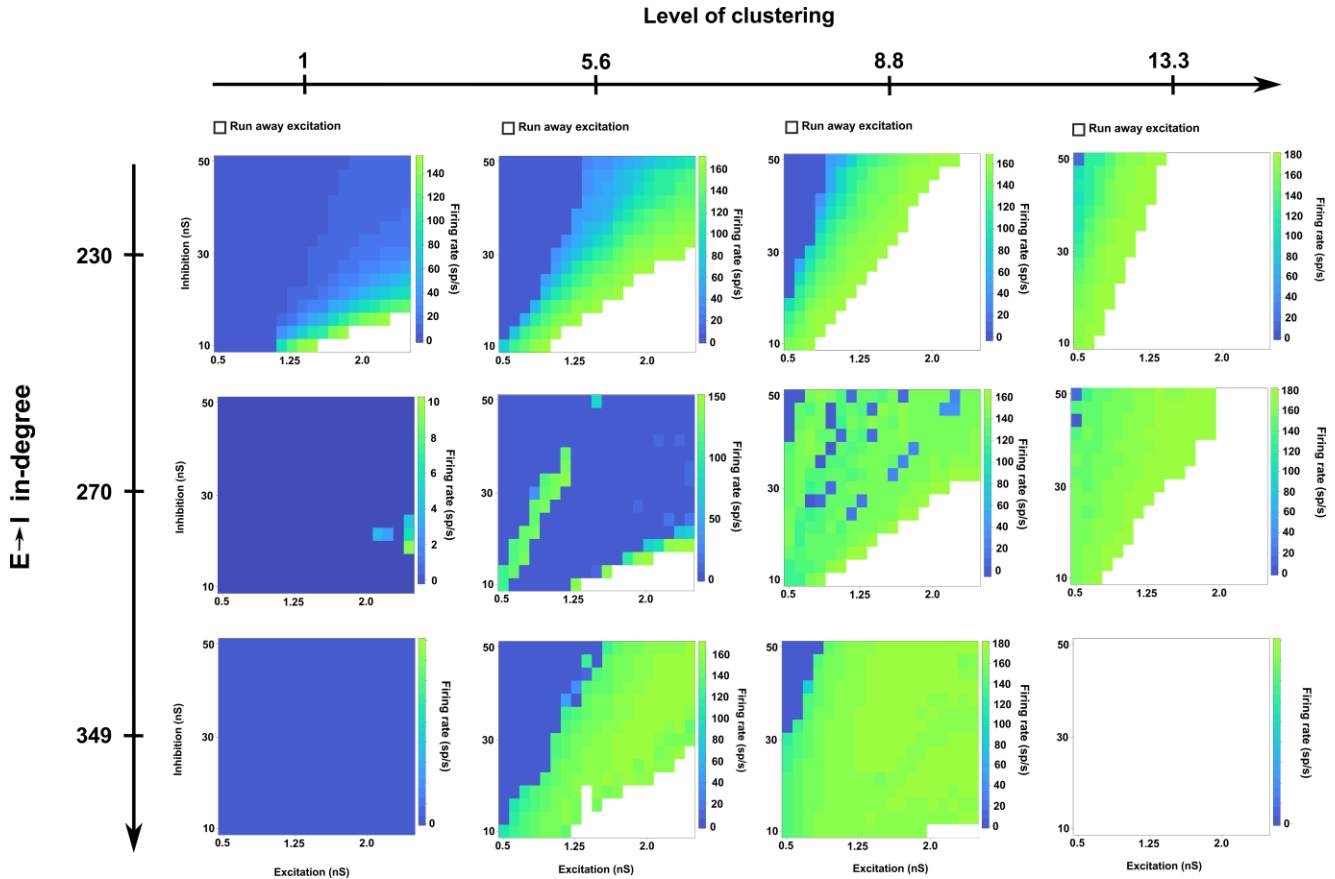


Figure 3-2. Persistent firing rate as a function of connections weights (Excitation/Inhibition), for varying level of clustering and $E \rightarrow I$ in-degree in clustered network of size $N=10,000$.

Grid of heatmaps showing the mean firing rates of persistent firing neuron observed in a clustered network of size 10,000 across the parameter space for varying level of clustering (1 (no clustering), 5.6, 8.8 and 13.3) and $E \rightarrow I$ in-degree (230 (unbiased), 270, 349).

This vanishing gradient indicates that for a given pair of g_e and g_i values, a more clustered network exhibits a significantly higher propensity for runaway excitation. The introduction of this topology alters the effective balance of conductances, which requires stronger inhibitory input to compensate for the increased excitatory gain within clusters and maintain stability.

The network combining clustered excitatory population and increasing the number of E→I connections can present a persistent bump of activity (Appendix 3), but in those conditions the neurons part of the persistent bump of activity present an unphysiologically high firing rate. In order to obtain a bump pattern of activity at low rate, we introduced additional excitatory conductance replicating the timescale of NMDA conductance ($g_{e_{slow}}$), with a time constant of $\tau_{NMDA} = 37 \text{ ms}$. In contrast, the other excitatory conductance ($g_{e_{fast}}$) has a time constant of $\tau_{AMPA} = 5 \text{ ms}$.

3.3.3 Large-scale clustered network model persistent activity in asynchronous irregular fluctuation-driven regime

For our simulations of network of size $N=50,000$, we employed Nevergrad, an open-source Python library for derivative-free optimization (Bennet et al., 2021). This tool is particularly effective as it runs multiple optimization algorithms in parallel. By using methods such as evolutionary strategies, covariance matrix adaptation, and random search, this library can dynamically allocate more computational resources to the algorithms performing best. This approach provides a more efficient way to explore a model’s parameter space.

Previous work has shown that clustered network topology can generate long-timescale dynamics in spiking neural networks (Gavenas et al., 2024; Klinshov et al., 2014; Litwin-Kumar & Doiron, 2012); however, these network architectures have not been explicitly used to model the generation of persistent spiking during WM. By exploring the parameter space of connectivity for the network, we found that organizing the excitatory neurons into 40 distinct clusters (with intra-cluster connection probability 5.94 times higher than random connectivity) can result in robust persistent activity following an input stimulus (Fig. 3-3B).

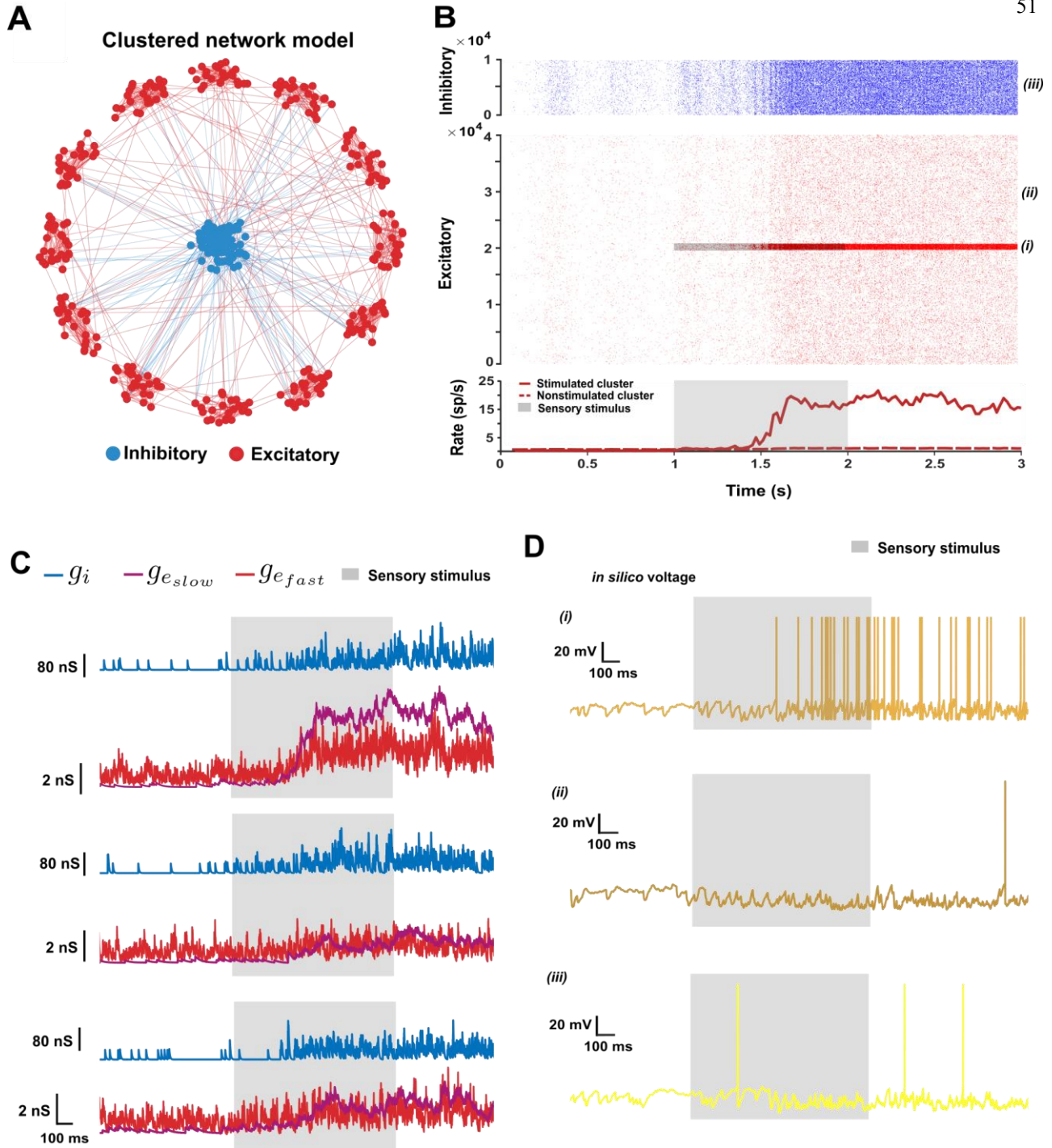


Figure 3-3. Large scale conductance based clustered network model present a plausible activity regime and synaptic signals dynamics.

(A) Graphic representation of our large-scale clustered model. The excitatory population is grouped in clusters. (B) Raster plot showing the spiking activity of the excitatory and inhibitory populations. One of 40 excitatory clusters receives a current stimulation representing sensory stimulus (grey square). The lower graph represents the evolution of firing rate of stimulated cluster and rest of the excitatory population. (C) Synaptic conductance signals of persistently firing neuron (top), a non-persistently firing excitatory neuron (middle) and an inhibitory neuron (bottom). (D) Voltage trace of persistently firing model neurons (top), a non-persistently firing excitatory neuron (middle) and an inhibitory neuron (bottom).

This clustered architecture connectivity represents a circuit motif that could plausibly occur within a single intrinsic strip pattern with dlPFC (Hoftman et al., 2017; Levitt et al., 1993; Melchitzky et al., 1998; Pucak et al., 1996). We consider our network model to represent a 300 μm wide and 1mm long stripe within the L2-3 of small primates comparable to a strip pattern in the marmoset dlPFC. Such section of cortical tissue would contain 50,000 neurons, for a neural density of 200,000 neurons/ mm^3 (Atapour et al., 2019).

Following stimulation of a localized patch of neurons, the stimulated cluster of neurons generates a persistent bump of activity, with spiking at physiologically realistic firing rates (17.28 sp/s average rate within the bump). Spiking activity within the bump exhibits variation of ISIs (mean CV = 0.88 ± 0.0118 SEM) that better match the irregularity of *in vivo* recordings from macaque dorsolateral prefrontal cortex during a standard WM task (mean CV = 0.99 ± 0.25 SEM) compared to the ring model (mean CV = 0.67 ± 0.02 SEM).

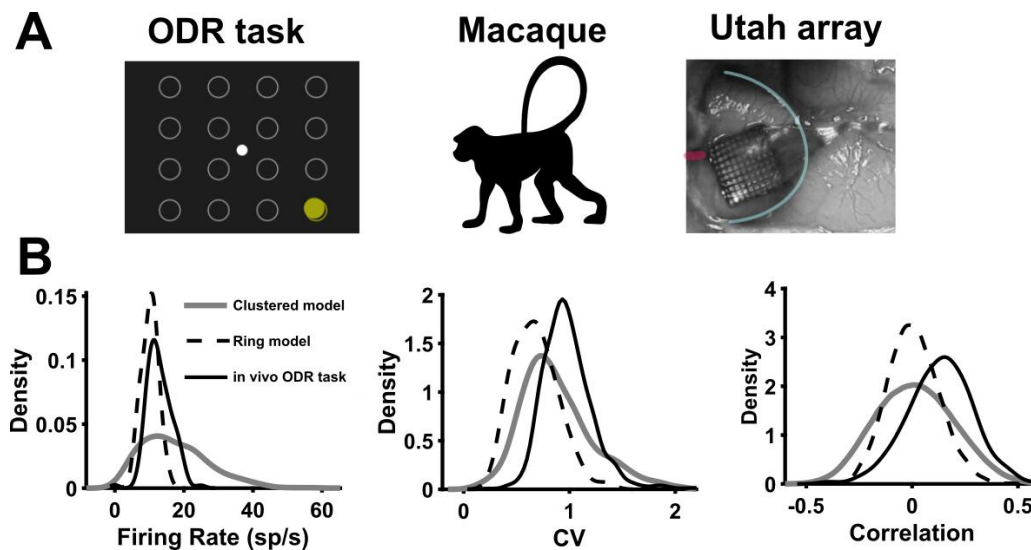


Figure 3-4. Clustered model better match *in vivo* higher order firing statistics.

(A) Neural activity, provided by Martinez lab, was recorded in the macaque prefrontal cortex using Utah array (right) while performing an ODR task (left). Adapted from Bush et al. (2024) (B) Comparison of the firing statistics distributions of persistently firing neurons in the ring network model, the clustered network model and the recorded *in vivo* neurons (when presented with preferred stimulus).

These results demonstrate that this large-scale clustered network can exhibit persistent spiking activity at a physiological rate, but with improved irregular firing pattern that better matches the activity observed *in vivo*.

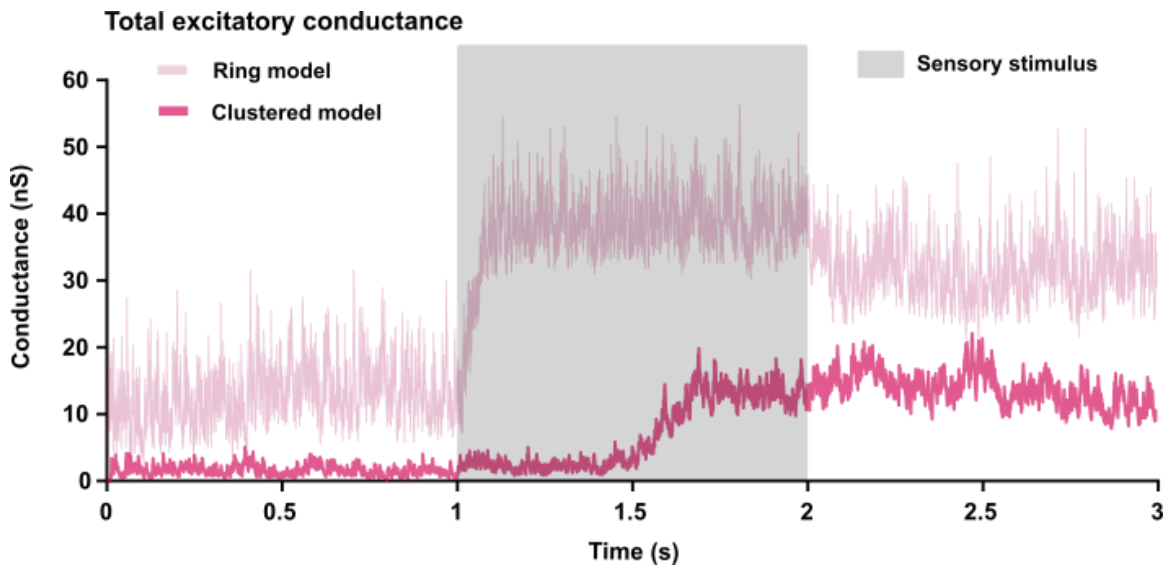


Figure 3-5. Comparison of excitatory input of persistent firing neurons in ring model and clustered model.

The AMPA and NMDA conductance traces received by persistent firing from the ring model (Fig. 3-1) were summed together (light pink). The AMPA and NMDA conductance traces received by persistent firing from our clustered model (Fig. 3-3) were summed together (dark pink).

Importantly, the fundamental mechanism underlying persistent spiking differs between these models. In the ring model, persistent activity strongly relies on a large increase in the slow NMDA conductance (Fig. 3-1). In contrast, the large-scale clustered network exhibits coordinated increases across fast AMPA, slow NMDA, and GABA inhibitory conductances (Fig. 3-3). In our model, the large, fluctuating inhibitory conductance suppresses activity globally. The high excitatory conductances, within the stimulated cluster, trigger spikes during transient decreases in inhibition (Fig. 3-7). Specifically, neurons in the stimulated cluster present a physiological firing rate (mean=17.28 sp/s), despite the increase in excitatory conductance being smaller than in the ring model (Fig. 3-6).

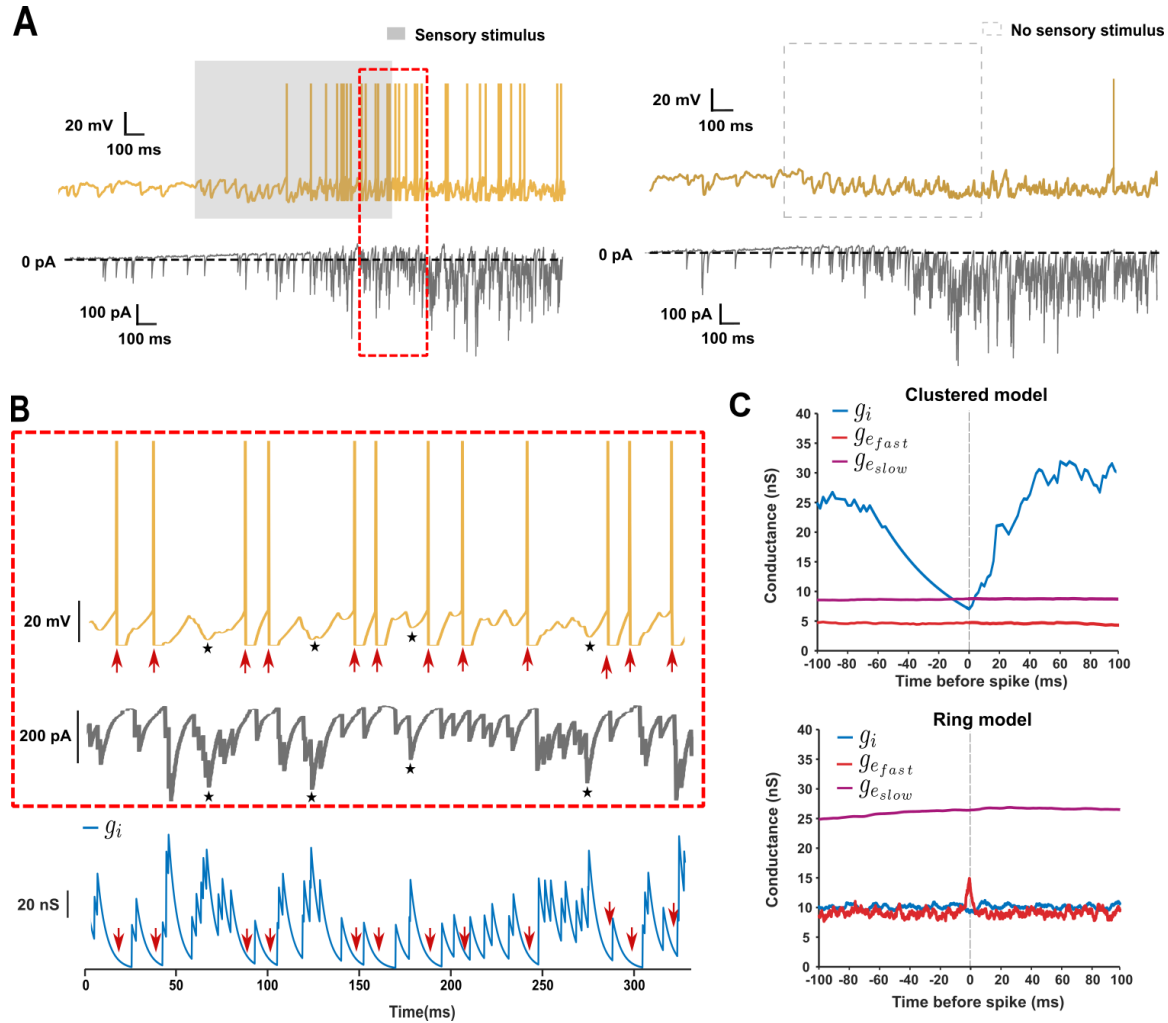


Figure 3-6. Inhibitory conductance fluctuations of clustered model drive strong negative stimulation and coincide with spike initiation.

(A) Voltage traces (top) and net current input (bottom) of a persistently firing neuron (left) and a non-stimulated neuron (right). (B) Zoom on the voltage and current input of persistently firing neuron from panel A, and the associated inhibitory conductance trace. Spike timing coincides with inhibitory conductance decrease (red arrows), while inhibitory conductance peak coincides with voltage repolarization (★). (C) Spike triggered average of the conductance signals in persistent firing neurons from our clustered model and the ring model.

The persistently firing neuron—shown in Fig. 3-3—presents, on average, a 2.35 nS AMPA increase and a 6.44 nS NMDA average increase. Meanwhile, the non-stimulated neuron presents a 0.63 nS AMPA average increase, and a 1.79 nS NMDA average increase. In contrast to excitation both neuron models present similar inhibitory increase (persistently firing neuron: $\Delta GABA = 26.21\text{nS}$; non-stimulated neuron: $\Delta GABA = 26.98\text{nS}$) starting from emergence of bump pattern of activity (Fig. 3-3 C). This inhibitory input results in the many downward peaks in the net current stimulation (Fig. 3-6 A).

The steady-state membrane potential (V_{ss}) provides a metric for the compounded excitatory inputs received by a neuron, it represents the value at which the neuron voltage would converge in the absence of action potential. In the ring model, a persistently firing neuron receives inputs that drive the voltage above the -50mV threshold ($V_{ss} = -24.3\text{mV}$). Meanwhile, the steady-state membrane potential for the persistent firing neuron in our clustered model, presenting the same threshold value, is below threshold ($V_{ss} = -60.9\text{mV}$), with spikes being initiated by transient decrease in inhibition (Fig. 3-6 C).

	Total voltage variance (mV^2)	$\text{Var}(V_{hyp} E)$ (mV^2)	$\text{Var}(V_{hyp} I)$ (mV^2)	$\text{Covariance}(V_{hyp} E, V_{hyp} I)$ (mV^2)	R (mV^2)
Ring model	5.79	6.57	3.4	-2.08	-2.11
Large-scale clustered model	83.58	6.18	20.52	-6.17	63.05

Table 1. Voltage variance decomposition.

Total voltage variance represents the voltage variance of a persistent firing neuron during persistent firing phase of the simulation. $V_{hyp} E$ represents a hypothetical voltage which consider effect of excitation while the inhibitory conductance is fixed at the mean. Times series of excitatory conductances were summed to calculate $V_{hyp} E$. $V_{hyp} I$ represents a hypothetical voltage which consider effect of inhibition and while the summed excitatory conductances are fixed at their summed mean.

To go beyond its effect of inhibition on spike timing, we wanted to quantify the contribution of inhibitory conductance fluctuations to the overall voltage variability in our model. To this end, performed a variance decomposition and compared the two models (Table 1). This analysis is a heuristic decomposition that provides insight into how excitation and inhibition contribute to the dynamics of the two models. This approach compares the actual voltage variance with that of two hypothetical voltage traces: one where only excitation fluctuates (with inhibition fixed at its mean), and one where only inhibition fluctuates (with excitation fixed at its mean).

Our large-scale clustered network exhibits substantially larger voltage fluctuations than the ring model: total variance of 83.58 mV^2 compared to 5.79 mV^2 . In the ring model, although excitation and inhibition do fluctuate, they are anticorrelated ($\text{Covariance}(V_{hyp} E, V_{hyp} I) = -2.08 \text{ mV}^2$). In our large-scale clustered model, inhibition has a strong individual effect on voltage variance ($V_{hyp} I = 20.52 \text{ mV}^2$). More strikingly, the residual nonlinear amplification is large and positive ($R = 63.05 \text{ mV}^2$). This indicates that the residual, nonlinear effect of excitatory and inhibitory conductances significantly amplifies voltage fluctuations beyond what a linear combination of their individual effects would predict.

3.4 Discussion

Our results demonstrate that a large-scale, clustered network model generates robust, self-sustained persistent activity at physiological rates in a more asynchronous irregular regime. The classical continuous ring attractor models established the necessity of recurrent connectivity, supported by NMDA receptor kinetics. Our model builds on these principles but implements a discrete, modular topology at large scale that more closely replicates the non-random structure of cortical microcircuits.

A hallmark of *in vivo* cortical activity is strongly irregular firing ($CV \sim 1$; Compte, et al., 2003; El Boustani et al., 2007; Shadlen & Newsome, 1998). While ring attractor models can sustain persistent activity at a plausible rate, they often do so with more regular spiking pattern (Barbieri, 2008). A primary finding of our model generates is that it generates persistent activity with a higher CV of 0.88; a value significantly higher than in previous ring models and closer to the *in vivo* irregularity in the awake brain (Fig. 3-3).

Fluctuation-driven regime, where firing is induced by membrane potential fluctuations from the mean rather than by integration of a mean input, has been suggested to be more plausible to explain the population behavior in cortical circuits in the AI regime (Ahmadian & Miller, 2021; Amsalem et al., 2024). While mean-driven regime activity has been shown to exist in persistently firing cortical neurons, it seems to rely more on intrinsic mechanisms (Egorov et al., 2002; Fransén et al., 2006).

In the ring attractor model, the persistent activity following a transient localized stimulus is evoked and maintained by the suprathreshold NMDA-dominated inputs. This mean-driven regime has been theorized to cause the more regular spiking patterns observed in these models, compared to the highly irregular firing seen *in vivo* (Barbieri, 2008). Additionally, experimental evidence shows that blocking of NMDA receptors does not fully abolish persistent activity (Van Vugt et al., 2020; M. Wang et al., 2013). This challenges models solely reliant on NMDA dynamics.

In contrast, applying a similar stimulus to our model causes a coordinated increase in both excitatory and inhibitory conductances, at the level of single cell synaptic inputs (Fig. 3-3). Such cohesive dynamic has been theorized to be a fundamental property of irregular *in vivo* activity (Renart et al., 2007).

In the ring model, the relatively small-scale and continuous topology requires strong excitatory synaptic transmission to maintain the bump attractor state through reverberation. In contrast, our clustered model is a discrete attractor featuring a modular topology, with high intra-cluster connection probability and low inter-cluster connection probability. Additionally, in our large-scale network, because reverberation relies on a larger number of neurons, the network can sustain a physiological firing rate from a smaller coordinated increase in both NMDA and AMPA conductance (Fig. 3-6).

In the ring model, inhibition is sufficiently strong to prevent runaway excitation at the network level. However, given the scale of the network where similarly tuned neurons share significant input overlap, it would likely not be possible to sustain fluctuations without risking collapse of the activity pattern. Yet experimental evidence, in the awake cat, shows that spike timing coincide with inhibitory decrease (Rudolph et al., 2007). This is coherent with our model, where the timing of action potential coincides with transient decrease in inhibitory conductance (Fig. 3-6 B-C).

To further explore how conductances dynamics influence voltage dynamics, we performed a variance decomposition to quantify how excitatory and inhibitory conductances contribute to the voltage fluctuations during persistent activity (Table 1). This method is not an exact variance partition but allows an interesting comparison between the operational regimes of the two models. In the ring model, excitatory and inhibitory conductances are anticorrelated. While their fluctuations nearly cancel, the strong excitatory input dominated by NMDA represents an important distinction between this model operational regime and the tightly balanced dynamic studied in random network (Ahmadian & Miller, 2021; Denève & Machens, 2016; Van Vreeswijk & Sompolinsky, 1996).

In our large-scale network, the framework is fundamentally different. Inhibition dominates the voltage variance, and strikingly, the nonlinear interaction between inhibition and excitation contributes more significantly to voltage fluctuations. This nonlinear synergy represents a loose balance regime that is more pronounced than previously described (Ahmadian & Miller, 2021; Denève & Machens, 2016).

This regime is enabled by sparse connectivity large-scale network, where neurons do not share extensive overlapping input; as a result large uncorrelated fluctuations can persist without bringing network into quiescence.

Therefore, by implementing biologically realistic clustered connectivity at a large scale, our model provides a more complete mechanistic framework for persistent activity, one that operates through coordinated conductance changes in a fluctuation-driven regime to produce an irregular firing pattern that more closely resembles those observed in the cortical activity of the awake brain. The suprathreshold NMDA-dominated inputs generated in the ring model can persist for model neurons *in silico* because voltage value is reset to hyperpolarized values after every spike, a mechanism that is not present in biological neuron. Thus, we expect that the conductance input signals received by persistently firing neurons in the ring model will not evoke physiological responses in biological neurons.

Chapter 4

4 Network clamp present models' operational regime *in vitro*

4.1 Introduction

Interfacing cells with computer-generated signals has enabled novel insights into cellular neurophysiology. With the introduction of dynamic clamp (Robinson & Kawai, 1993; Sharp et al., 1993), researchers have been able to simulate conductance-based interactions at the membrane of biological neurons *in vitro* (Prinz et al., 2004). Through this, it has been possible to create artificial ion channels to study the effect of random noise process recapitulating spontaneous activity (Chance et al., 2002; Destexhe et al., 2001; Fellous et al., 2003). These studies have presented neurons *in vitro* with stochastic conductance signals to test the impact of such noisy inputs on the integrative properties of neurons.

Network clamp goes beyond recreating synaptic input statistics, as we use dynamic clamp to inject biological cells *in vitro* with synaptic conductance inputs derived from spiking network models. While classical dynamic clamp studies have used synthetic signals, network clamp allows us to ask a more nuanced question. Because our network clamp approach bridges computational models of neural circuits and biological systems, it can be used to experimentally test the interdependence of individual neurons and their synaptic connectivity and allow rapid iterative testing of models. By providing synaptic inputs whose scale and dynamics fall within the physiological range a biological neuron can integrate, this interface enables the study of intracellular voltage responses and demonstrate direct link between the proposed network dynamic and the responses evoked in biological neurons.

We used network clamp to test, for the first time, the physiological plausibility of synaptic conductance inputs generated by two competing models of persistent activity: the classical ring attractor model and our novel large-scale clustered model.

Inputs derived from the ring model failed to evoke persistent activity in biological neurons; instead, the stimulation drove neurons into unphysiological state in the form of a

depolarization block. Conversely, conductance signals from our large-scale clustered model successfully evoked persistent firing within range of the model’s prediction. We then further explore the ability of our large-scale clustered network model to evoke predicted response in biological neurons by injecting inputs from neuron models in different contexts: injection of inputs from non-persistent neuron models or persistent firing interrupted by distractors also evoked predicted responses. The model derived inputs appropriately evoked spiking or absence of spiking and transition between both states in biological neuron voltage response. In a small set of network clamp experiments, we also replicated our result in marmoset PFC pyramidal neurons.

The network clamp method offers a powerful approach to study the interaction between single neurons and network dynamics in the controllable environment of *in vitro* intracellular recording. We next used this method to study pharmacological effects in such interaction. Dopamine is a key modulator of working memory (WM) and persistent firing (Arnsten et al., 1994; Vijayraghavan et al., 2007; M. Wang et al., 2019; Williams & Goldman-Rakic, 1995). This neuromodulator has been shown to modulate excitability at both synaptic and intrinsic levels in PFC pyramidal neurons (Cools & Arnsten, 2022; Henze et al., 2000; C. Yang & Seamans, 1996). We performed exploratory network clamp experiments to demonstrate the potential of the method to study this effect in the context of plausible voltage dynamics.

More broadly, we establish a powerful new method that has the potential to bridge theory and experiments for studying a wide range of emergent network dynamics across neural systems.

4.2 Methods

4.2.1 Electrophysiology and in vitro Recordings

The patch-clamp recordings were performed on pyramidal neurons of the medial prefrontal cortex of a 5 months old C57BL/6 mouse. All experimental procedures were performed in accordance with the Canadian Council on Animal Care guidelines and

approved by the University of Western Ontario Animal Use Subcommittee (AUP: 2018-130).

Acute brain slice preparation. Animals were intraperitoneally injected with pentobarbital (100 mg/kg) and intracardially perfused with an ice cold oxygenated N-methyl-D-glucamine (NMDG) solution containing 92 mM NMDG, 2.5 mM KCl, 1.2 mM NaH₂PO₄, 30 mM NaHCO₃, 20 mM HEPES, 25 mM D-Glucose, 5 mM sodium ascorbate, 2 mM thiourea, 3 mM sodium pyruvate, 10 mM MgCl₂, 0.5 mM CaCl₂. pH was adjusted to 7.2-7.4 with HCl. The brain was quickly removed, submerged into NMDG solution and coronally sectioned (thickness 250 μ M) using a vibratome (VT1200S, Leica Microsystems). The slices recovered 15 min at 32°C into NMDG solution oxygenated using carbogen (95% O₂, 5% CO₂). Subsequently, the slices were placed for 45 min into HEPES holding solution (92 mM NaCl, 2.5 mM KCl, 1.2 mM NaH₂PO₄, 30 mM NaHCO₃, 20 mM HEPES, 25 mM D-Glucose, 5 mM sodium ascorbate, 2 mM thiourea, 3 mM sodium pyruvate, 2 mM MgCl₂, 2 mM CaCl₂).

Electrophysiological recordings. Brain slices were placed in a recording chamber and continuously perfused with oxygenated aCSF solution (1 ml/min, 32°C) containing 126 mM NaCl, 2.5 KCl, 1.2 mM NaH₂PO₄, 26 mM NaHCO₃, 12.5 mM D-Glucose, 1 mM MgCl₂, 2 mM CaCl₂. Borosilicate glass pipettes (Sutter Instruments) were pulled in a P1000 Micropipette Puller (Sutter Instruments) and filled with an intracellular solution containing 108 mM potassium gluconate, 2 mM MgCl₂, 8 mM sodium gluconate, 8 mM KCl, 1 mM K₂-EGTA, 4 mM K₂-ATP, 0.3 mM Na₃-GTP and 10 mM HEPES. The resistance of the pipettes was between 3 and 6 M Ω and the access resistance was less than 20 M Ω . Synaptic transmission was blocked using picrotoxin (0.1 mM) and kynurenic acid (1mM).

Tissue slices were visualized using a Scientifica SliceScope microscope. Recordings were performed in whole-cell patch-clamp configuration using a Multiclamp 700B amplifier (Molecular Devices) and digitized using Digidata 1550B (Molecular Devices). After baseline recording, only neurons with a stable resting membrane potential below -50 mV, cell capacitance and membrane resistance were estimated. Only neurons with a

stable resting membrane potential below -50 mV, series resistance between 50–130 M Ω , and input resistance above 50 M Ω were included for analysis. Membrane resistance and access resistance were monitored throughout the recordings. Recordings were discarded if the access resistance increased above 20 M Ω during the experiment. In the current-clamp set-up, the neuron membrane voltage was adjusted to -70mV if necessary. Holding current was injected to maintain voltage at -70mV. To characterize basic electrophysiological properties, a series of 1-second square current pulses were injected, starting with a hyperpolarizing current sufficient to bring the membrane potential to -100 mV. The amplitude of subsequent current pulses step increments, until the firing pattern shows signs of saturation. The rheobase current and voltage threshold for action potential initiation were measured from these recordings. Responses to hyperpolarizing current injections were obtained to measure the sag ratio and rebound delay. The sag ratio was calculated as the steady-state voltage held at -55mV divided by the peak voltage deflection in response to a hyperpolarizing current step bringing the membrane potential to approximately -100 mV.

4.2.2 Network Clamp Experimental Technique

Windows based dynamic clamp software Stdpc was used for network clamp experiments (Kemenes et al., 2011). The Digidata 1550B was replaced by the National Instruments data acquisition (NIDAQ)-enabled National Instruments (NI) board. Files containing the conductance signals inserted in the Stdpc software as SimulDAQ conductance time series files.

The sensory cue square-pulse current stimulation was produced with the Wire module. Network clamp stimulation and recording were performed in current-clamp mode, no holding current was maintained. Network clamp stimulation was repeated 4 times consecutively with a 1 second interval.

At each time step, t , the Stdpc software calculated the current stimulation for each signal $I_{\{e,i\}}(t)$ is calculated based on the target conductance $g_{\{e,i\}}(t)$ and the driving force (Fig.1 A) following equation:

$$I_{\{e,i\}}(t) = g_{\{e,i\}}(t)(V_m(t) - E_{\{e,i\}})$$

The neuron voltage and the effective current stimulation were collected using the Clampex software. Data was analyzed using MATLAB 2024B custom scripts.

4.2.3 Oculomotor delayed response (ODR) task

In vivo data was collected by Dr. Martinez-Trujillo Lab (Busch et al., 2024) . The neural activity was recorded during ODR task, a 96-channel Utah array was implanted in two macaque left LPFC between principal sulcus and arcuate sulcus. The oculomotor delayed response task performed was separated into four epochs: fixation, stimulus presentation, delay, and response. The cue was presented for 1000 ms at one of 16 grid locations. The cue then disappeared and the subject maintained fixation for an additional 3000 ms or 1500 ms, after which the fixation point disappeared, and the subjects responded by making a saccade to the remembered location.

4.2.4 Neuronal tuning and persistent firing

We assessed neuronal spatial tuning by comparing firing rates across stimulus locations using a Kruskal–Wallis one-way ANOVA (criterion: $p < 0.05$). A neuron was deemed tuned if firing rates varied significantly across locations, with the preferred location defined by maximal firing rate. Persistent firing was identified based on two criteria: (1) significant spatial tuning during the delay period, and (2) significantly higher firing rate during delay than during baseline inter-trial intervals. ANOVA tests with adjustable significance thresholds were used. Relaxing these thresholds progressively included additional neurons into the persistent-firing category. Using these criteria, persistent neurons were identified in NHP B performing the ODR task, aligning with previous findings. The number of persistent neurons slightly exceeded previous literature; stricter thresholds reduced this number to conventionally reported levels.

4.3 Results

4.3.1 Network clamp allows the injection of network model derived inputs into biological neurons *in vitro*

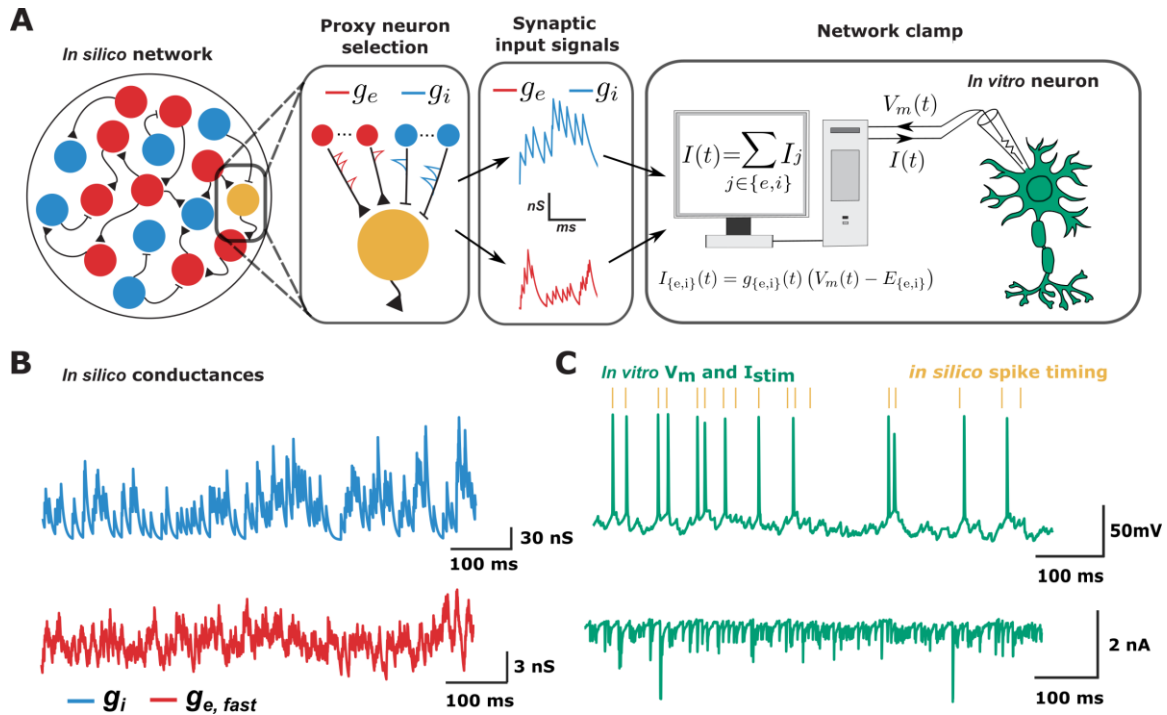


Figure 4-1. Network clamp replicates presynaptic input of an artificial neuron *in vitro*.

(A) A proxy neuron is selected from the conductance based neural network model of interest. The synaptic inputs are compiled into separate time series of conductances. For each time step, these conductance values are converted into current by multiplying them by the driving force, which is calculated from the biological neuron's membrane potential in the previous time step. The resulting currents are summed and injected into a biological neuron via whole-cell patch clamp electrophysiology. (B) Example excitatory and inhibitory conductance signals. (C) Voltage trace (top) and current stimulation (bottom) during a network clamp injection of the conductance signals presented in panel B.

The network clamp method allows us to test model-derived predictions in biological neurons (Fig. 4-1), validating the biological realism of computational models. Beyond validation, network clamp enables us to identify key biophysical properties underlying the emergence of neural activity states—features that are not readily detected using conventional approaches.

For example, widely used square-pulse current-clamp protocols facilitate mechanistic characterization under simplified conditions but fail to capture the complexity of *in vivo* input dynamics. In contrast, network clamp applies theoretically-grounded stimuli that reflect state-generating network inputs, enabling us to probe neuron-network interactions in biological neurons *ex vivo*.

In network clamp, the currents injected, calculated in real time, result from the interaction of the synaptic conductances generated *in silico* that reflect realistic synaptic input dynamics that neurons might experience *in vivo* and the properties of the biological neuron in patch-clamp that determine the voltage response of the neuron. This interaction enables the intracellularly recorded biological neuron to respond with realistic membrane potential dynamics that reflect a natural operating regime (Fig. 4-1B, right).

Critically, the network models used in network clamp are not tuned to produce synaptic conductances of a specific amplitude, but rather to generate hypothesis-driven network-level dynamics. The resulting conductance signals are injected into *in vitro* neurons without scaling or adjustment, preserving their model-derived dynamics while allowing the biological neuron's voltage-dependent responses to shape the current integration naturalistically.

4.3.2 Network clamp injection of the ring model causes inactivation

We first tested injecting synaptic inputs from the ring model of Compte et al. (2000), an important computational model for understanding the mechanism of persistent activity in PFC. Conductance traces from replication of the Compte ring model were injected into pyramidal neurons of the mouse prelimbic (PL) area of mPFC using StdpC dynamic clamp software (Kemenes et al., 2011). Prior to network clamp stimulation, properties of pyramidal neurons were characterized using the classical square-pulse stimulation protocol (Fig. 4-2).

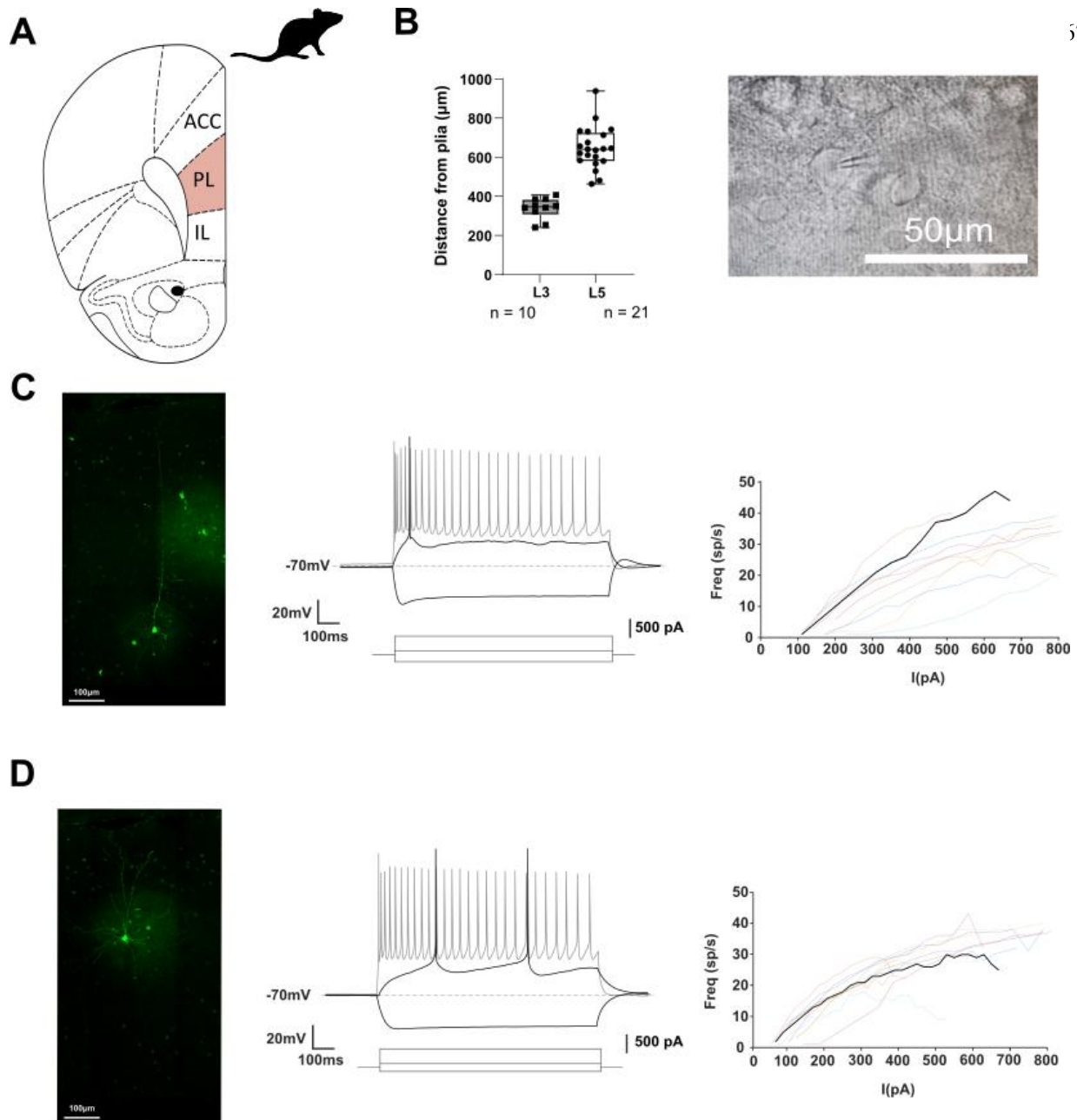


Figure 4-2. Electrophysiological characterization of pyramidal neurons in the mouse medial prefrontal cortex (mPFC).

(A) Schematic of mouse mPFC subregions. Patch clamp recordings were made in the prelimbic area (PL). (B) (Left) Classification of neurons from layer 2-3 and 5 based on their soma's distance from the pial surface. (Right) Representative differential interference contrast (DIC) image showing whole-cell patch clamp recording from a pyramidal neuron in an acute mPFC slice (scale bar: $100\mu\text{m}$). (C) Confocal image of a biocytin-filled layer 5 pyramidal neuron visualized with streptavidin-Alexa Fluor 568 (pseudo-colored green). scale bar: $100\mu\text{m}$ (left). Example voltage responses to square current injections used for intrinsic property characterization (middle). Frequency-current (F-I) relationship, with the black curve representing the average across $n = 21$ cells (right). (D) Confocal image of a biocytin-filled layer 3 neuron visualized with streptavidin-Alexa Fluor 568 (pseudo-colored green). Scale bar: $100\mu\text{m}$ (left). Example voltage responses to square current injections used for intrinsic property characterization (middle). Frequency-current (F-I) relationship, with the black curve representing the average across $n=10$ cells (right).

The variability in properties among pyramidal neurons test did not cause any significant layer-specific difference in network clamp experiments. Thus, the data of those two populations were pooled together in subsequent analysis.

We selected a neuron from the group of persistently firing cells in this ring model and injected its conductance inputs into individual L2-3 and L5 pyramidal neurons in the PL of mouse medial PFC (Fig.4-2). The entire 3-second simulation was injected into the neuron, including the pre-stimulus baseline, the stimulus period, and the post-stimulus phase.

During stimulus presentation, when the network generates the persistent bump in the computer simulation, the strong NMDA conductance pushes the membrane potential of the biological neuron into an inactivation state (Fig. 4-3D). We replicated the experiment with ring simulation matching the original framework and report similar results (Fig. 4-5).

	Capacitance (pF)	Resistance (M Ω)	Resting (mV)	Threshold (mV)	Sag ratio	AP amplitude (mV)	AP half-width (ms)
L5 (n=22)	140.93 $\pm$ 9.6	138.35 $\pm$ 14.43	-78.25 $\pm$ 1.4	-32.92 $\pm$ 1.2	0.0996 $\pm$ 0.0146	85.16 $\pm$ 1.5	1.22 $\pm$ 0.079
L2-3 (n=12)	116.94 $\pm$ 6.2	134.17 $\pm$ 11.38	-77.58 $\pm$ 2.5	-37.41 $\pm$ 0.6	0.0510 $\pm$ 0.0086	85.53 $\pm$ 1.1	1.25 $\pm$ 0.084
P value	0.1180	0.8731	0.8659	0.8623	0.1166	0.8557	0.3391

Table 2. Electrophysiological comparison between L5 and L2-3 pyramidal neurons in the prelimbic area (PL) of the mouse mPFC. Value mean $\pm$ SEM. P values were determined using the Mann-Whitney test.

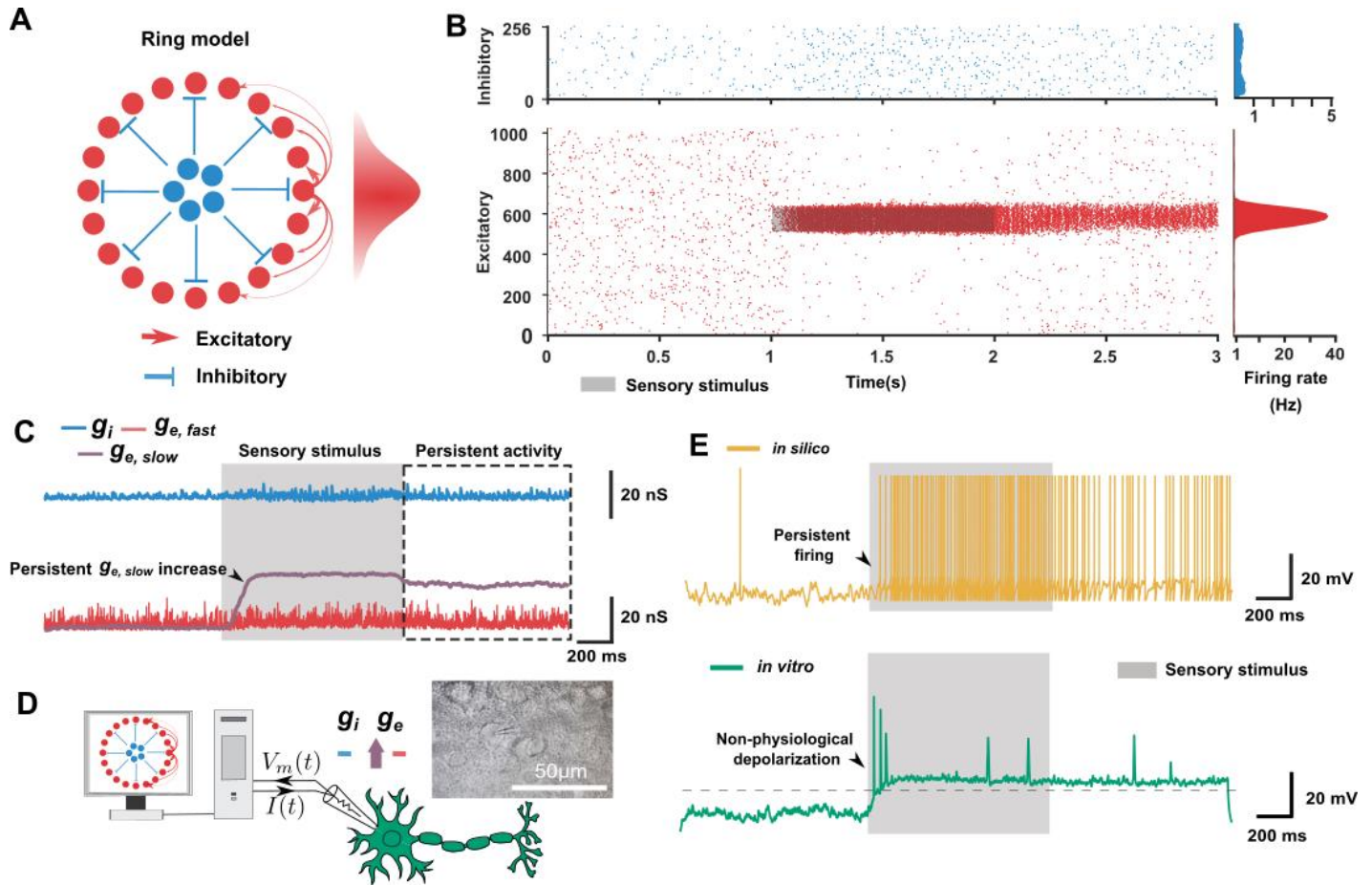


Figure 4-3. NMDA-dominated synaptic conductance signals from the ring model induce a non-physiological *in vitro* voltage response.

(A) The ring network model presents local E-E connections. The connection weights decrease with distance following a Gaussian distribution. (B) Raster plot representing inhibitory and excitatory spiking activity. Activity profile averaged over the self-sustained period (Right). (C) Conductance signals of a persistently firing proxy neuron model. (D) Network clamp experimental set up. Synaptic conductance inputs from the ring model were injected into the soma of a neuron *in vitro* under whole-cell patch-clamp configuration. (Top right) Bright-field micrograph of mouse mPFC brain slice during whole-cell patch clamp configuration. (E) (Top) Voltage trace of the persistently firing proxy neuron model receiving the conductance inputs presented in C. (Bottom) Voltage response of the *in vitro* neuron evoked by network clamp injection of conductance signals from panel C. Recording shows signs of overstimulation and inactivation concurrent to slow excitatory conductance increase (grey dotted line represents initial threshold value).

This inactivation, observed in all neurons tested ($n = 5$, Fig. 4.4), demonstrates that while a NMDA-dominated state can generate persistent firing in model neurons *in silico*, the same model-derived inputs drive biological cells into a non-physiological activity regime where they do not exhibit persistent firing. This result has been replicated in pyramidal neurons of area 46 in the marmoset PFC (Fig. 4-11 C).

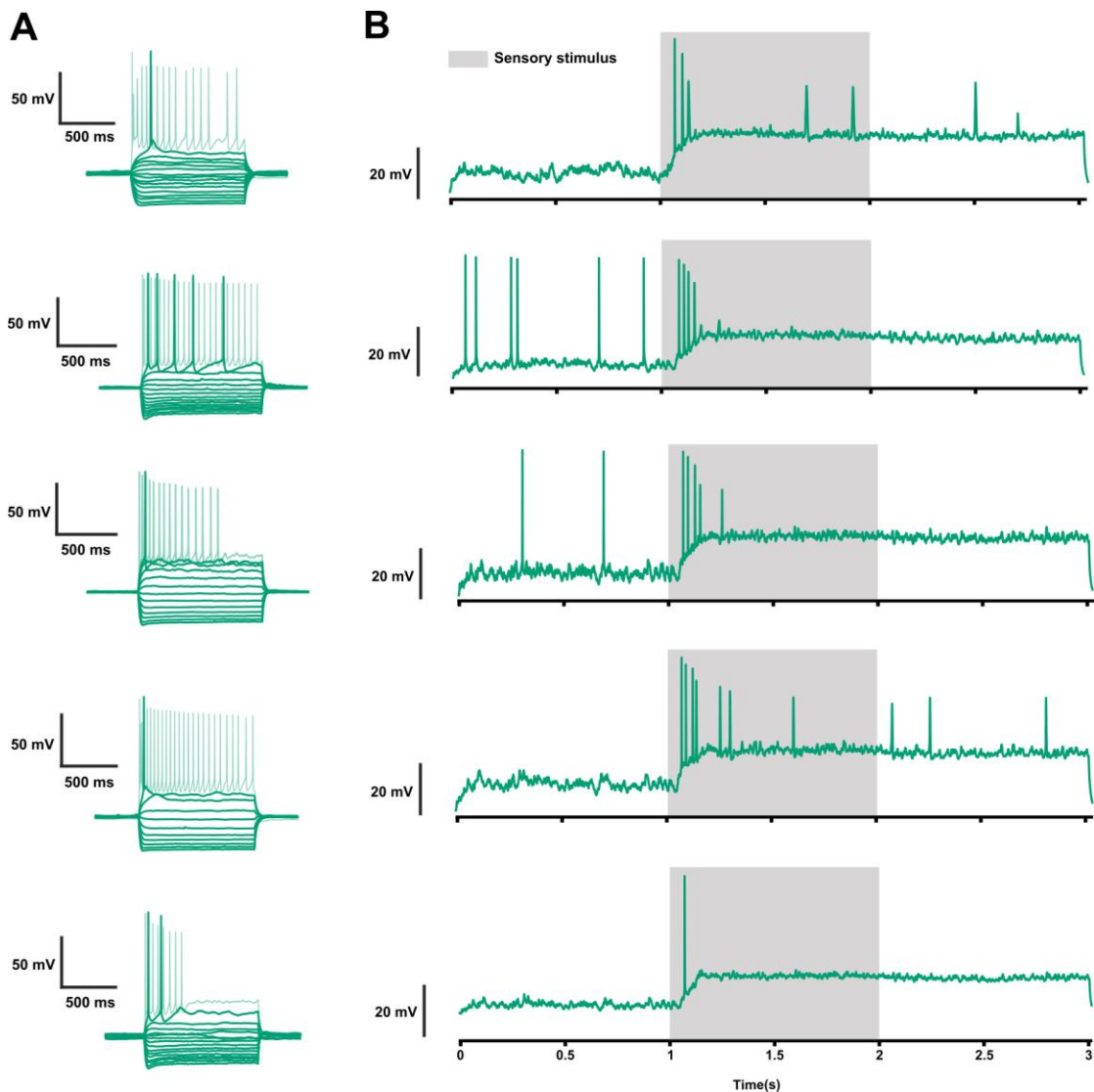


Figure 4-4. Replications ring model network clamp stimulation on mPFC pyramidal neurons.

(A) Example traces of voltage response to square current injection. (B) Example traces of voltage response to network clamp injection of synaptic conductance signals of persistently firing model neuron from ring attractor network model in the respective biological neurons shown in A.

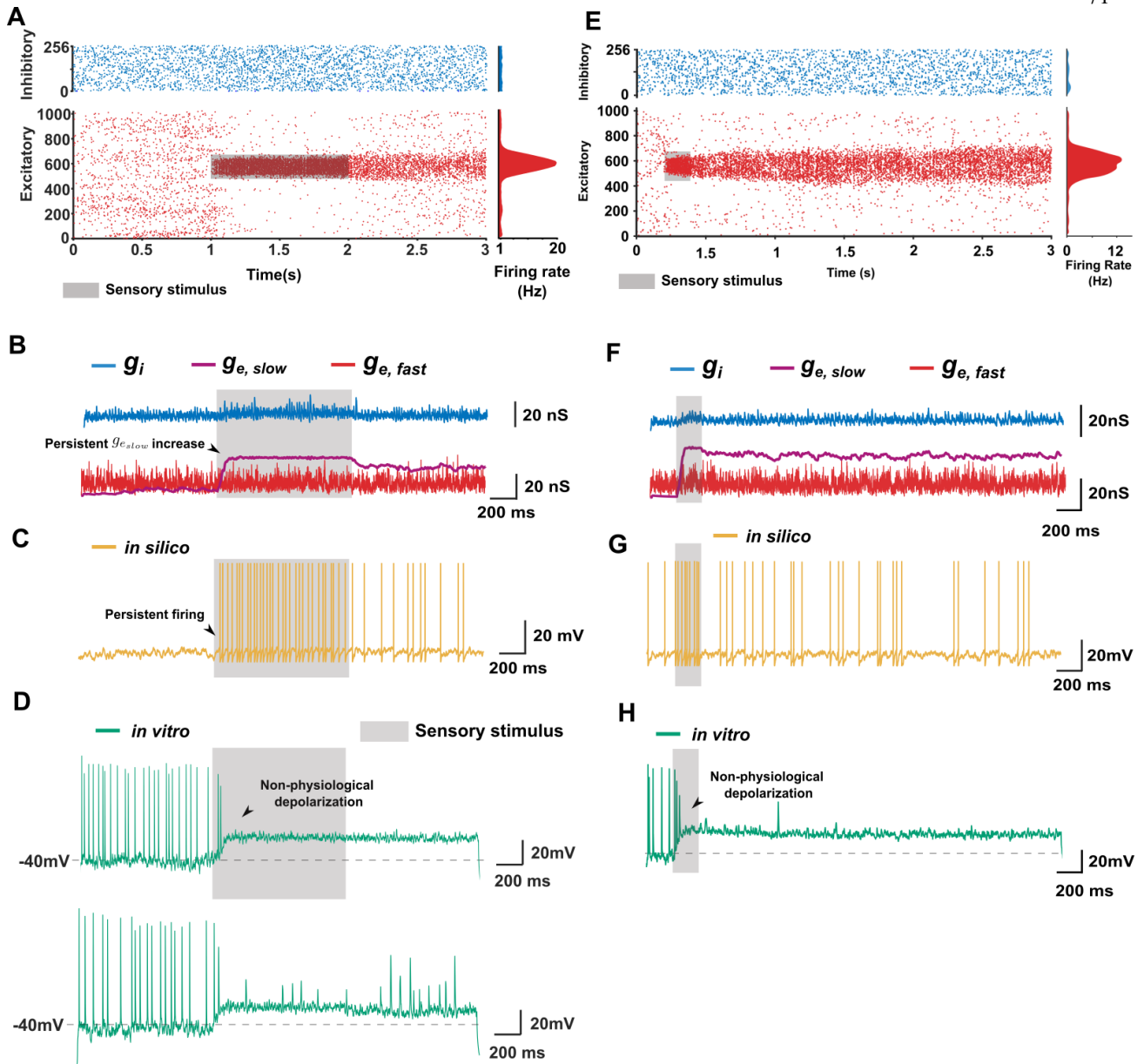


Figure 4-5. Synaptic conductances from Compte et al. (2000) induce non-physiological voltage response *in vitro*.

(A) Raster plot of a ring model replication of Compte et al. (2000) with the original resistance and capacitance. (B) Conductance signals from a representative persistently firing model neuron. (C) Voltage trace of the selected model neuron. (D) Injection of synaptic conductance signals from B into biological neurons generates action potential during the pre-stimulus phase when model neuron rarely fires. During the stimulus and post-stimulus phases, the biological neurons respond with non-physiological response with sustained depolarization and a loss of action potential firing. (E) Raster plot of ring model replication of (Compte,2000) with the original neuron properties and the original 200ms transient stimulation. (F) Conductance signals from a representative persistently firing neuron model. (G) Voltage trace of the representative model neuron. (H) Injection of synaptic conductance signals from F into the biological neuron generates action potentials during the pre-stimulus phase. During the stimulus and post-stimulus phases, the biological neurons respond with non-physiological response with sustained depolarization and a loss of action potential firing.

We modified the model neurons intrinsic properties of resistance and capacitance from the original publication to better match the biological pyramidal neurons in the mouse PL area; we used the values from our own model, which were picked for the same reason.

But we also tested the conductance inputs from ring attractor simulation with intrinsic properties and transient stimulation proposed in the original publication (Fig. 4-5). This adjustment caused similar nonphysiological responses in the biological neurons as the persistent activity state remained driven by strong NMDA-dominated conductance inputs (Fig. 4-5 D,H).

4.3.3 Large scale clustered network model evokes physiologically plausible persistent activity within range of prediction

To validate the relevance of the network model scale and topology and the resulting synaptic operational regime, we used network clamp to inject the conductance traces from model neurons into pyramidal neurons of the mouse PL area (Fig. 4-2). Strikingly, we found that when inputs received by a persistently firing model neuron, from within our clustered network, were injected into biological cells via network clamp, that stimulation evoked persistent spiking with realistic membrane potential dynamics (Fig. 4-6 C). This voltage dependent stimulation accurately drove the voltage dynamics of the biological neuron to follow the model's prediction, with the exception of refractory periods and numerous successive spikes (Fig. 4-6 D). *In vitro* neurons were injected with inputs from neuron models persistently firing at different rates (Fig. 4-7; Fig. 4-8 A). While the *in vitro* neurons presented a firing rate lower than predicted, the injection of input from neuron models with higher firing rates caused a higher firing rate in the *in vitro* neurons (n = 25; Fig. 4-6 E). These results were consistently observed both in L2/3 and L5 pyramidal neurons.

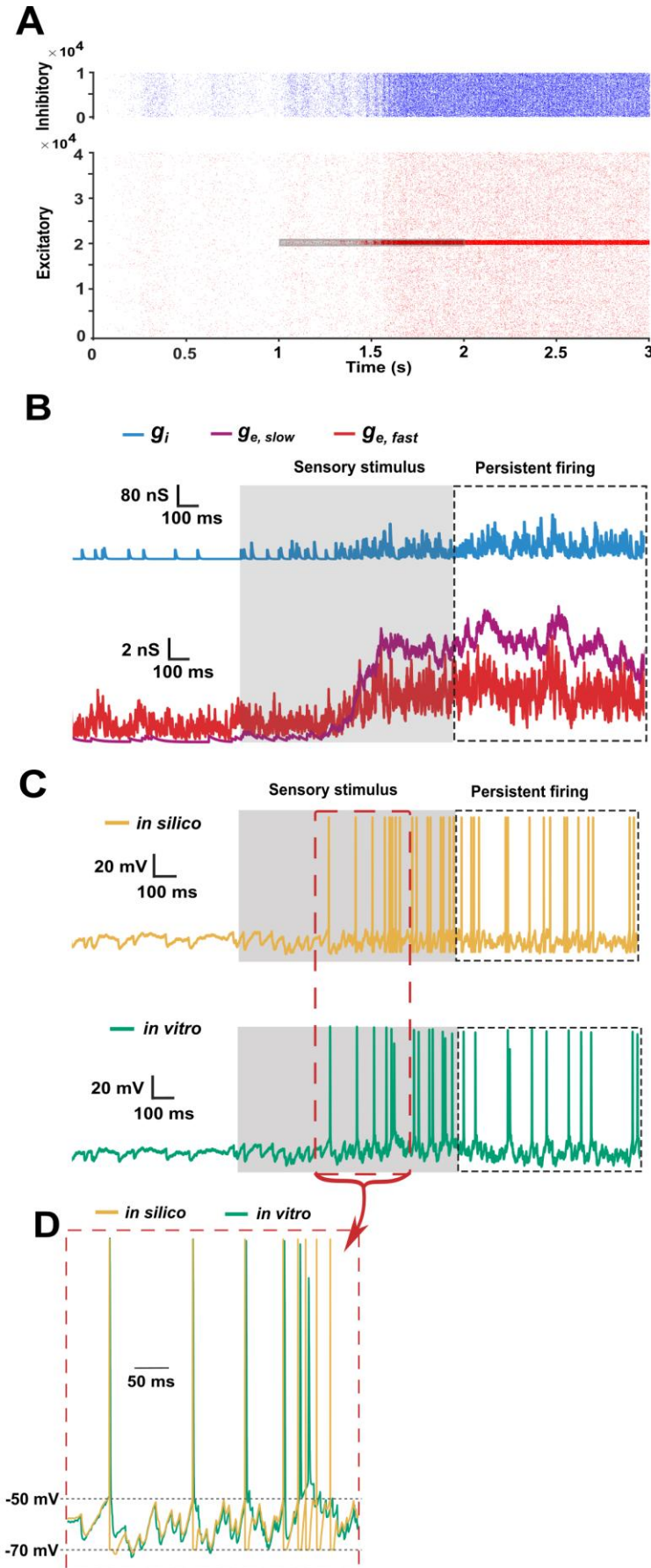


Figure 4-6. Conductance signals generated within large-scale model with clustered topology is able to evoke plausible persistent spiking activity *in vitro*.

(A) Raster plot of the spiking activity of our clustered network model presented in Fig 3-3. (B) Conductance input signals received by persistently firing proxy neuron model. (C) (Top) Voltage trace of the proxy persistently firing neuron model receiving conductance inputs from panel B. (Bottom) Biological neuron *in vitro* voltage response to network clamp injection of conductance inputs from panel B (D) Zoom in and superposition of the voltage traces of neuron model and *in vitro* neuron receiving inputs from panel C.

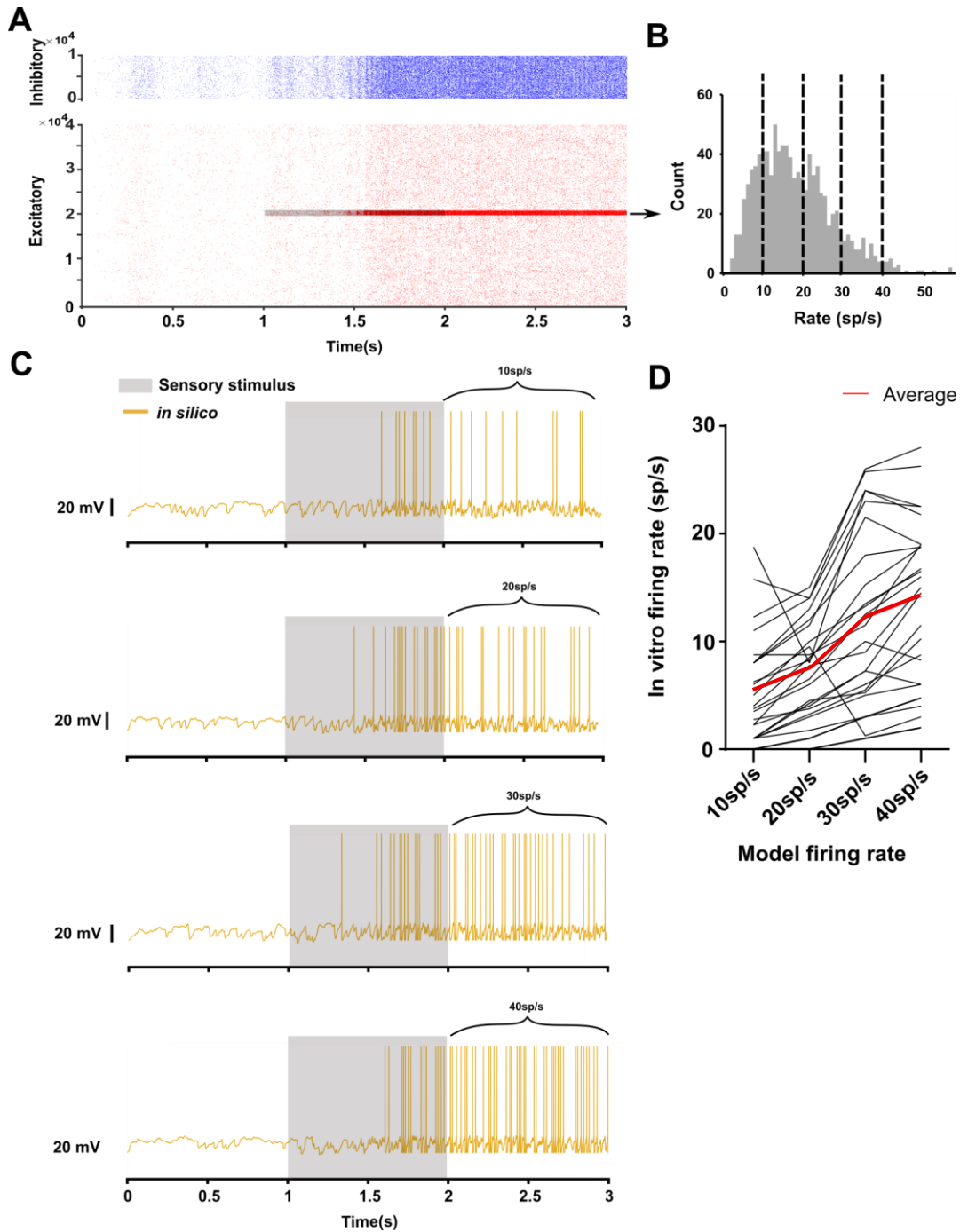


Figure 4-7. Persistently firing neurons with representative firing rates were selected for network clamp experiments.

(A) Raster plot of the spiking activity of our clustered network model presented in Fig 3. (B) Distribution of firing rate during the persistent firing phase stimulated cluster of neurons. (C) Four representative model neurons were selected based on the firing rate (10, 20, 30, 40 sp/s) indicated in the firing rate distribution graph shown in B. The synaptic conductance signals of these representative model neurons were used as network clamp stimulation in Fig. 3 and 4. (D) Firing rate of biological neuron *in vitro* in response to network clamp injection of inputs from neuron models presenting different rates of persistent firing.

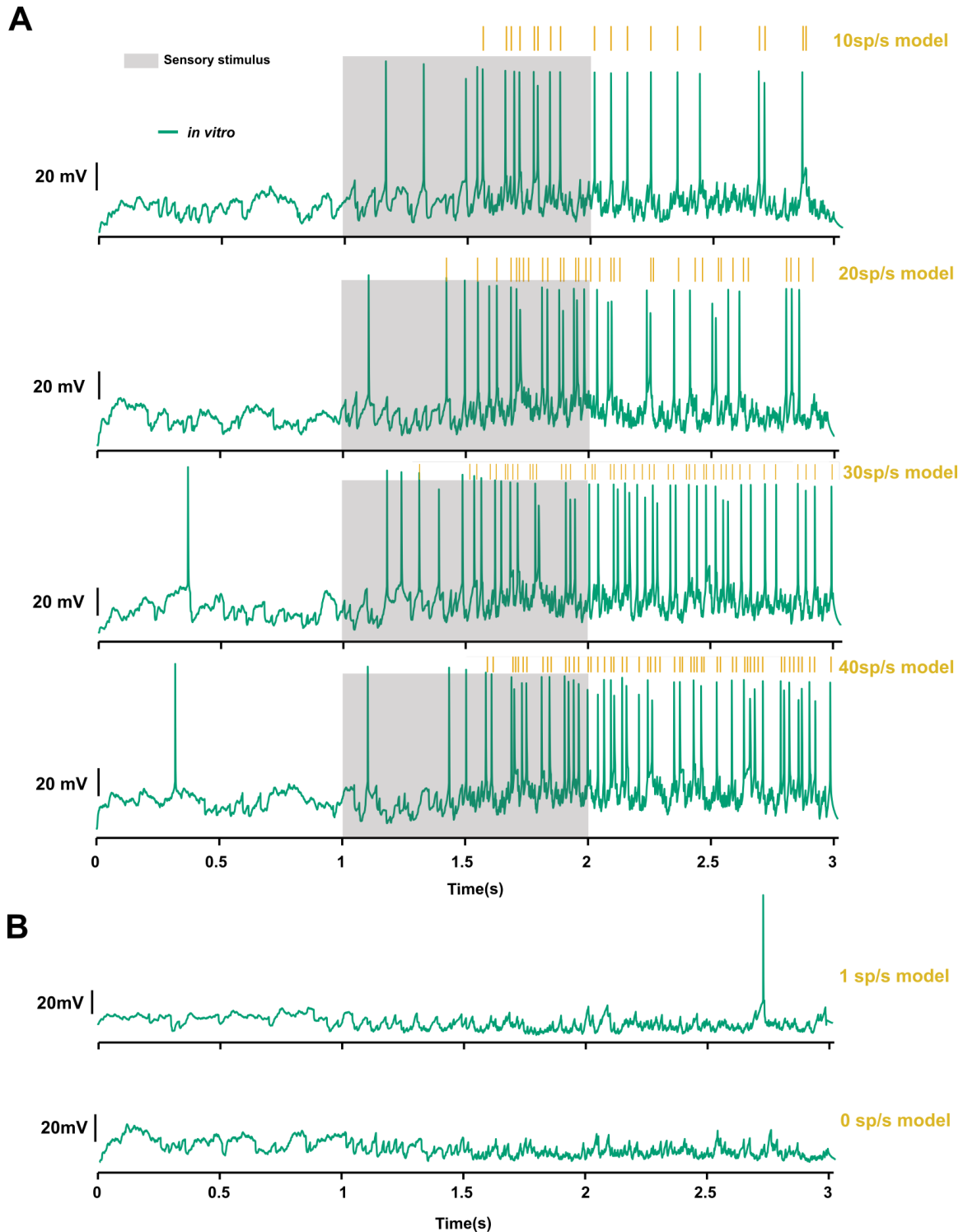


Figure 4-8. Example voltage trace of *in vitro* neuron in the PL area of the mouse mPFC during network clamp stimulation.

(A) Example voltage traces of mouse mPFC pyramidal neurons during network clamp injection of synaptic conductance signals of persistently firing model neurons firing around 10, 20, 30 and 40 sp/s. **(B)** Example voltage traces from mouse mPFC pyramidal neurons during network clamp injection of synaptic conductance input signals from model neurons that do not show persistent firing.

4.3.4 Network clamp validation of large-scale dynamics across contexts

In addition to injecting *in vitro* neurons with inputs received by persistently firing neuron we performed control network clamp stimulations—where we injected inputs from model neurons that do not exhibit persistent spiking—did not evoke persistent firing in these cells (Fig. 4-8 B).

A key feature of persistent activity models is that the activity can be robust to distractors while the bumps can also be flexible to shift when driven with an effective stimulus. We next validated that the large-scale clustered network could exhibit these more complex dynamics, and that these dynamics could also be followed by biological neurons in network clamp. To do this, we simulated the effect of distracting stimuli in the network model by applying current stimulation to a separate excitatory cluster after the initial stimulus. Persistent bumps in the clustered network were robust, with distractors much larger than the initial stimulus required to interrupt the persistent bump (5nA, Fig. 4-9). When successful, however, the distractor stimulus not only interrupts the initial bump but also generates a new persistent bump at the stimulated location. When we injected conductance signals from model neurons participating in the initial bump into biological neurons via network clamp, they exhibited persistent activity as predicted. Similarly, when we injected signals from neurons in the cluster receiving distractor stimulation, the biological neurons accurately followed the predicted transition from the initial persistent state to the new distractor-evoked state. These results demonstrate that network clamp can capture complex network state transitions, further validating the biological relevance of our large-scale clustered model.

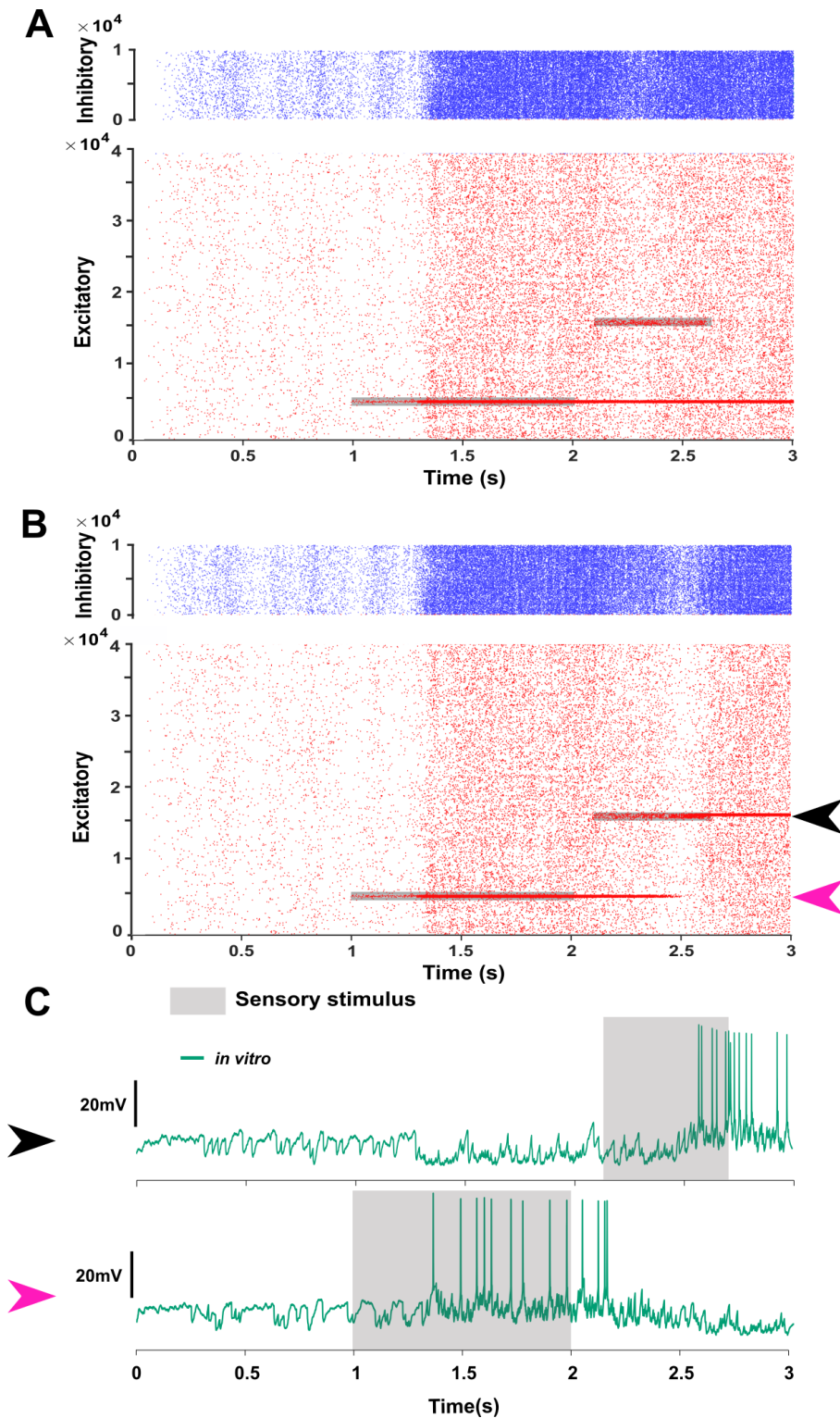


Figure 4-9. Network clamp stimulation replicates switch in activity in response to injection of synaptic input from distractor.

(A) Raster plot of the clustered network model activity. One cluster receives a 200 pA and a subsequent 4.8 nA distractor in a different cluster. (B) Raster plot of the clustered network model activity. One cluster receives a 200 pA and a subsequent 5 nA distractor stimulus in a separate cluster. (C) Network clamp injection of synaptic inputs of one neuron part of the cluster receiving the initial stimulus (bottom) and from a neuron receiving the distractor stimulus (top).

4.3.5 Network clamp enables the study of dopamine effect on neuron excitability in an *in vivo* like context

Network clamp thus allows studying in detail how biological neurons respond to synaptic inputs generated by spiking neural networks while performing computations. Crucially, this hybrid technique offers a platform to investigate how pharmacological agents interact with active network states: by pharmacologically manipulating the intrinsic properties of the biological 'node', we can determine how changes in the intrinsic properties shape the integration of network-level synaptic inputs.

To examine this, we next tested how dopamine affects neuronal responses to the same network-derived synaptic input. We performed network clamp stimulation before and after bath application of dopamine (20 μ M) to assess how neuromodulatory effects on intrinsic neuronal properties influence responses to *in vivo*-like stimulation. Standard current-clamp characterization confirmed that dopamine depolarized resting membrane potential and facilitated firing response to depolarizing current steps, increasing overall neuronal excitability (Fig. 4-10. A-D), replicating previously reported results *in vitro* (Henze et al., 2000; Seamans & C. Yang, 2004). When we injected network-derived synaptic inputs from our clustered network into neurons following administration of dopamine, we observed increased firing rates compared to pre-drug conditions (Fig. 4-10. E-F), demonstrating that neuromodulatory changes at the single-neuron level tune their network-generated activities. Further, network clamp revealed that the facilitatory actions of dopamine were more prominent during the stimulus phase (i.e. the onset of persistent firing) than the subsequent sustained phase (Fig. 4-10. G), suggesting specific cellular mechanisms through which dopamine promotes persistent firing.

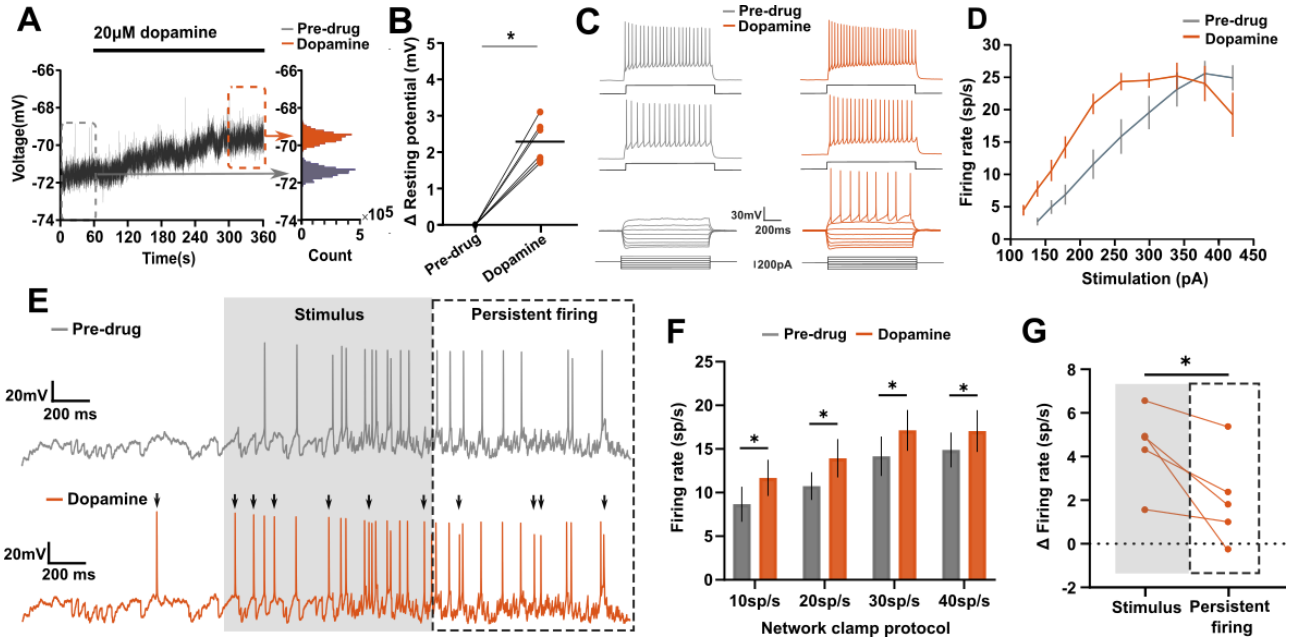


Figure 4-10. Dopamine facilitates mPFC pyramidal neurons persistent activity during network clamp.

(A) Example voltage trace of mPFC pyramidal neuron before and after dopamine treatment (20 μ M) bath application. (B) Summary graph for dopamine-induced resting membrane potential increase wilcoxon * $p=0.0312$ (C) Example voltage traces in response to square-pulse current injections under a baseline condition and after dopamine application. Bottom, 1 s-current injection starting with -190 pA and +40 pA steps. Middle trace, +210 pA stimulation. top trace, +250 pA stimulation. (D) Firing-Current injection (F-I) curve (n=6) under baseline condition (grey) and after dopamine application (orange). (E) Example voltage traces of an *in vitro* neuron receiving the same 3 s-network clamp stimulation before and after dopamine application. Arrows indicate additional spikes in dopamine condition which were not present in pre-drug condition (F) Summary graph for firing rate of *in vitro* neurons before (grey) and after dopamine application (orange). A two-way repeated measures ANOVA revealed a significant main effect of dopamine treatment (n = 5, $F(1, 4) = 24.89$, $p = 0.0075$). Šídák's multiple comparison test showed that dopamine significantly increased firing at 10 sp/s ($p = 0.0103$), 20 sp/s ($p = 0.0122$), 30 sp/s ($p = 0.0245$) and 40 sp/s ($p = 0.0289$). (G) Firing rate increase by dopamine treatment is significantly higher (paired t-test, $p = 0.0449$) during the stimulus phase (grey), during the emergence of bump activity pattern than during post stimulus phase (self-sustained persistent activity).

Importantly, we found that the increase in persistent firing can occur through interactions with multiple intrinsic factors: some neurons showed reduced firing thresholds while others exhibited increased sensitivity to depolarizing events with maintained inhibitory responses (Fig. 4-11), an observation in line with dopamine's multifaceted actions facilitating neural excitability (Seamans & C. Yang, 2004). While multiple intrinsic factors shaped the change in responses across biological cells after dopamine, the consistent overall effect demonstrates how network clamp can dissect contributions of single-cell and network-level properties, offering a novel approach to directly test how neuromodulation influences neural computation.

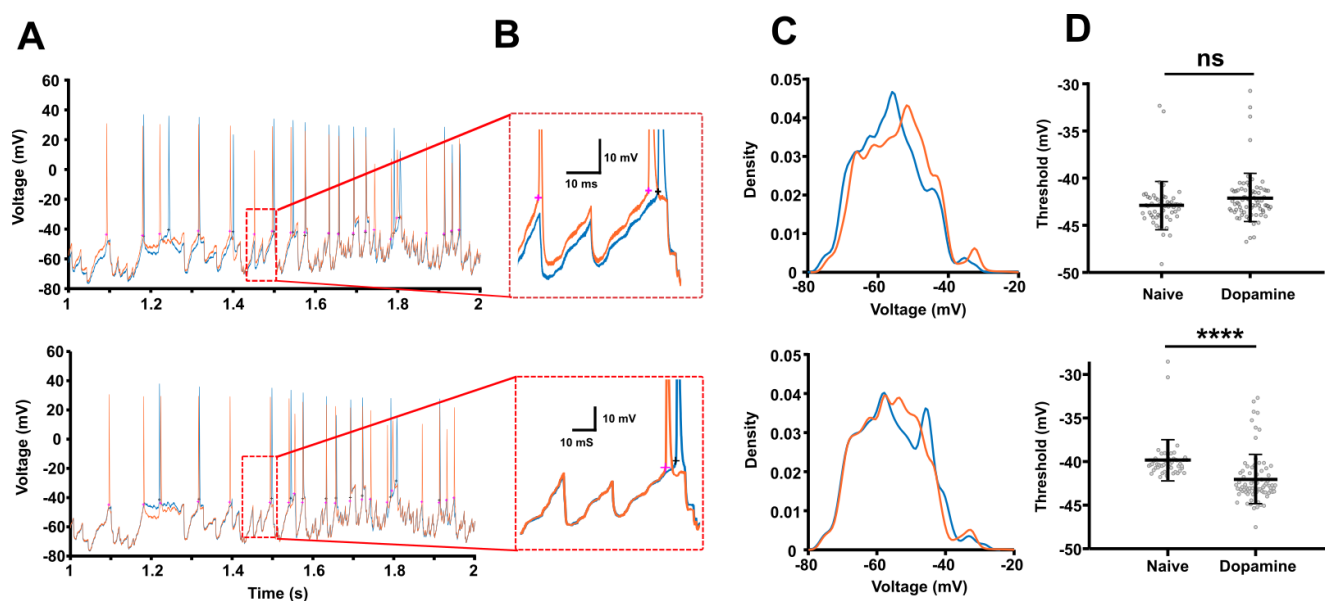


Figure 4-11. Subthreshold analysis of dopamine effect on intrinsic neural excitability.

(A) Two example voltage traces of network clamp stimulation during emergence of persistent bump phase, injected in two L5 pyramidal neurons in mouse prelimbic before (baseline) and after (dopamine) 20 μ M dopamine treatment. (B) Zoom in on the section of network clamp stimulation where dopamine treatment evokes generation of a new action potential. (C) Voltage distribution during bump emergence phase of network clamp stimulation in baseline (grey) and dopamine (orange) condition. (D) Threshold values during bump emergence phase of network clamp stimulation in baseline (grey) and dopamine (orange) condition. (paired t test ns $p = 0.0991$, **** $p < 0.0001$).

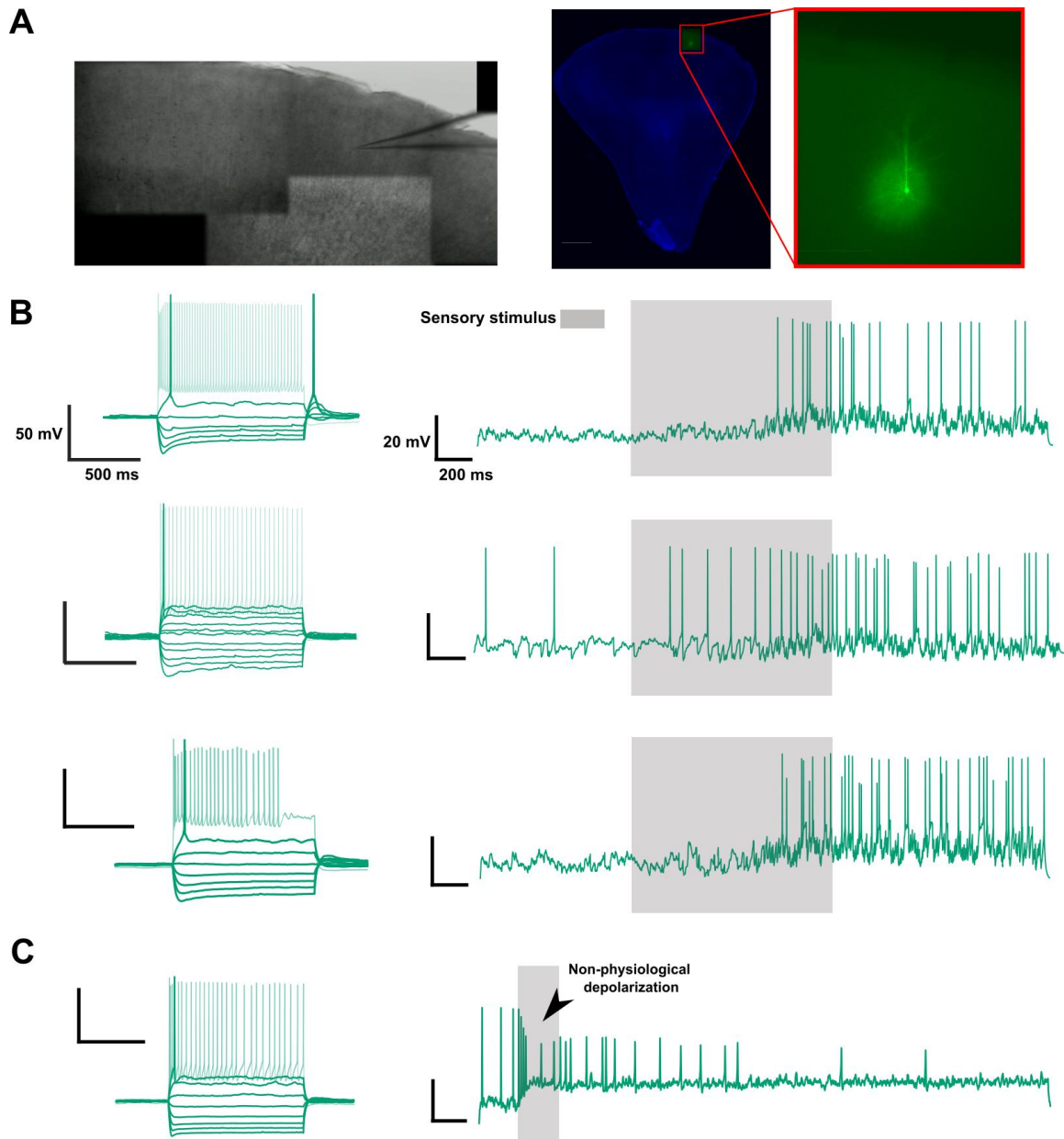


Figure 4-12 Network clamp recordings from pyramidal neurons in the marmoset PFC *in vitro*

(A) (Left) Bright-field microscope image during whole-cell patch-clamp recording of pyramidal neuron in area 46 of marmoset PFC slice. (Middle) Confocal image of a brain slice following post hoc labeling (blue DAPI, green streptavidin-Alexa Fluor568). (Right) Magnified image of the same neuron. (B) (Left) Representative voltage response from 3 pyramidal neurons to square current-step stimulation. (Right) Voltage response of the corresponding neurons to network clamp injection of clustered model conductance signals. (C) (Left) Voltage response traces of a pyramidal neuron receiving square current-step stimulation. (Right) Voltage response to network clamp injection of synaptic conductance signals from the ring model.

4.4 Discussion

Previous dynamic clamp studies have established a fundamental principle: the integrative properties of biological neurons are modulated in the context of spontaneous background activity (Chance et al., 2002; Destexhe et al., 2001; Fellous et al., 2003). However, these approaches typically rely on statistically defined noise processes, such as Ornstein-Uhlenbeck, to recapitulate the statistical properties of background activity. However, these synthetic processes are inherently limited: they model synaptic input as stationary noise, failing to capture the rich, non-stationary temporal structure and coordinated conductance fluctuations that emerge when a network actively performs a computation.

We go a step further by introducing network clamp; instead of using mathematical approximation, we use the signals generated by functional spiking neural network models presenting persistent activity. While previous studies have measured intrinsic properties of neurons in the presence of noise, network clamp addresses a different question. With network clamp, we ask how a network's architecture—tuned for a specific population dynamic—shapes the synapse signals generated in that network, and whether those signals remain within the range that a biological neuron can integrate. Indeed, with our network clamp method, signals features are not pre-defined by the experimenter but emerge directly from the model's dynamics and connectivity. We then use these validated emergent model-derived signals to ask how neuromodulatory action of dopamine reshapes neuronal firing within the context of a specific circuit state.

A key challenge in development of such a method has been generating network signals with an amplitude that biological neuron can integrate in a network that presents the dynamic of interest. Ring attractors models are typically implemented at relatively small scale (a few thousand neurons), which impose a constraint: with fewer synaptic inputs per neuron, each individual synapse must carry more weight to sustain recurrent excitation. To maintain sufficient recurrent activity at physiological rate, the ring model requires an overreliance on high amplitude slow conductance to maintain persistently firing neurons in a mean-driven regime, leading to a depolarization block in *in vitro* neurons (Fig. 4-3; Fig.4-4).

In this work, we injected biological neurons with inputs generated by two different spiking network models of persistent activity. In the established ring model, the scale and network connectivity require a strong increase of slow excitatory conductance to maintain a mean driven activity regime. In such regime, persistently firing neurons receive suprathreshold input. Using the network clamp method, we show that such strong stimulation causes biological *in vitro* neurons to enter sustained depolarization block and fail to generate action potentials. In contrast, the conductance signals generated in our large-scale clustered network model evoked subthreshold voltage fluctuations in biological neurons close to prediction (Fig. 4-6 D). Such injections of model-derived inputs from our model not only evoked persistent activity in *in vitro* neurons, but also correctly evoked absence of spiking and transition between states when injecting inputs of non-persistent neuron models or persistent firing interrupted by a distractor. Therefore, this appropriate functional contrast produced by our model and validated by network clamp *in vitro*, supports a mechanism for persistent activity where reverberation within discrete clusters in a large-scale network gives rise to a biologically plausible operational regime.

Our network clamp method combines the environmental control of *in vitro* experiments and the ability to rapidly iterate over different versions of network models. This makes it ideal to further our understanding of the interplay between single neurons and network dynamics within the PFC microcircuit. The neuromodulatory actions of dopamine exemplify this interaction, given its pivotal role in WM performance (Arnsten et al., 1994; Cai & Arnsten, 1997; Murphy et al., 1996; Zahrt et al., 1997) and delay persistent firing (Arnsten et al., 2015; Vijayraghavan et al., 2007; M. Wang et al., 2019; Williams & Goldman-Rakic, 1995).

Dopamine modulates PFC function at multiple levels. At the synaptic level, it enhances excitatory postsynaptic potentials (EPSPs) (Rotaru et al., 2007; Seamans et al., 2001). At the intrinsic level, dopamine alters the global excitability of neurons. *In vitro* studies in both primates and rodents have shown a shift in the input-output relationship of PFC neurons (Henze et al., 2000; Seamans & C. Yang, 2004).

Some studies attribute the increase in excitability and firing to a hyperpolarized action potential threshold (Henze et al., 2000; C. Yang & Seamans, 1996), while others report changes in input resistance, particularly in response to depolarizing current injection (Lavin & Grace, 2001; Penit-Soria et al., 1987; C. Yang & Seamans, 1996). Dissecting the specific contribution of neuromodulators like dopamine to WM has proven difficult *in vivo*. Because dopamine acts primarily through volume transmission, it is challenging to isolate its effects at the single-neuron level from its broader action on network dynamics (Özçete et al., 2024). Network clamp resolves this challenge by ‘isolating’ the network input *in silico*, limiting the effect of drugs to the postsynaptic neurons. Our findings show that dopamine enhances the gain of biological neurons to identical network inputs suggest a key role for the postsynaptic neuromodulation in dopamine’s action in WM. This isolation of variables provides a new avenue for testing the cellular mechanisms of drugs independent of circuit-level compensation.

Chapter 5

5 General Discussion and Future Directions

5.1 Summary of a translational investigation

This thesis has provided evidence for advance of network models of persistent activity by enhancing their biological realism in terms of scale, structure, and critically, their operational regime. In Chapter 2, we established the biological foundation of our work, reviewing physiological evidence that highlights the unique anatomy, histology, and electrophysiology of the dorsolateral prefrontal cortex (dlPFC), including the recurrent connectivity within “stripes” patterns, and the NMDA-receptor-expressing synapses thought to underlie working memory (WM) function. We also surveyed the computational models in the literature, noting that while seminal biophysical models like the ring attractor provided crucial insights, they are limited by their simplified connectivity and their circuit mechanism dominated by slow excitatory signal in a mean-driven regime. At the scale of those models, this results in activity that is overly regular compared to *in vivo* recordings (Barbieri, 2008; Compte, et al., 2003). We hypothesized that this mechanism requires unphysiologically strong excitatory conductance that may be biologically implausible.

In Chapter 3, we addressed the limitations of the ring attractor model by developing a large-scale network of 50,000 neurons with a clustered excitatory population. This model not only robustly generates persistent activity in a bump pattern but does so with firing statistics that better match *in vivo* recordings from the dlPFC during WM tasks. The elevated coefficient of variation ($CV \approx 0.88$) in our model, compared to the ring model ($CV \approx 0.59$), provides strong support for the hypothesis that a subthreshold stimulation supporting fluctuations driven activity is more plausible mechanism for network-driven persistent activity in the cortex.

In Chapter 4, we introduced and deployed the network clamp method, a novel methodology that creates a direct bridge between *in silico* network models and *in vitro* electrophysiology. This allowed us to perform the first direct test of a model’s synaptic operational regime in a biological neuron.

The response evoked by the two models clearly show that synaptic signals from the ring model evoked unphysiological responses, in the form of sustained depolarization block, when presented to *in vitro* neurons, while our clustered model evoked persistent spiking at plausible rates. Our control experiments confirmed that our model successfully captures the different scales, as inputs from non-stimulated clusters did not evoke persistent firing, as expected. Furthermore, we demonstrated the fidelity of network clamp stimulation, as biological neurons accurately followed predicted state transitions when we injected signals from model neurons responding to a distractor stimulus.

In a crucial proof-of-concept, we used our network clamp method to investigate the effects of dopamine, a key neuromodulator in PFC-dependent WM, within the context of realistic, network-derived synaptic input. We found that dopamine consistently increased persistent activity rates, and our subthreshold analysis revealed that this enhancement can arise through diverse changes in intrinsic neuronal properties, highlighting the strong potential of our method to dissect pharmacological dynamics in a unique context.

5.2 An iterative dialogue: Using experiments to validate models and inform models refinements

An important point to be considered about the network clamp method is that the model-derived synaptic inputs are injected directly into the soma of biological neurons, via whole-cell patch-clamp configuration. To study the integrative properties of neurons in an “*in vivo* like” context, it would be possible to pass the synaptic signals through a compartment model of dendritic arborescence before injecting the signal in the biological neurons.

In the present study, network clamp is primarily used to test the model derived signals against biological reality rather than study the dynamic of dendritic integration. Network clamp experiments can be used as a central part of a dialogue between experimental and computational neuroscience, as this testing of models can provide feedback and direction to refine the model and improve its plausibility. The failure of ring model inputs to evoke persistent activity (Chapter 4, Fig. 4-3) indicated that a regime dominated by sustained suprathreshold NMDA excitation is not viable. This motivated us to develop a network model that maintains persistent activity through a more cohesive synaptic regime—one that generates fluctuation-driven voltage dynamic rather than mean-driven suprathreshold depolarization. The resulting large-scale clustered model achieved this: injection of its synaptic signals successfully evoked persistent activity in biological neurons. However, notable differences between the model's prediction and the biological response remain.

In chapter 4, figure 4-7 D shows there is a discrepancy in firing rate between model neurons and firing rate of biological neurons receiving network clamp injection of the inputs received by model neuron. Figure 4-6 D shows instance where model neuron fires 4 action potentials in rapid succession but the biological neuron failed to fire two of these action potentials. This dynamic could likely be replicated in the model neuron by implementing a spike frequency adaptation mechanism to replicate this behavior. Implementation of this could be informed by the characterization of the biological neuron response. The implementation of such mechanisms would likely change the network level dynamics. It might increase the irregularity of persistent activity to even higher levels or compromise the attractor dynamics of the network entirely.

This iterative dialogue represents a challenge by incentivizing researchers to implement biological constraints but provides justification for the choices made and direction of development of models of cortical dynamics.

The successful implementation of the network clamp method in this work presents a promising potential application beyond model validation. Specifically, network clamp provides a framework to dissect pharmacological mechanisms in the controlled environment of patch-clamp electrophysiology.

We showcased this by studying how dopamine increases intrinsic excitability of neurons when they receive the same repeatable model-derived synaptic input. This approach offers a unique advantage because we can isolate the effect of a neuromodulator on the post-synaptic neuron from its effect on the network as a whole. Indeed, with network clamp the conductance input to the neuron is held constant, so any changes in firing must arise from changes in the neuron's intrinsic properties or its integration of those inputs.

As part of the iterative dialogue between experiment and computation, biological discoveries made through network clamp can be applied back to the model to study their effect at the network level. A future direction for this project is to implement the observed dopaminergic effects into our large-scale clustered model to study how they shape network-level dynamics. This would provide a rapid way of bringing experimentally supported evidence to the computational study of WM circuits. We predict that incorporating dopamine modulation will sharpen the bump activity profile and enhance the network's robustness to distractor as theorized by previous studies.

Finally, as the effectiveness of large-scale simulations increases, it will open the possibility to run simulation in parallel to network clamp experiments. In this approach, the spike output of biological neurons would be fed back into large scale clustered model in real time, creating a true hybrid network. This would allow us to study how a single biological neuron can influence the dynamics of a large-scale model, offering unprecedented insight into the interplay between single-cell properties and emergent network computation.

5.3 On the novelty of network model and the network clamp method

Dynamic clamp has been used for decades to inject artificial conductances into real neurons (Robinson & Kawai, 1993; Sharp et al., 1993; Prinz et al., 2004). It has enabled the creation of artificial synaptic and electrical coupling between cells, the addition or subtraction of voltage-dependent channels to study their impact on excitability or firing patterns.

Single synapse conductances have been used to study synaptic integration, short-term plasticity rules, and shunting inhibition in a highly parametric way (Hughes et al., 1998; Kemenes et al., 2011).

A major application has been to simulate the effect of synaptic background activity. Foundational studies have used stochastic signals that recapitulate the statistical properties of spontaneous synaptic background activity. These studies have employed Ornstein–Uhlenbeck (OU) processes, which reproduce *in vivo* like high conductance states and demonstrate how synaptic background activity modulates gain, input resistance, and firing statistics (Chance et al., 2002; Destexhe et al., 2001; Fellous et al., 2003). However, because these prescribed signals are set by experimenters rather than generated by a network of interacting neurons, they are limited in the questions they can address about mechanisms such as state transitions. In the context of persistent activity: how do the amplitude, balance, or temporal structure of conductance inputs differ between baseline and persistent-activity states ? OU noise, being stationary by construction, cannot answer this question.

In contrast, with network clamp the dynamic signals emerge from a spiking neural network that actively performs a computation. In the present work network clamp allows us to test the plausibility of synaptic regime of two models. This allows us to compare the effect of mean-driven synaptic regime (Chapter 4, Fig. 4-2C) and fluctuation driven synaptic regime (Chapter 4, Fig. 4-3C). Studies have shown that cortical activity is thought to be driven by inputs below threshold on average driving voltage fluctuations, with spikes being triggered by rapid fluctuations (Kuhn et al., 2004; Nowak, 1997; Rosenbaum et al., 2017).

It had been theorized that an attractor model could present such operational regime if excitation and inhibition both showed an increase transition between baseline and persistent activity state (Barbieri, 2008; Renart et al., 2007). We showed that by implementing a large scale, a discrete attractor network can present cohesive increase of both excitatory ($g_{e_{fast}}, g_{e_{slow}}$) and inhibitory conductance inputs, resulting in more irregular firing.

With smaller excitatory amplitude and higher amplitude inhibition model neurons receive inputs that maintain voltage below threshold on average, with voltage showing an important variance. Particularly fluctuation in inhibitory conductance coinciding with spike timing, in line with experimental estimate (Rudolph et al., 2007). By injecting these model-derived conductance signals in biological neurons, we show that this operational regime is coherent with biological scale.

Our model is the first attractor model, to our knowledge, to be able to evoke plausible response. We used network clamp in previous work (Ichiyama et al., 2022). In future applications, other models could be used in network clamp experiments to study variety of questions.

Dynamic clamp has also been used in previous work in what has been described as hybrid networks (Le Masson et al., 1995). But those approaches have used at most one neuron model, similar to cell coupling or using simpler mathematical approximation as gating function (Le Masson et al., 2002; R. D. Pinto et al., 2000; Varona et al., 2001). With network clamp, by contrast, biological neuron receives input from an effective network model actively performing computation, this requires the plausibility of the synaptic operational regime to be considered during the development of the model. Furthermore, network clamp approach as presented does not constitute a full interface between network model and biological neuron.

But as simulation hardware and dynamic clamp methods continue to advance, it will become possible to simulate complex networks—such as our large-scale attractor—during network clamp experiments, and to feed the output of the biological neuron back into the simulation in a closed-loop approach. Similarly, increasingly detailed biological constraints will be implemented, even in such large-scale networks. The results presented in this thesis can encourage researchers to adopt the synaptic operational regime as a metric that can be tested directly *in vitro* and to pursue questions at the dynamic interface between experiment and computational theory.

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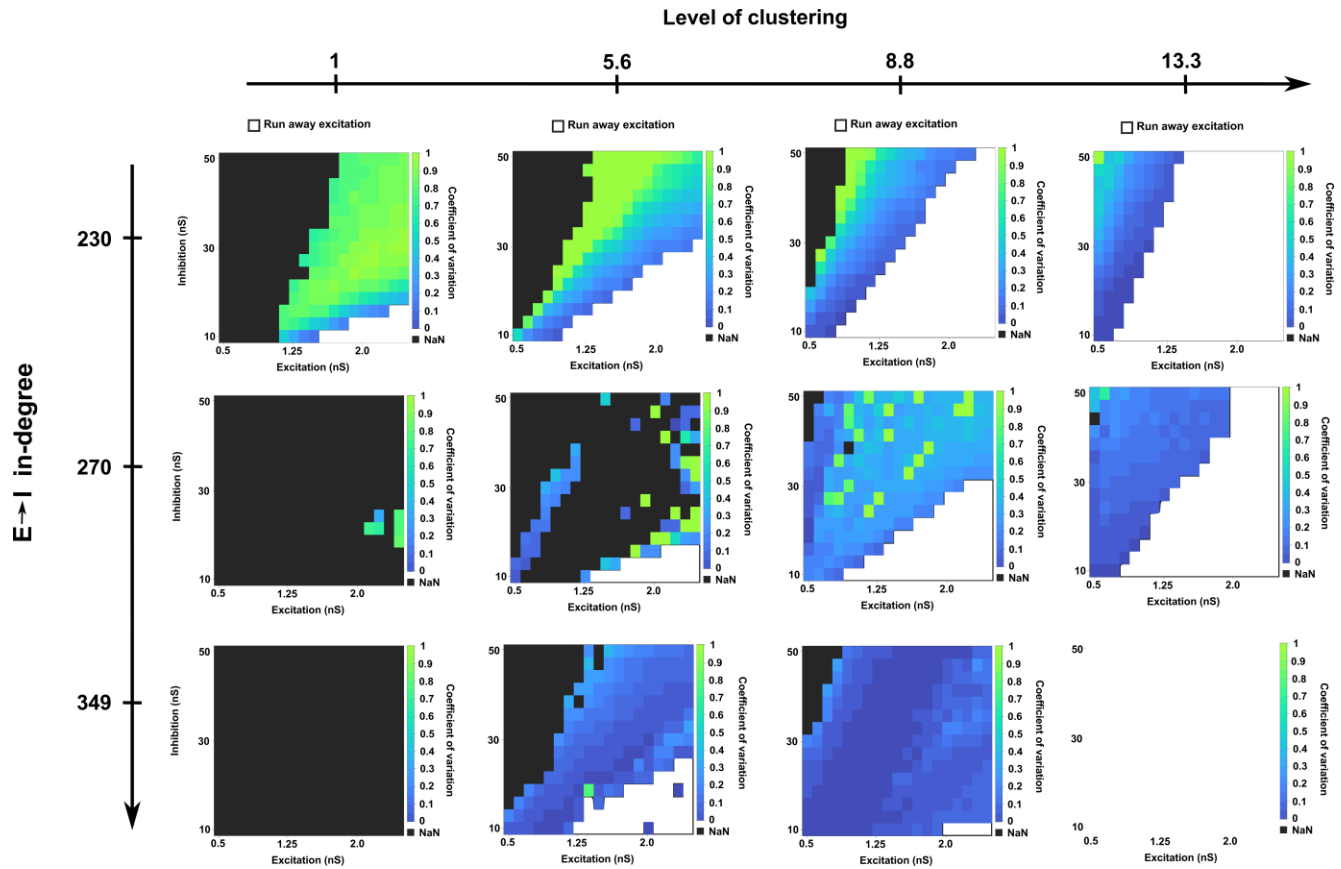
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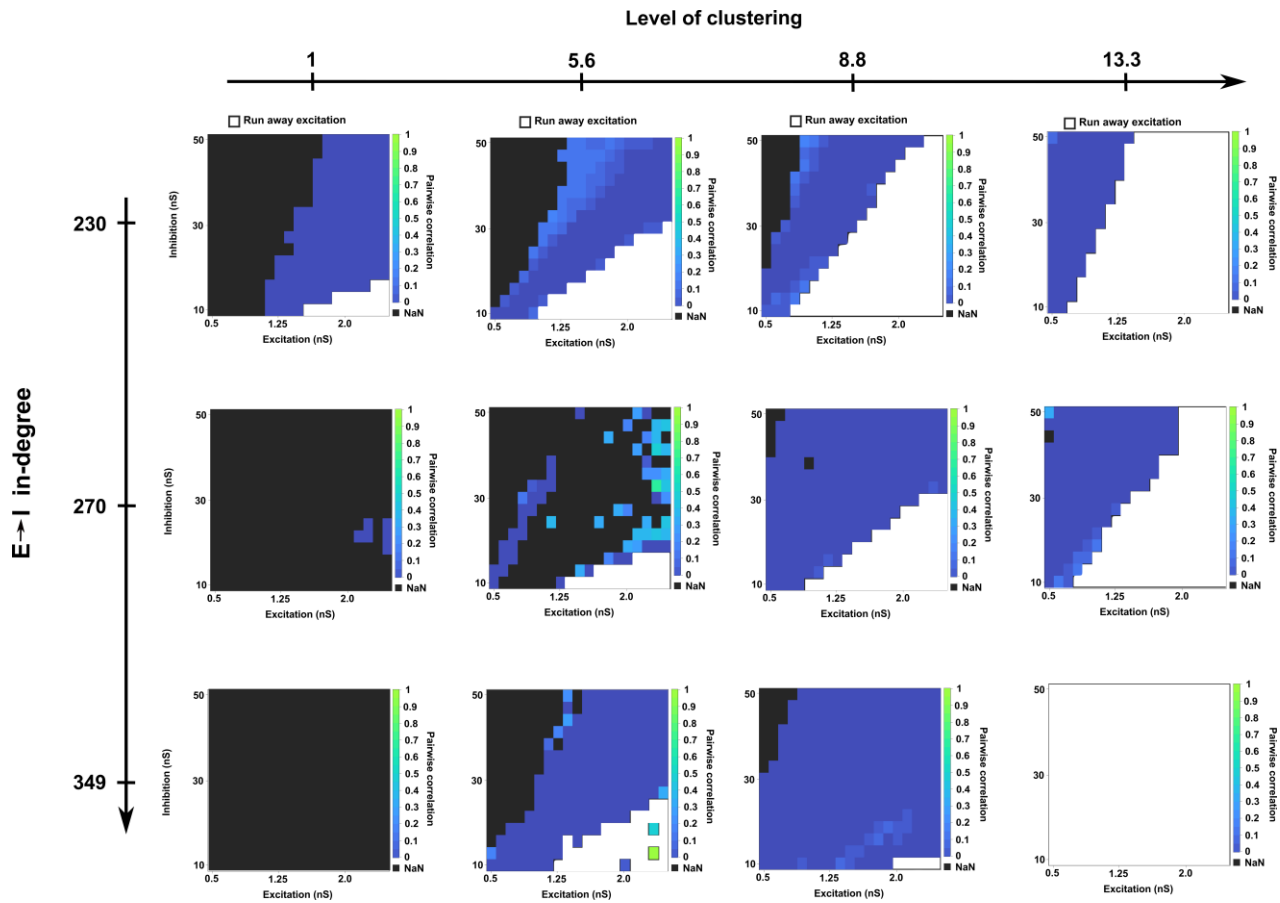
Review of Neuroscience, 40(1), 603-627. <https://doi.org/10.1146/annurev-neuro-070815-014006>

Appendix



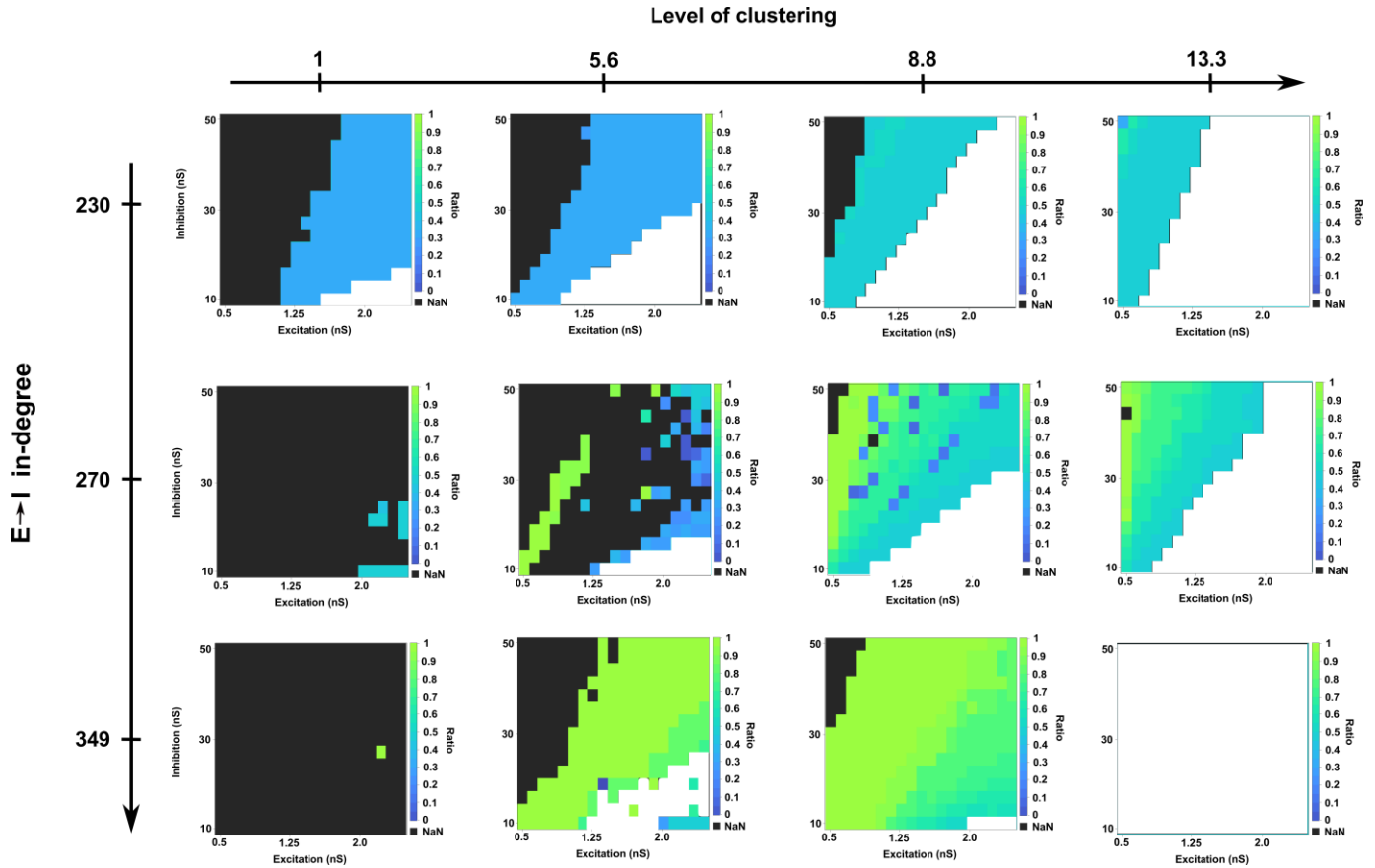
Appendix A. Coefficient of variation (CV) of the interspike intervals (ISIs) of persistently firing neurons as a function of connections weights (Excitation/Inhibition), for varying level of clustering and $E \rightarrow I$ in-degree in clustered network of size $N=10,000$.

Grid of heat maps presenting the CV of ISIs of persistent firing neurons observed in a clustered network of size 10,000 across the parameter space for varying level of clustering (1(no clustering), 5.6, 8.8 and 13.3) and $E \rightarrow I$ in-degree (230 (unbiased), 270, 349).



Appendix B. Pairwise correlation of persistently firing neurons as a function of connections weights (Excitation/Inhibition), for varying level of clustering and $E \rightarrow I$ in-degree in clustered network of size $N=10,000$.

Grid of heat maps presenting the pairwise correlation between persistent firing neurons observed in a clustered network of size 10,000 across the parameter space for varying level of clustering (1 (no clustering), 5.6, 8.8 and 13.3) and $E \rightarrow I$ in-degree (230 (unbiased), 270, 349).



Appendix C. Activity pattern ratio as a function of connections weights (Excitation/Inhibition), for varying level of clustering and $E \rightarrow I$ in-degree in clustered network of size $N=10,000$..

Grid of heat maps presenting the activity pattern ratio between persistently firing neurons activity rate and background neurons observed in a clustered network of size 10,000 across the parameter space for varying level of clustering (1 (no clustering), 5.6, 8.8 and 13.3) and $E \rightarrow I$ in-degree (230 (unbiased), 270, 349).

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